ThreatModel for Azure Storage

Introduction

Read the blog: The last Azure Storage security document that we'll ever need.

Content

This publication includes:

- overall data flow diagram of Azure Storage

- overview of the Mitre ATT&CK matrix for Azure Storage

- prioritized list of all threat scenarios

- list of all the control activities and testing procedures

- risk-based prioritized list of control implementation

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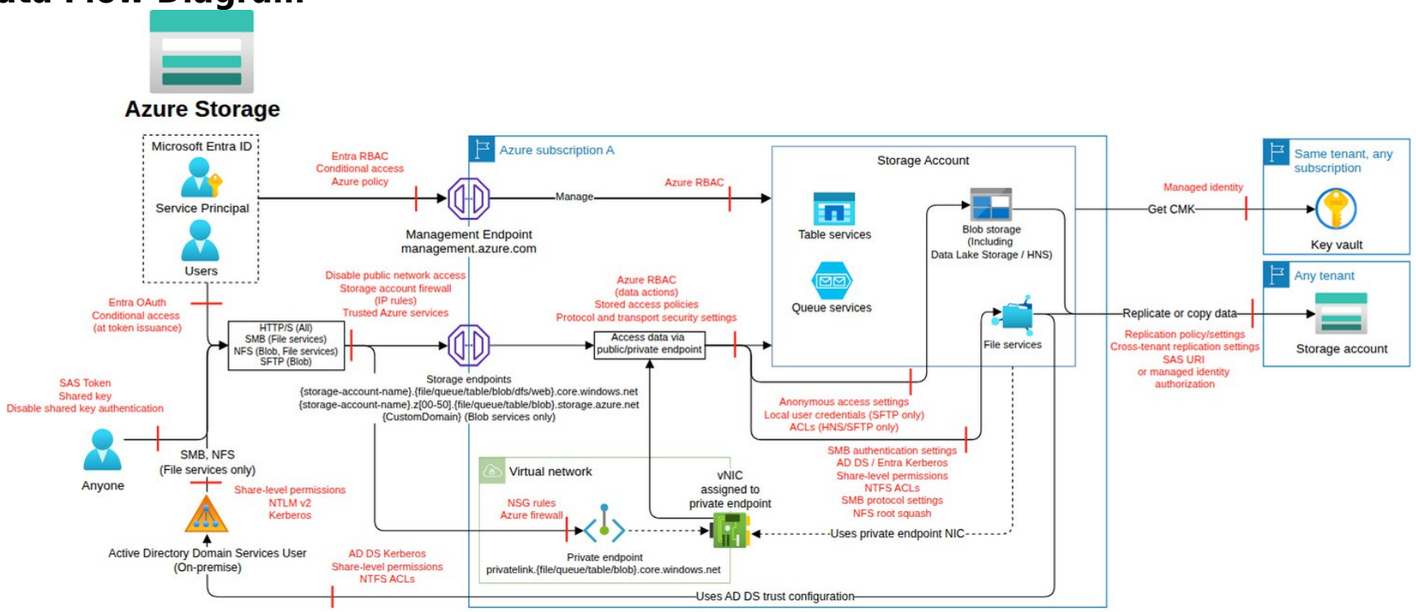
Source

The latest version of this work is hosted on GitHub.

Contact

If you have any questions, please contact chatbot@trustoncloud.com.

Data Flow Diagram



Security Scorecard

Security in the Cloud	
Number of Actions*	206
Identity management	Azure RBAC, storage account key, local SFTP users, Entra ID, AD DS, Azure AD Kerberos
Number of IAM permissions*	161
Resource-based access	file share ACL, queue ACL, table ACL, storage account access keys, SAS tokens
Logging coverage for APIs	56.7% (missing 78)
Number of Logging Event Names*	116
Network Filtering	VNET security, Storage Account Firewall, Private endpoint
Encryption-at-rest	Yes
Encryption-in-transit	Yes

* See details in Appendixes

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Mitre ATT&CK matrix for Azure Storage

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
		Unauthorized access path through virtual network rule changes [Storage.T50]	Data manipulation through reuse of a deleted storage account name [Storage.T4]	Service impact through unauthorized modification of blob content [Storage.T12]	Privilege escalation by modifying Azure file share permissions [Storage.T17]			Credential access by listing or regenerating storage account access keys [Storage.T1]	Blob inventory discovery through inventory access [Storage.T24]				Unauthorized access to storage data using a compromised access key or SAS token [Storage.T3]	Service impact through regeneration of storage account access keys [Storage.T2]
		File share access through SMB Shared Key [Storage.T65]		Distribute malicious content via unauthorized upload [Storage.T20]	Data exposure through CORS configuration changes [Storage.T26]			Credential access through SMB Shared Key retrieval [Storage.T64]					Unauthorized access to blob data by enabling anonymous public access [Storage.T5]	Service impact through recursive deletion of hierarchical namespace content [Storage.T7]
		Public access path through account configuration [Storage.T68]		Persistent access through SFTP local user credentials [Storage.T44]	Privilege escalation through queue access policy modification [Storage.T27]								Data exposure through storage account replication [Storage.T13]	Data manipulation through unauthorized modification of blob data [Storage.T8]
		Shared Key access path through the account configuration [Storage.T69]			Privilege escalation through table access policy modification [Storage.T28]								Unauthorized access to Azure file share contents [Storage.T15]	Encrypt blob contents for ransom using authorized access [Storage.T9]
					Privilege escalation through hierarchical namespace ACL modification [Storage.T66]								Unauthorized access to blob data through overly permissive Azure RBAC assignments [Storage.T48]	Financial impact through unauthorized upload of Azure file share contents [Storage.T16]
													Unauthorized access to blob data by renaming a path used by Azure RBAC assignments [Storage.T55]	Service impact through recursive deletion of Azure file share contents [Storage.T18]
													Unauthorized access to blob data through overly permissive hierarchical namespace ACLs [Storage.T58]	Impact through unauthorized encryption of Azure file share contents [Storage.T19]
													Data exposure through file share encryption changes [Storage.T60]	Publish malicious content by modifying static website blobs [Storage.T22]
													Data exposure through legacy file share protocols [Storage.T61]	Service impact through lifecycle management changes [Storage.T25]
													Data exposure through unauthorized replication policy [Storage.T63]	Service impact through queue message modification or deletion [Storage.T31]

													Unauthorized access to sensitive queue message contents [Storage.T67]	Financial impact through query acceleration usage [Storage.T34]
														Service impact through encryption scope modification [Storage.T38]
														Service impact through disabling blob soft delete [Storage.T39]
														Impact of disabling blob versioning [Storage.T40]
														Service impact through replication configuration changes [Storage.T42]
														Impact by modifying blob container immutability protections [Storage.T56]
														Unauthorized failover through configuration change [Storage.T59]
														Service impact through private endpoint deletion [Storage.T62]

Feature Classes

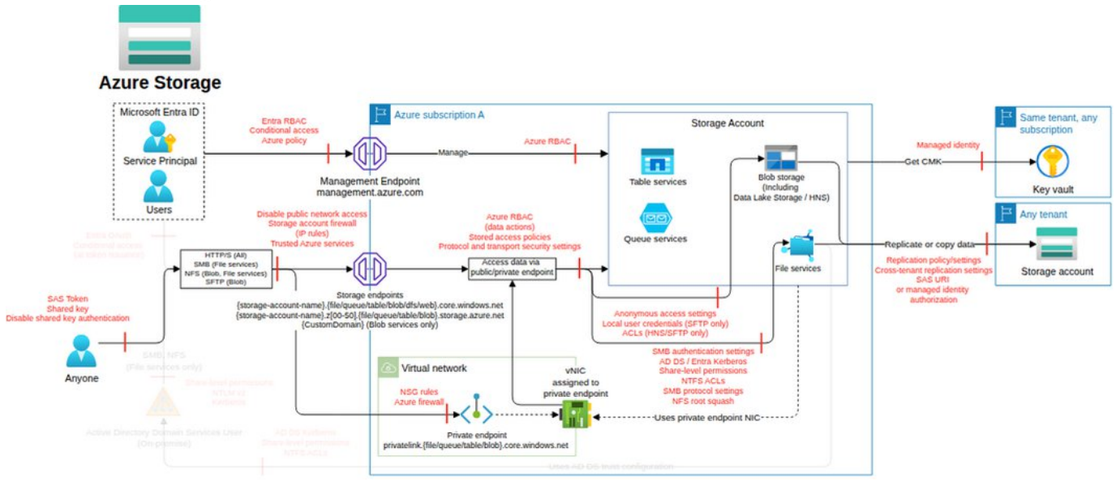
Azure Storage has the following feature classes and subclasses (i.e. dependent on the usage of its class) that can be activated, restricted, or blocked using Microsoft Azure Identity and Access Management.

Feature	Relation	Description
Storage account (FC1)	class	Azure Storage is Microsoft's Cloud Storage solution for modern data storage scenarios. Azure Storage offers a scalable object store for data objects, a file system service for the cloud, a messaging store for reliable messaging, and a NoSQL store.
Blob storage (FC2)	subclass of Storage account	Object storage solution for storing unstructured data (blobs) accessible via HTTP/S and optionally via the Network File System (NFS) v3 and SFTP protocols.
Blob inventory (FC10)	subclass of Blob storage	The Azure Storage blob inventory feature provides an overview of your containers, blobs, snapshots, and blob versions within a storage account. Use the inventory report to understand various attributes of blobs and containers, such as the total data size, age, encryption status, immutability policy, and legal hold.
File shares (FC3)	subclass of Storage account	Azure Files offers fully governed file shares in the cloud accessible via Server Message Block (SMB) protocol, Network File System (NFS) v4.1 protocol, or the in-preview Azure Files REST API.
Queues (FC4)	subclass of Storage account	Azure Queue Storage is a service for storing messages. Access messages via HTTP/S calls.
Tables (FC5)	subclass of Storage account	Azure table storage is a service that stores non-relational structured data (or structured NoSQL data) in the cloud, providing a key/attribute store with a schemaless design.
Local users (FC11)	subclass of Storage account	Blob storage supports the SSH File Transfer Protocol (SFTP). This support lets you securely connect to blob storage via an SFTP endpoint, allowing you to use SFTP for file access, file transfer, and file management.
Private endpoints (FC12)	subclass of Storage account	Azure Storage private endpoint is a network interface that enables secure, private connectivity between Azure Storage and virtual networks using Azure Private Link, preventing exposure to the public internet and enhancing security.

Storage account *(class, FC1)*

Azure Storage is Microsoft's Cloud Storage solution for modern data storage scenarios. Azure Storage offers a scalable object store for data objects, a file system service for the cloud, a messaging store for reliable messaging, and a NoSQL store.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

Action	IAM Permission
Asynchronously creates a new storage account with the specified parameters. If an account is already created and a subsequent create request is issued with different properties, the account properties will be updated. If an account is already created and a subsequent create or update request is issued with the exact same set of properties, the request will succeed. The update operation can be used to update the SKU, encryption, access tier, or tags for a storage account.	Microsoft.Storage/storageAccounts/write
Creates or updates the diagnostic setting for the resource.	Microsoft.Storage/storageAccounts/providers/Microsoft.Insights/diagnosticsettings/write
Creates or updates the diagnostic setting for the resource.	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/diagnosticsettings/write
Creates or updates the diagnostic setting for the resource.	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/diagnosticsettings/write
Creates or updates the diagnostic setting for the resource.	Microsoft.Storage/storageAccounts/fileServices/providers/Microsoft.Insights/diagnosticsettings/write
Creates or updates the diagnostic setting for the resource.	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/diagnosticsettings/write
Create or update the object replication policy of the storage account.	Microsoft.Storage/storageAccounts/objectReplicationPolicies/write

Threat List

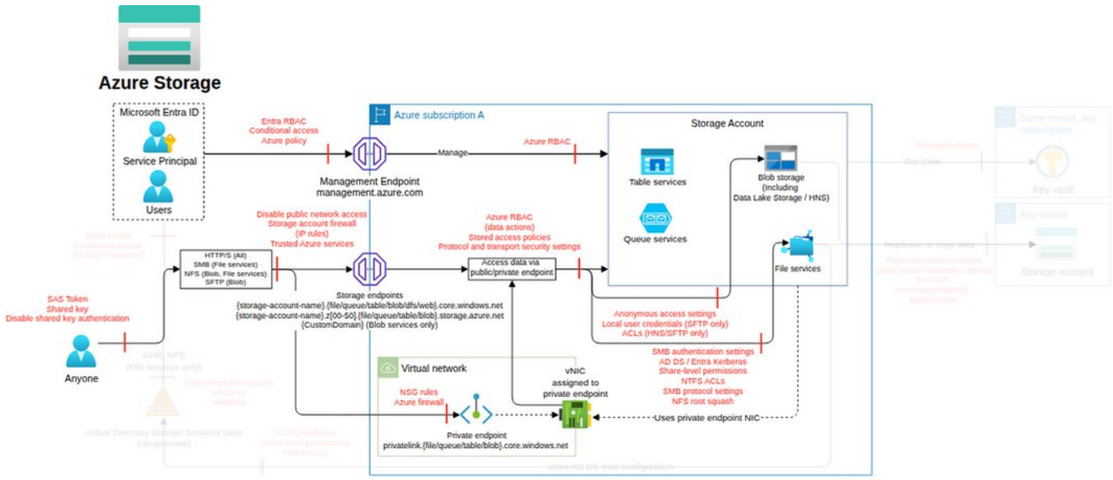
Name	CVSS
Credential access by listing or regenerating storage account access keys	High (8.1)
Unauthorized access to storage data using a compromised access key or SAS token	High (7.5)
Data exposure through unauthorized replication policy	Medium (5.7)
Data exposure through storage account replication	Medium (5.2)
Data manipulation through reuse of a deleted storage account name	Medium (5.2)
Service impact through lifecycle management changes	Medium (5.2)
Service impact through disabling blob soft delete	Medium (5.2)

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Service impact through encryption scope modification	Medium (4.8)
Service impact through regeneration of storage account access keys	Medium (4.5)
Data exposure through CORS configuration changes	Medium (4.0)
Service impact through replication configuration changes	Low (3.5)
Unauthorized failover through configuration change	Low (3.5)
Unauthorized access path through virtual network rule changes	Low (2.4)
Public access path through account configuration	Low (2.4)
Shared Key access path through the account configuration	Low (2.4)

Credential access by listing or regenerating storage account access keys

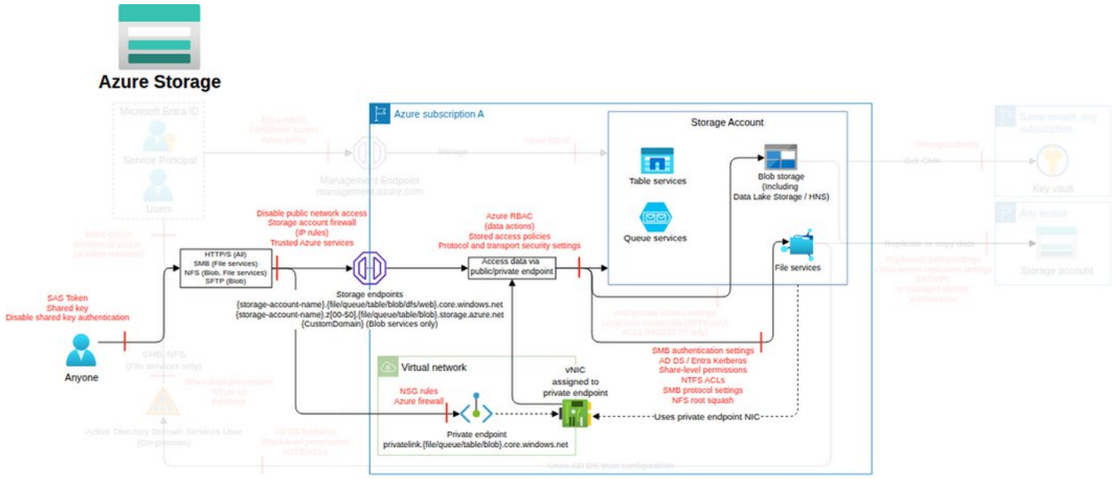
Threat Id	Storage.T1
Name	Credential access by listing or regenerating storage account access keys
Description	Storage account access keys provide broad authorization to storage account resources. An attacker can list, regenerate, or rotate storage account access keys to obtain a valid Shared Key and enable later unauthorized access to storage data.
Goal	Launch another attack
MITRE ATT&CK®	TA0006
CVSS	High (8.1)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/listKeys/action", "Microsoft.Storage/storageAccounts/regenerateKey/action", "Microsoft.Storage/storageAccounts/rotateKey/action"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C44 - Prevent storage account firewall rules from allowing unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode). C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	Very High	8	1	-

Unauthorized access to storage data using a compromised access key or SAS token

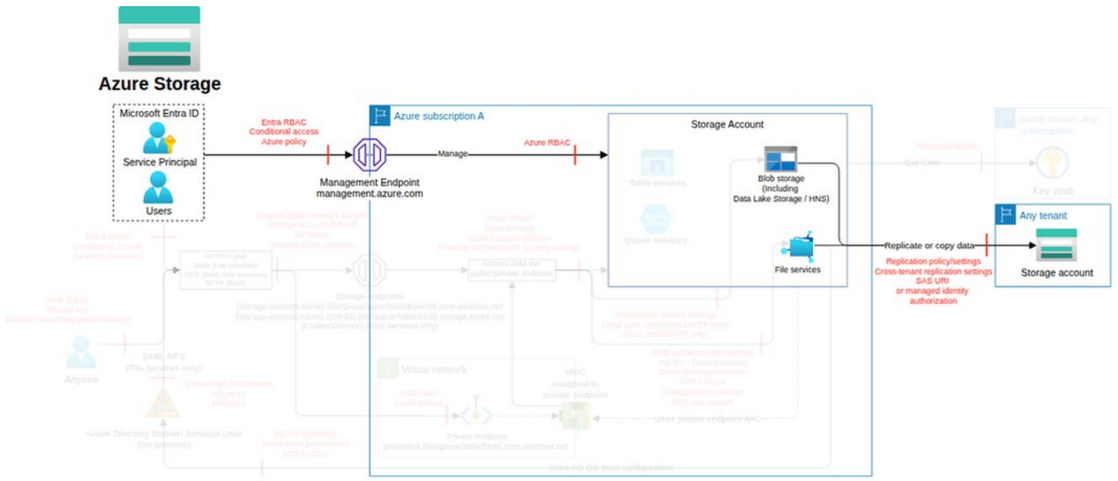
Threat Id	Storage.T3
Name	Unauthorized access to storage data using a compromised access key or SAS token
Description	Storage account access keys and Shared Access Signature tokens authorize access to storage resources based on the credential's effective scope. An attacker can use a compromised storage account access key, connection string, or SAS URL to read storage data outside the intended authorization path.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	High (7.5)
IAM Access	{}



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Medium	4	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	1	-	2

Data exposure through unauthorized replication policy

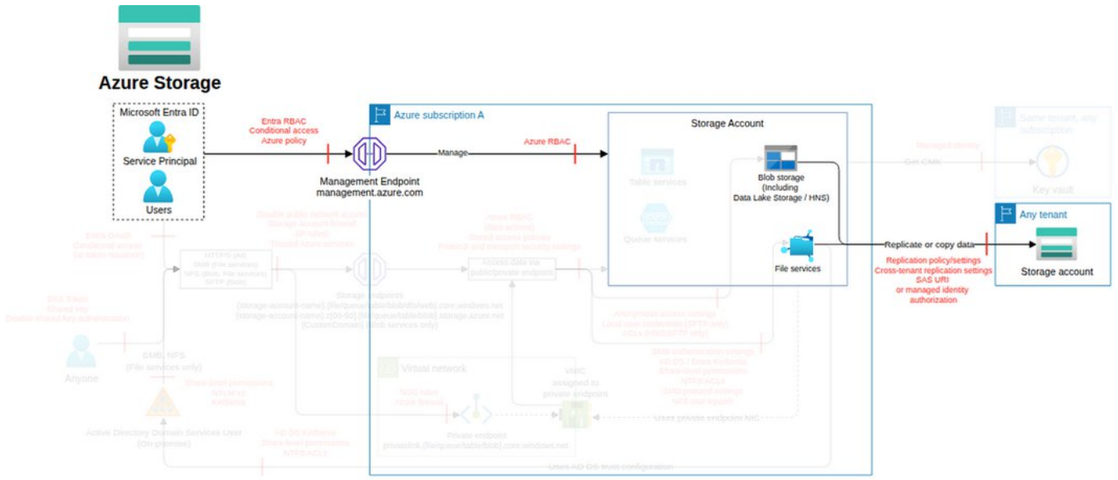
Threat Id	Storage.T63
Name	Data exposure through unauthorized replication policy
Description	Storage replication policies can continuously replicate blob data, metadata, and tags from a source storage account to a destination storage account. An attacker can configure an unauthorized replication policy to continuously copy storage data to an unauthorized destination, including a storage account in another subscription or tenant.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.7)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C107 - Maintain the list of storage accounts required to use Azure resource locks, following the Azure Resource Manager ThreatModel. C108 - Ensure required storage accounts use Azure resource locks. C110 - Prevent modification or deletion of required storage accounts by using a resource lock set to read-only and/or delete, using the Azure Resource Manager ThreatModel.	Very High	2	1	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C47 - Maintain a list of storage accounts that require cross-tenant replication and their allowed destination tenants. C48 - Ensure only required storage accounts have cross-tenant replication enabled and the destination storage accounts are authorized.	Medium	2	-	-

Data exposure through storage account replication

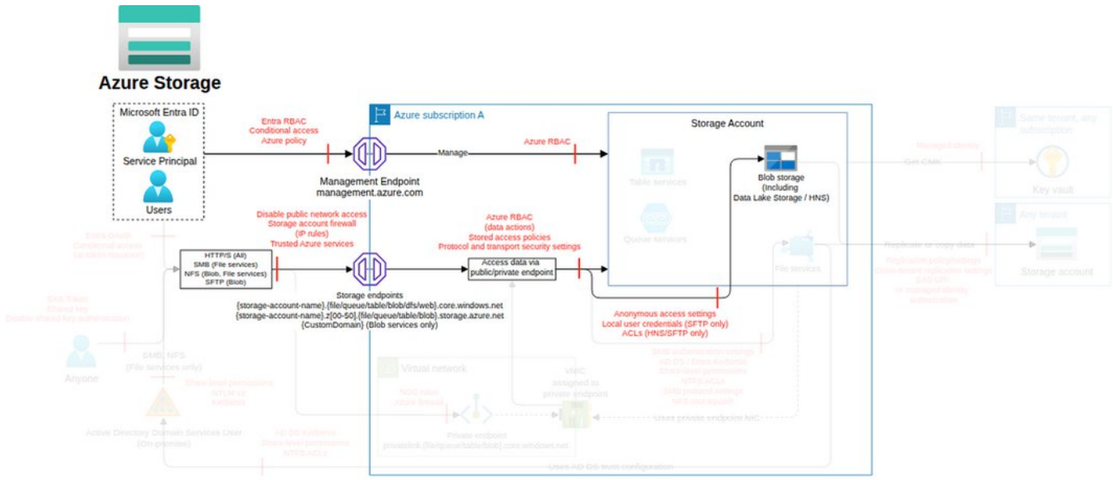
Threat Id	Storage.T13
Name	Data exposure through storage account replication
Description	Azure Storage object replication can copy blobs, metadata, and tags from a source storage account to a destination storage account. An attacker can configure replication to copy storage data to an unauthorized destination, exposing replicated content outside the intended storage boundary.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C47 - Maintain a list of storage accounts that require cross-tenant replication and their allowed destination tenants. C48 - Ensure only required storage accounts have cross-tenant replication enabled and the destination storage accounts are authorized. C147 - Maintain a list of storage blobs that require Azure Backup. C150 - Maintain a list of authorized replication policies and their destination storage accounts. C151 - Ensure replication policies and their destination storage accounts are authorized.	Very High	5	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO11 - Enable enhanced monitoring and notifications for storage accounts C50 - Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations. C51 - Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Very Low	2	-	-

Data manipulation through reuse of a deleted storage account name

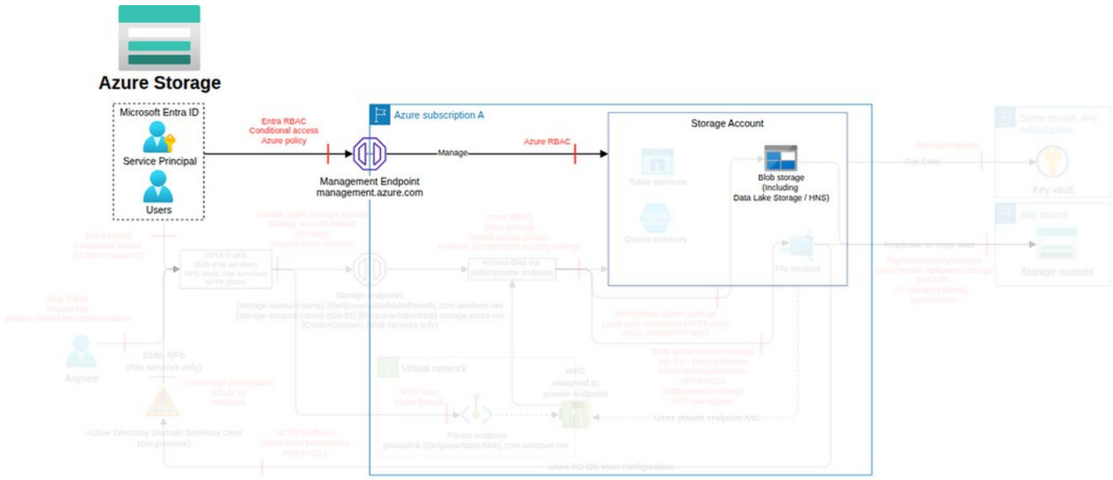
Threat Id	Storage.T4
Name	Data manipulation through reuse of a deleted storage account name
Description	Azure Storage account names are globally unique and can be referenced by applications, users, or third parties. An attacker can delete a storage account and recreate a new storage account with the same name to host misleading, malicious, or attacker-controlled data at a previously trusted endpoint.
Goal	Data manipulation
MITRE ATT&CK®	TA0002
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/delete" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C107 - Maintain the list of storage accounts required to use Azure resource locks, following the Azure Resource Manager ThreatModel. C108 - Ensure required storage accounts use Azure resource locks.	Very High	2	-	-

Service impact through lifecycle management changes

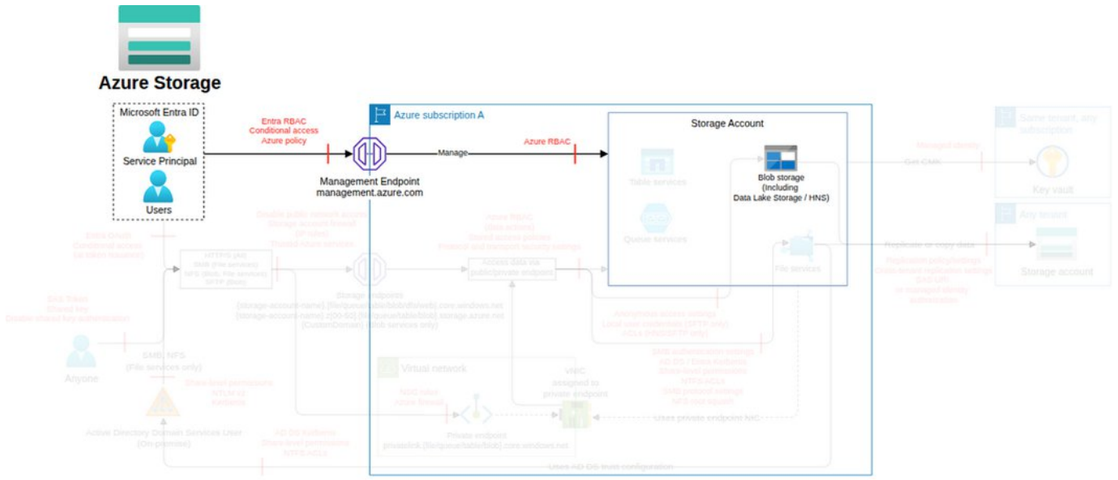
Threat Id	Storage.T25
Name	Service impact through lifecycle management changes
Description	Blob Storage lifecycle management applies rules that transition or expire blob data based on age, access, or metadata conditions. An attacker can modify lifecycle management rules to delete blob data or move it to colder tiers, reducing data availability or increasing recovery time.
Goal	Data manipulation
MITRE ATT&CK®	TA0040
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/managementPolicies/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C15 - Maintain a list of the blob storage containers that are required to have a minimum retention period enabled. C17 - Prevent the creation of required storage accounts without the blob soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/deleteRetentionPolicy.enabled": false} in Deny mode). C19 - Ensure required storage accounts have the soft-delete option enabled for the containers. C20 - Prevent the creation of containers without the soft-delete option enabled (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/blobServices/containers/softDelete" in Deny mode).	Low	2	2	-

Service impact through disabling blob soft delete

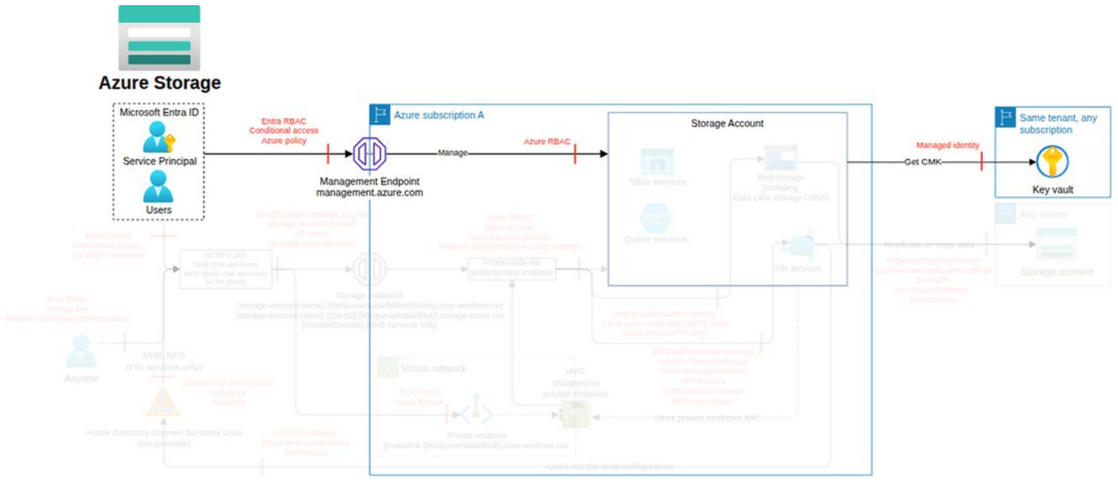
Threat Id	Storage.T39
Name	Service impact through disabling blob soft delete
Description	Blob soft delete allows deleted blob data to be recovered during a configured retention period. An attacker can disable blob soft delete to reduce recovery options after later unauthorized deletion or overwrite of blob data.
Goal	Launch another attack
MITRE ATT&CK®	TA0040
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C15 - Maintain a list of the blob storage containers that are required to have a minimum retention period enabled. C17 - Prevent the creation of required storage accounts without the blob soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/deleteRetentionPolicy.enabled": false} in Deny mode). C19 - Ensure required storage accounts have the soft-delete option enabled for the containers. C20 - Prevent the creation of containers without the soft-delete option enabled (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/blobServices/containers/softDelete" in Deny mode).	Very Low	2	2	-

Service impact through encryption scope modification

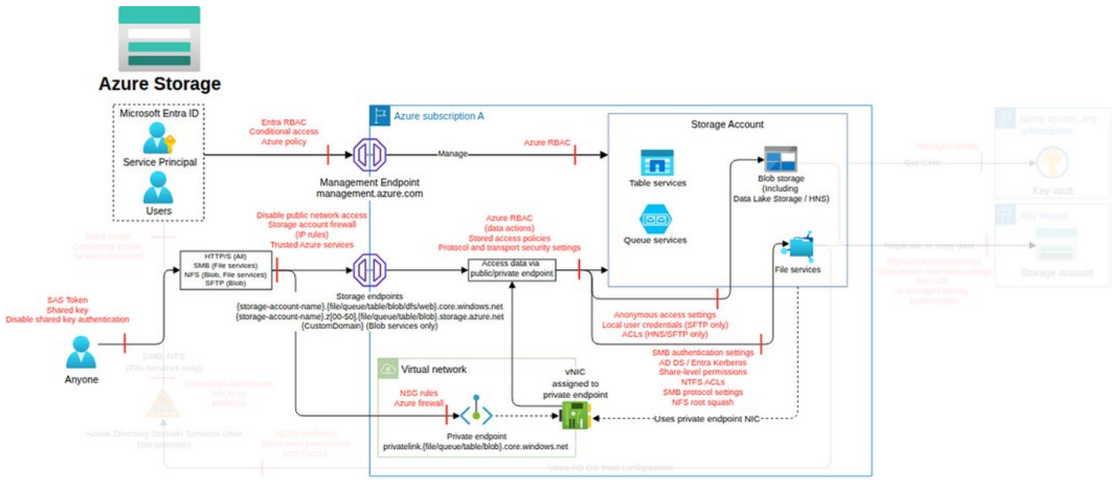
Threat Id	Storage.T38
Name	Service impact through encryption scope modification
Description	Azure Storage encryption scopes define the encryption settings and keys used to protect blob data. An attacker can modify an encryption scope to use a different key, causing future blob writes or updates under that scope to become inaccessible to users or resources that do not have access to the new key.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Medium (4.8)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/encryptionScopes/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO12 - Enforce encryption on data at rest and protect encryption keys C60 - Maintain a list of authorized customer-managed keys used by storage accounts. C61 - Ensure only authorized customer-managed keys are configured for storage accounts. C63 - Prevent storage accounts that require customer-managed keys from using service-managed keys (e.g., by using built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Deny mode). C173 - Maintain a list of storage accounts that require Azure infrastructure encryption (i.e., double encryption). C174 - Ensure required storage accounts use Azure infrastructure encryption.	High	4	1	-

Service impact through regeneration of storage account access keys

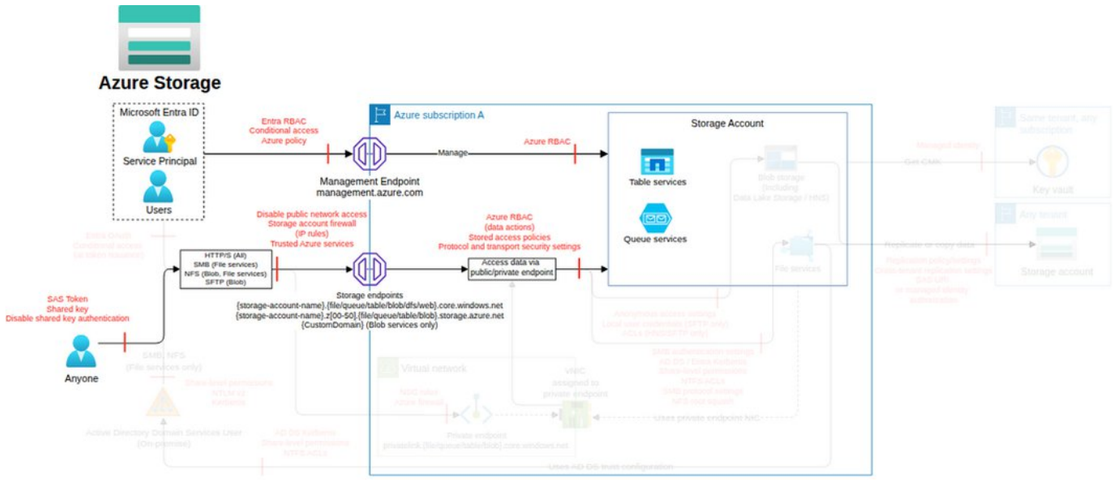
Threat Id	Storage.T2
Name	Service impact through regeneration of storage account access keys
Description	SAS tokens can be signed with a storage account access key, enabling applications to access storage account data. An attacker can rotate or regenerate a storage account access key to invalidate SAS tokens derived from that key, blocking access for applications that depend on those tokens.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Medium (4.5)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/regenerateKey/action", "Microsoft.Storage/storageAccounts/rotateKey/action"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Data exposure through CORS configuration changes

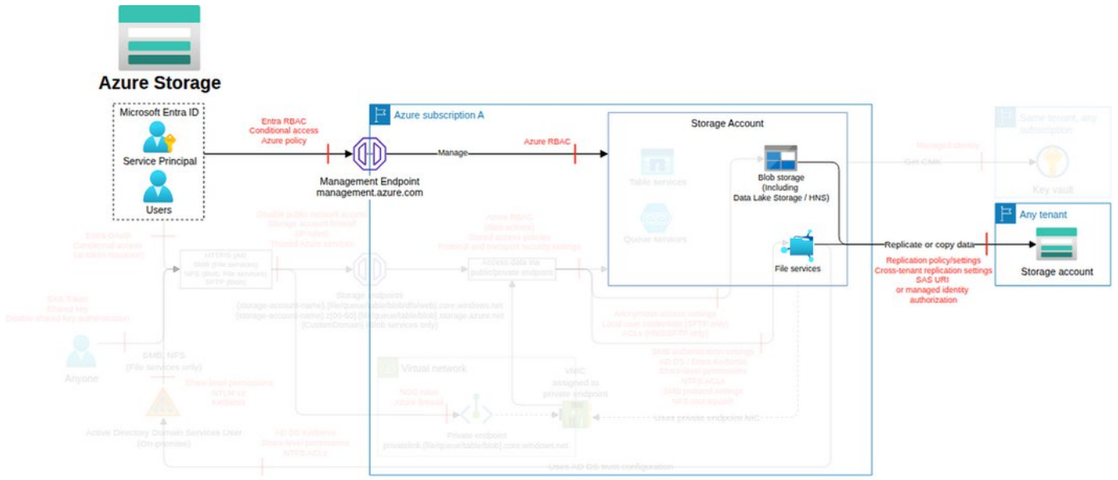
Threat Id	Storage.T26
Name	Data exposure through CORS configuration changes
Description	CORS settings control which browser-based origins can access Azure Storage service responses. An attacker can modify CORS rules for blob, queue, file, or table services to allow untrusted origins to access storage responses through a browser-based application flow.
Goal	Launch another attack
MITRE ATT&CK®	TA0004
CVSS	Medium (4.0)
IAM Access	{ "UNIQUE": "Role:Storage Account Contributor" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO14 - Restrict and manage external websites (origins) that can make browser-based requests to storage accounts C94 - Maintain a list of authorized CORS per endpoint, trusted origins, and corresponding settings. C95 - Ensure only authorized storage accounts have CORS-trusted origins and corresponding settings configured. C96 - Prevent unauthorized storage accounts from using CORS trusted origins and corresponding settings (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/cors.allowedOrigins[*]": ["https://myapp.contoso.com"]} in Deny mode).	High	2	1	-

Service impact through replication configuration changes

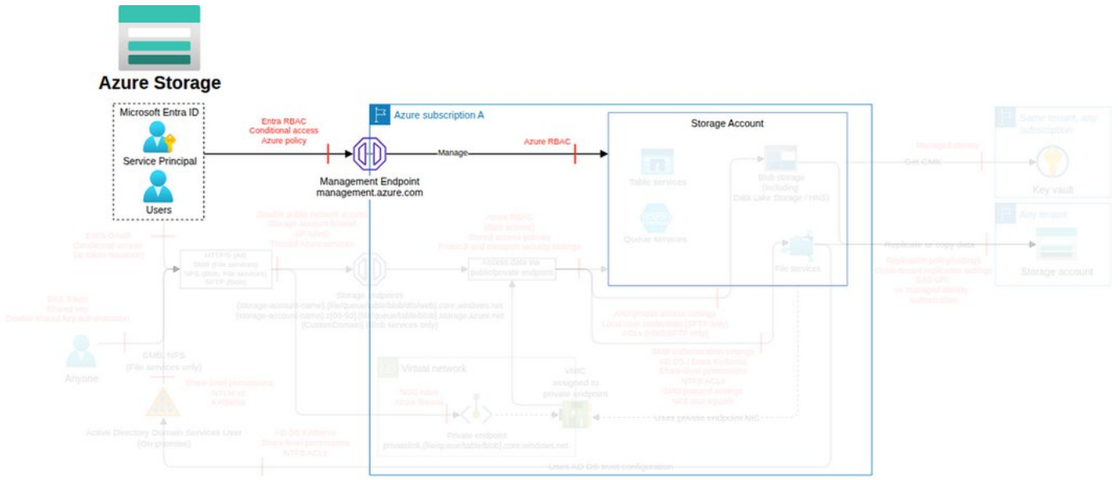
Threat Id	Storage.T42
Name	Service impact through replication configuration changes
Description	Azure Storage replication helps preserve data availability and durability across storage locations. An attacker can modify replication configuration to reduce redundancy, weaken recovery options, or increase the impact of a regional failure.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Low (3.5)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C47 - Maintain a list of storage accounts that require cross-tenant replication and their allowed destination tenants. C48 - Ensure only required storage accounts have cross-tenant replication enabled and the destination storage accounts are authorized. C71 - Maintain a list of storage accounts that require redundancy. C72 - Ensure required storage accounts use redundancy.	Very High	4	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO11 - Enable enhanced monitoring and notifications for storage accounts C50 - Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations. C51 - Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Very Low	2	-	-

Unauthorized failover through configuration change

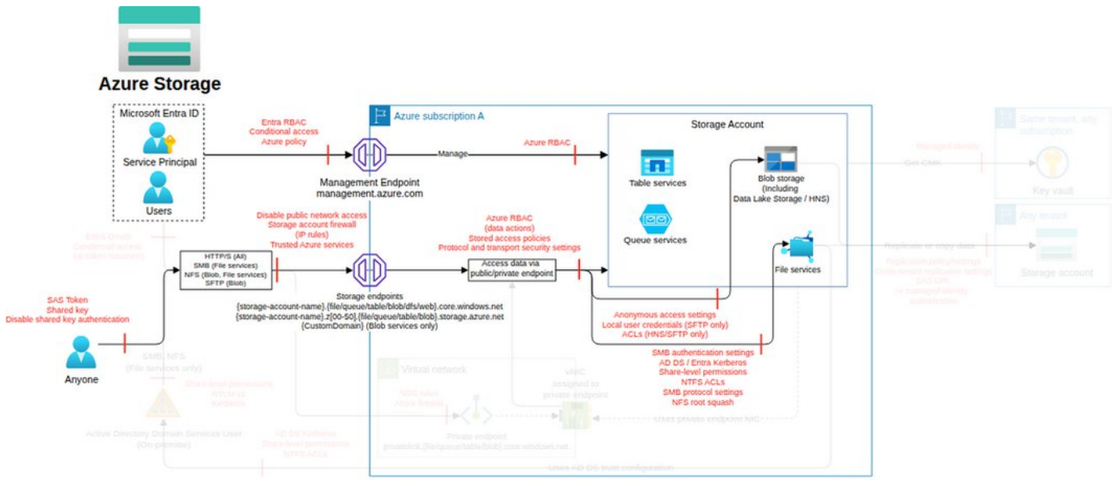
Threat Id	Storage.T59
Name	Unauthorized failover through configuration change
Description	Azure Storage accounts support failover, which transitions geo-redundant storage accounts from their primary region to the secondary, deleting the original primary data and converting the account to locally redundant storage. An attacker can initiate an unauthorized failover to disrupt availability, cause loss of unreplicated data, degrade redundancy, and introduce inconsistencies due to asynchronous replication.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Low (3.5)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/failover/action" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C75 - Maintain a list of authorized storage account regions that can be used for redundancy. C76 - Ensure the authorized storage account region used for redundancy is configured. C77 - Ensure only the authorized storage account region is set for redundancy (e.g., by using a custom Azure Policy on {"location": ["eastus","westeurope"]} in Deny mode).	High	2	1	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-

Unauthorized access path through virtual network rule changes

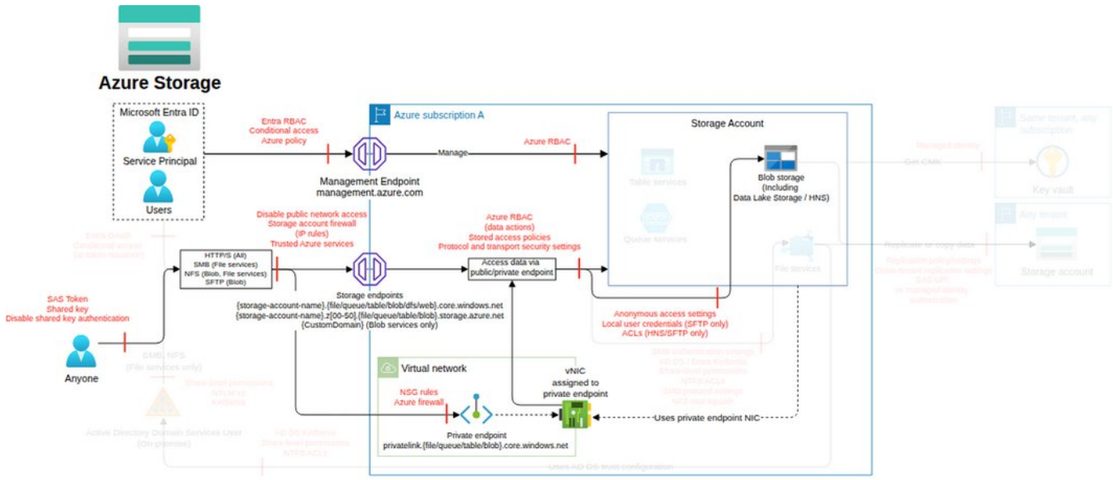
Threat Id	Storage.T50
Name	Unauthorized access path through virtual network rule changes
Description	Storage account network settings can restrict access through public network access, firewall rules, IP rules, virtual network rules, and trusted service bypass settings. An attacker can modify storage account network configuration to allow access from unauthorized network paths, enabling later access to storage account resources.
Goal	Launch another attack
MITRE ATT&CK®	TA0001
CVSS	Low (2.4)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C44 - Prevent storage account firewall rules from allowing unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode). C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	High	8	1	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C6 - Maintain the list of storage accounts authorized to have public network access enabled. C7 - Ensure only authorized storage accounts have public network access enabled. C8 - Prevent unauthorized storage accounts from having public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Deny mode). C55 - Maintain a list of blobs and containers that require anonymous access. C56 - Ensure anonymous access is set only for required blobs and containers. C57 - Ensure only required blobs and containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Deny mode).	Medium	4	2	-

Public access path through account configuration

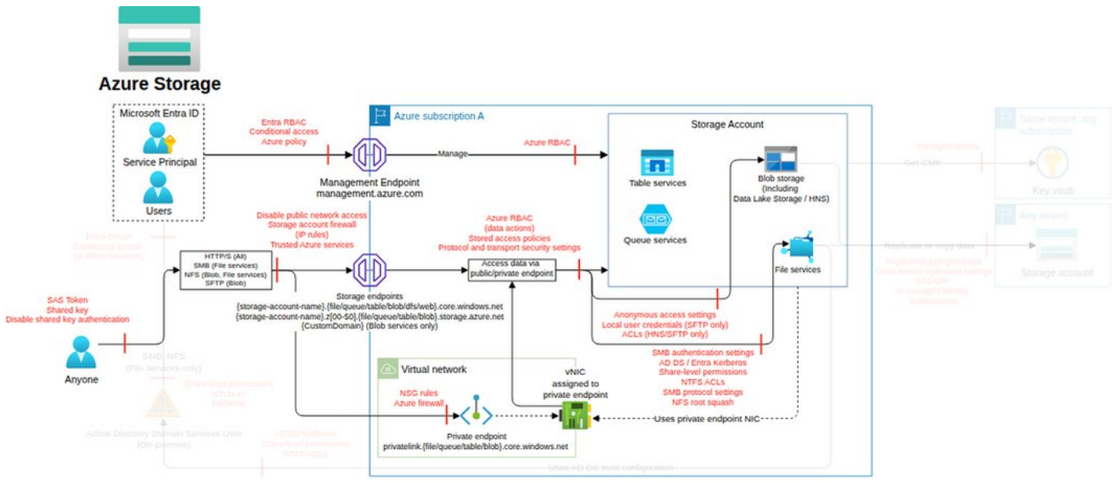
Threat Id	Storage.T68
Name	Public access path through account configuration
Description	Storage accounts can control whether blob containers are allowed to expose data through anonymous public access. An attacker can enable account-level public blob access, allowing containers to be configured for anonymous access and enabling later exposure of blob data.
Goal	Launch another attack
MITRE ATT&CK®	TA0001
CVSS	Low (2.4)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C6 - Maintain the list of storage accounts authorized to have public network access enabled. C7 - Ensure only authorized storage accounts have public network access enabled. C8 - Prevent unauthorized storage accounts from having public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Deny mode). C55 - Maintain a list of blobs and containers that require anonymous access. C56 - Ensure anonymous access is set only for required blobs and containers. C57 - Ensure only required blobs and containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Deny mode).	Medium	4	2	-

Shared Key access path through the account configuration

Threat Id	Storage.T69
Name	Shared Key access path through the account configuration
Description	Storage accounts can control whether requests are authorized with account access keys through Shared Key authorization. An attacker can enable Shared Key access on a storage account, allowing access keys or key-signed SAS tokens to authorize requests outside the intended identity-based authorization path.
Goal	Launch another attack
MITRE ATT&CK®	TA0001
CVSS	Low (2.4)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/write" }

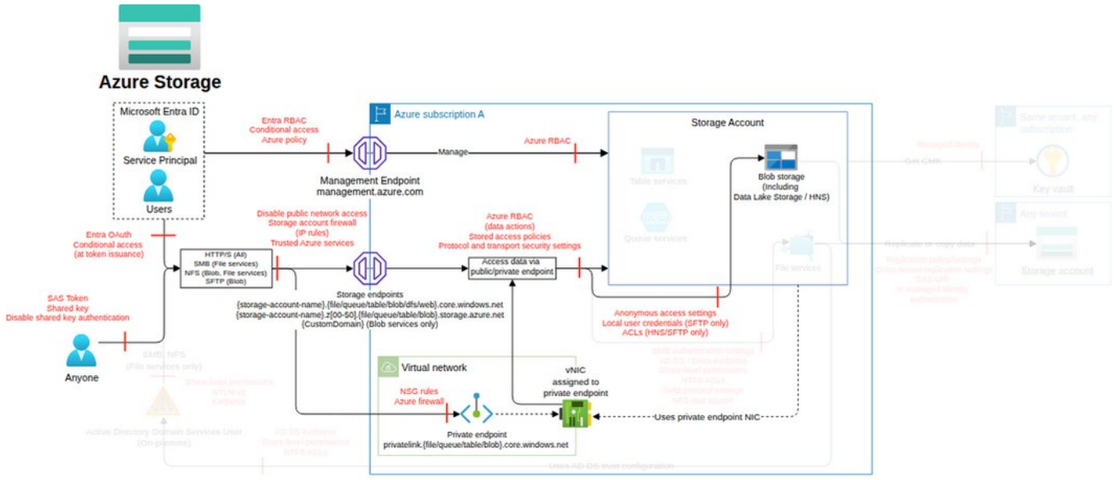


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Medium	4	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode).	Medium	1	-	1

Blob storage *(subclass of Storage account, FC2)*

Object storage solution for storing unstructured data (blobs) accessible via HTTP/S and optionally via the Network File System (NFS) v3 and SFTP protocols.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

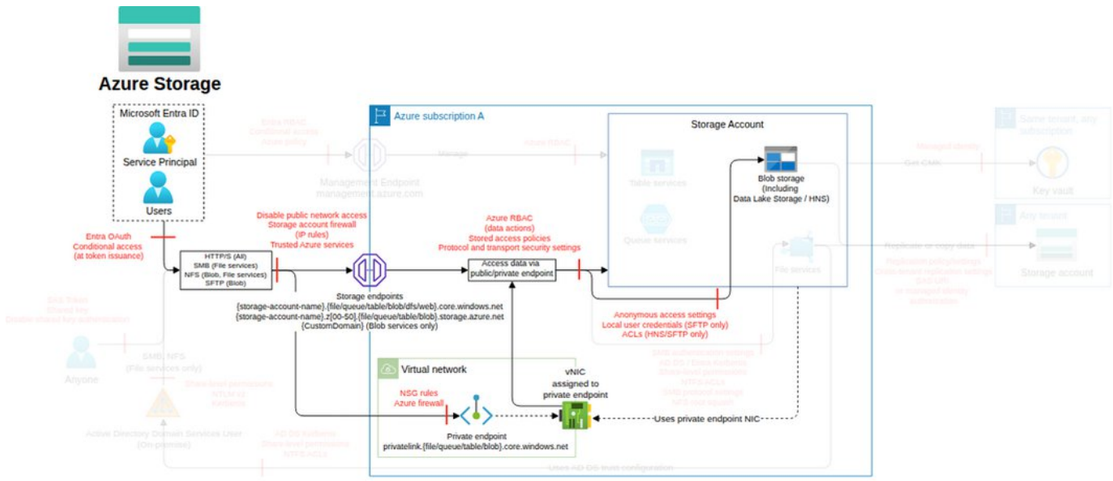
Action	IAM Permission
Creates a new container under the specified account. If the container with the same name already exists, the operation fails returns all user-defined metadata and system properties for the specified container. The data returned does not include the container's list of blobs operation marks the specified container for deletion. The container and any blobs contained within it are later deleted during garbage collection restype.	Microsoft.Storage/storageAccounts/blobServices/containers/write

Threat List

Name	CVSS
Privilege escalation through hierarchical namespace ACL modification	High (8.1)
Encrypt blob contents for ransom using authorized access	Medium (6.1)
Unauthorized access to blob data through overly permissive Azure RBAC assignments	Medium (5.7)
Unauthorized access to blob data through overly permissive hierarchical namespace ACLs	Medium (5.7)
Data manipulation through unauthorized modification of blob data	Medium (5.7)
Service impact through recursive deletion of hierarchical namespace content	Medium (5.2)
Unauthorized access to blob data by renaming a path used by Azure RBAC assignments	Medium (5.2)
Impact of disabling blob versioning	Medium (5.2)
Impact by modifying blob container immutability protections	Medium (5.2)
Service impact through unauthorized modification of blob content	Medium (5.1)
Financial impact through query acceleration usage	Medium (4.6)
Publish malicious content by modifying static website blobs	Medium (4.6)
Unauthorized access to blob data by enabling anonymous public access	Medium (4.5)

Privilege escalation through hierarchical namespace ACL modification

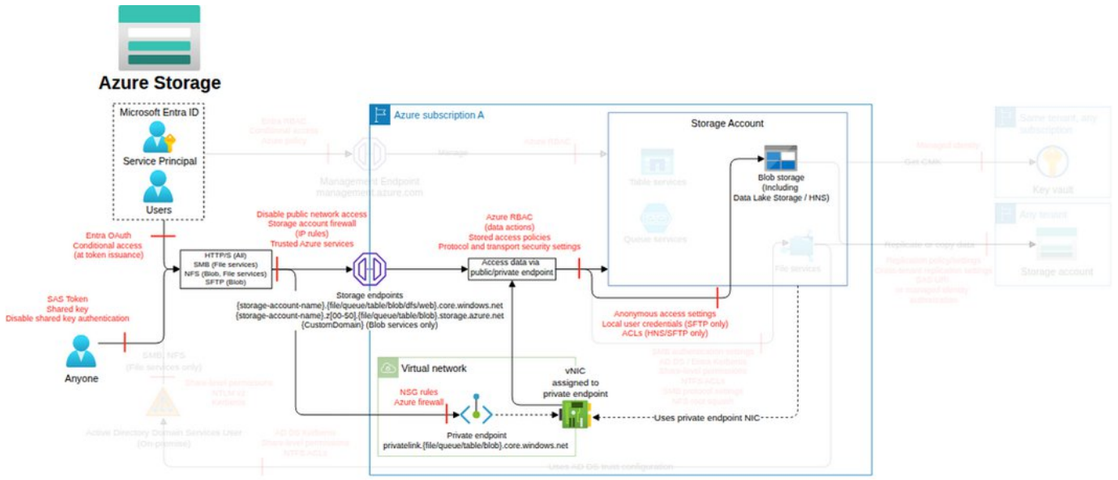
Threat Id	Storage.T66
Name	Privilege escalation through hierarchical namespace ACL modification
Description	Azure Storage accounts with hierarchical namespace enabled can authorize file and directory access using POSIX-like Access Control Lists. An attacker can modify hierarchical namespace ACLs to grant broader access to protected blob data.
Goal	Launch another attack
MITRE ATT&CK®	TA0004
CVSS	High (8.1)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write", "Microsoft.Storage/storageAccounts/blobServices/containers/write"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO7 - Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model C158 - Maintain a list of authorized Access Control Lists (ACLs) on Data Lake Storage containers. C159 - Ensure all Access Control Lists (ACLs) on Data Lake Storage containers are authorized.	Medium	2	-	-

Encrypt blob contents for ransom using authorized access

Threat Id	Storage.T9
Name	Encrypt blob contents for ransom using authorized access
Description	Azure Blob Storage can store file and object data that may be replaced through authorized blob access. An attacker can replace blob contents with encrypted data under their control and demand payment for recovery.
Goal	Direct Financial Gain
MITRE ATT&CK®	TA0040
CVSS	Medium (6.1)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write" }



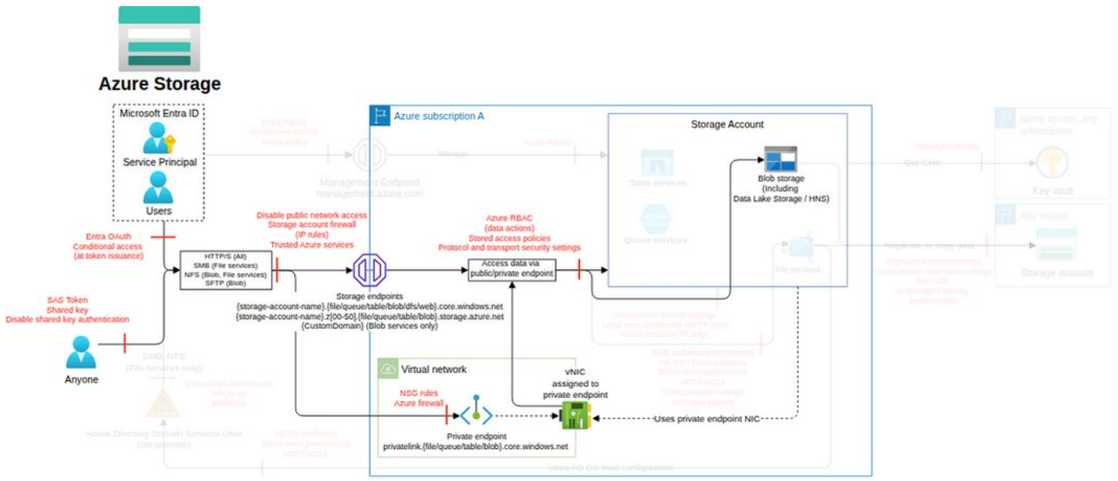
Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C32 - Maintain a list of storage container blobs that require immutability or soft delete. C33 - Ensure required storage container blobs have immutability or soft delete enabled.	Very High	2	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C15 - Maintain a list of the blob storage containers that are required to have a minimum retention period enabled. C16 - Ensure required storage accounts have the soft-delete feature for blobs enabled for the defined minimum retention period. C17 - Prevent the creation of required storage accounts without the blob soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/deleteRetentionPolicy.enabled": false} in Deny mode). C19 - Ensure required storage accounts have the soft-delete option enabled for the containers. C20 - Prevent the creation of containers without the soft-delete option enabled (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/blobServices/containers/softDelete" in Deny mode). C46 - Ensure storage blobs are using Azure Backup, following the requirements in the Azure Backup ThreatModel. C147 - Maintain a list of storage blobs that require Azure Backup. C150 - Maintain a list of authorized replication policies and their destination storage accounts.	High	6	2	-
CO12 - Enforce encryption on data at rest and protect encryption keys C60 - Maintain a list of authorized customer-managed keys used by storage accounts. C61 - Ensure only authorized customer-managed keys are configured for storage accounts. C63 - Prevent storage accounts that require customer-managed keys from using service-managed keys (e.g., by using built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Deny mode). C173 - Maintain a list of storage accounts that require Azure infrastructure encryption (i.e., double encryption). C174 - Ensure required storage accounts use Azure infrastructure encryption.	Medium	4	1	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods.	Medium	4	-	-

C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.				
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	1	-	2
CO11 - Enable enhanced monitoring and notifications for storage accounts C50 - Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations. C51 - Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Very Low	2	-	-

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Unauthorized access to blob data through overly permissive Azure RBAC assignments

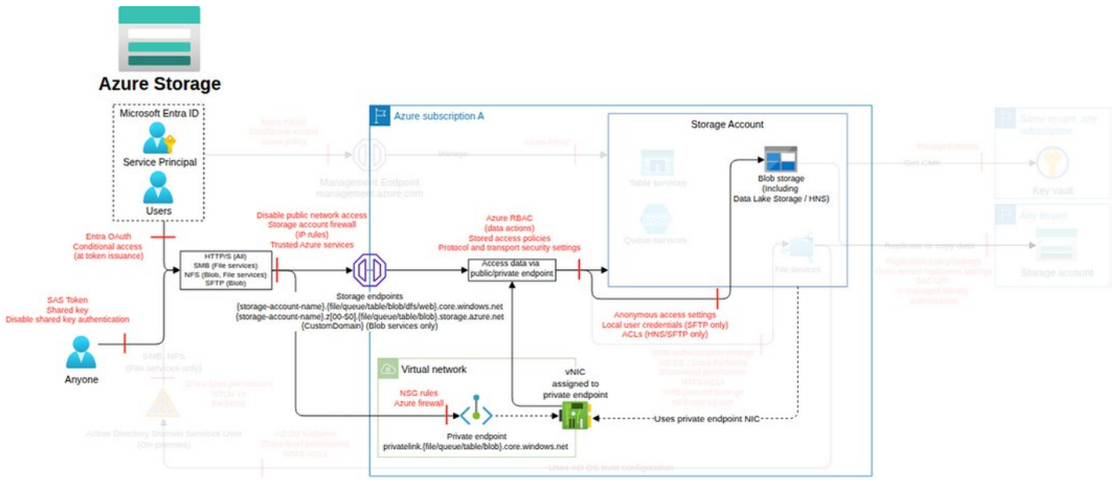
Threat Id	Storage.T48
Name	Unauthorized access to blob data through overly permissive Azure RBAC assignments
Description	Azure Storage can authorize blob data access through Azure RBAC assignments that evaluate request, resource, environment, or principal attributes. An attacker can satisfy overly permissive or incorrectly scoped assignments to read blob data outside the intended authorization scope.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.7)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Unauthorized access to blob data through overly permissive hierarchical namespace ACLs

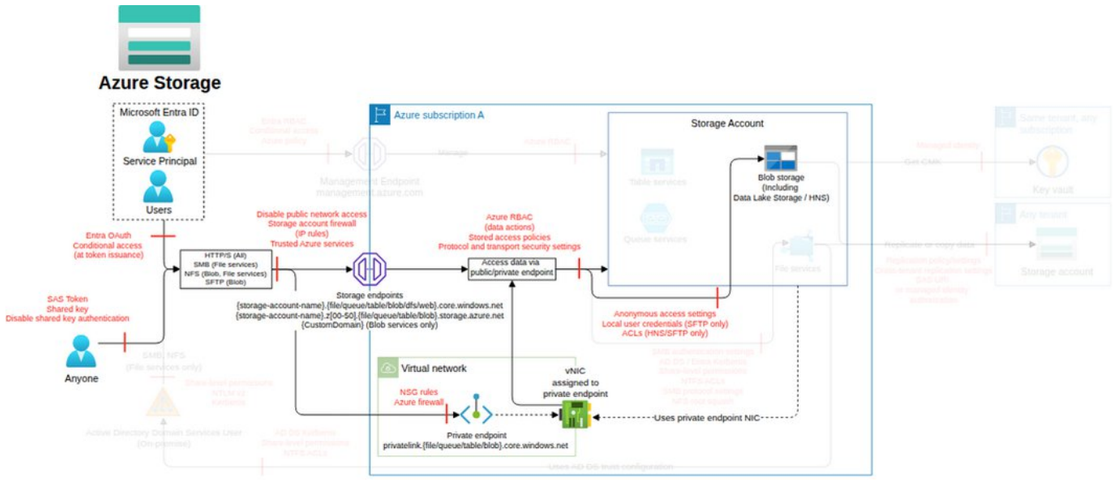
Threat Id	Storage.T58
Name	Unauthorized access to blob data through overly permissive hierarchical namespace ACLs
Description	Azure Storage accounts with hierarchical namespace enabled can authorize file and directory access using POSIX-like Access Control Lists. An attacker can use overly permissive hierarchical namespace ACLs to read blob data outside the intended authorization scope.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.7)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO7 - Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model C158 - Maintain a list of authorized Access Control Lists (ACLs) on Data Lake Storage containers. C159 - Ensure all Access Control Lists (ACLs) on Data Lake Storage containers are authorized.	Medium	2	-	-

Data manipulation through unauthorized modification of blob data

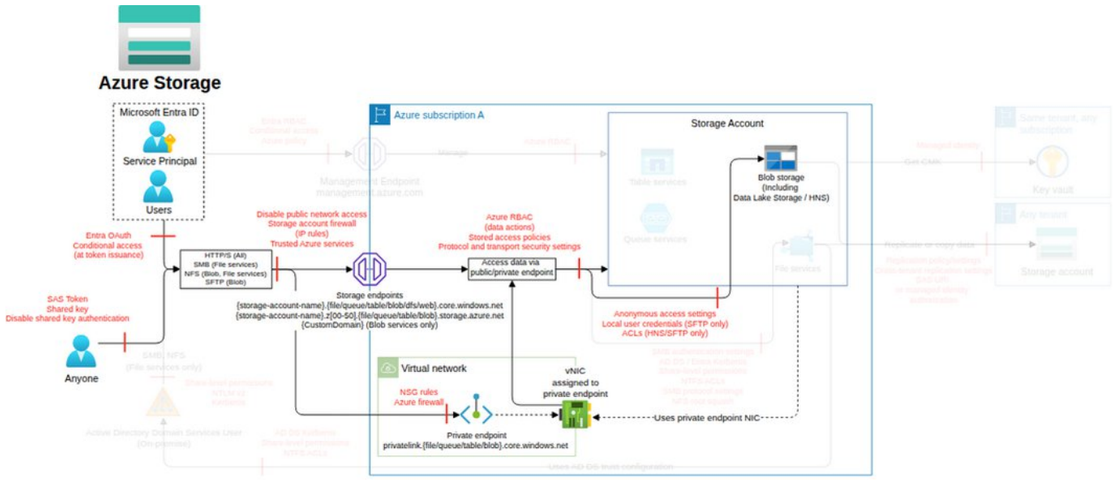
Threat Id	Storage.T8
Name	Data manipulation through unauthorized modification of blob data
Description	Azure Blob Storage supports authorized access for creating, replacing, or modifying blob content. An attacker can modify blob data to alter stored content outside the intended authorization path.
Goal	Data manipulation
MITRE ATT&CK®	TA0040
CVSS	Medium (5.7)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write", "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/add/action"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C32 - Maintain a list of storage container blobs that require immutability or soft delete. C33 - Ensure required storage container blobs have immutability or soft delete enabled.	Very High	2	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO11 - Enable enhanced monitoring and notifications for storage accounts C50 - Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations. C51 - Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Very Low	2	-	-

Service impact through recursive deletion of hierarchical namespace content

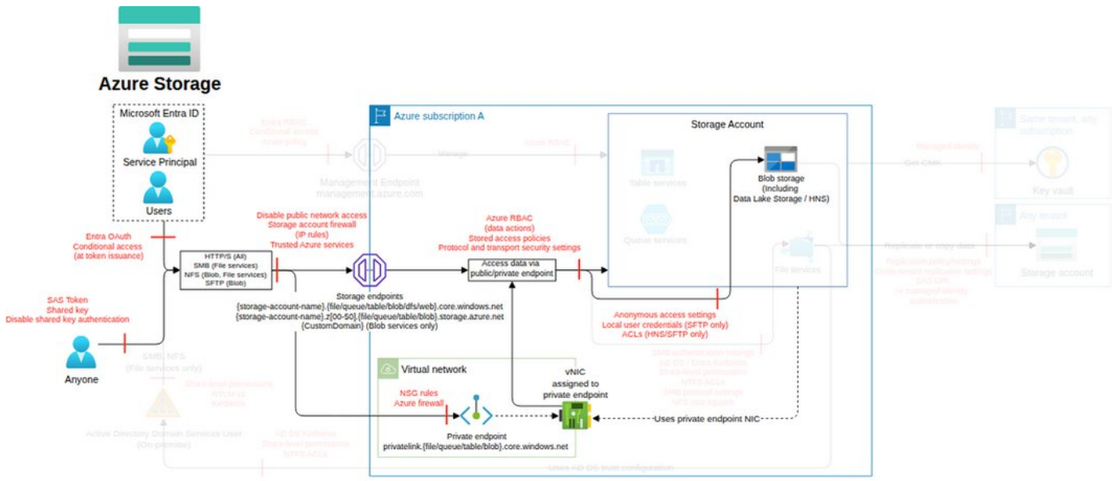
Threat Id	Storage.T7
Name	Service impact through recursive deletion of hierarchical namespace content
Description	Azure Storage accounts with hierarchical namespace enabled can organize blob data into directories and files. An attacker can recursively delete directories or files to make stored content unavailable to legitimate users.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Medium (5.2)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete", "Microsoft.Storage/storageAccounts/blobServices/containers/delete"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO21 - Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model C22 - Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled. C23 - Ensure all required storage accounts have hierarchical namespace (i.e., Data Lake Storage) enabled.	Low	2	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C10 - Ensure versioning is enabled on required containers. C12 - Ensure snapshots are enabled for required file shares. C15 - Maintain a list of the blob storage containers that are required to have a minimum retention period enabled. C16 - Ensure required storage accounts have the soft-delete feature for blobs enabled for the defined minimum retention period. C19 - Ensure required storage accounts have the soft-delete option enabled for the containers. C46 - Ensure storage blobs are using Azure Backup, following the requirements in the Azure Backup ThreatModel. C145 - Maintain a list of Azure Files that require snapshots. C147 - Maintain a list of storage blobs that require Azure Backup. C150 - Maintain a list of authorized replication policies and their destination storage accounts.	Low	9	-	-
CO11 - Enable enhanced monitoring and notifications for storage accounts C50 - Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations. C51 - Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Very Low	2	-	-
CO24 - Protect the integrity of blob storage data C32 - Maintain a list of storage container blobs that require immutability or soft delete. C144 - Maintain the list of storage accounts or blob containers that require blob versioning.	Very Low	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token).	Very Low	1	-	-

Unauthorized access to blob data by renaming a path used by Azure RBAC assignments

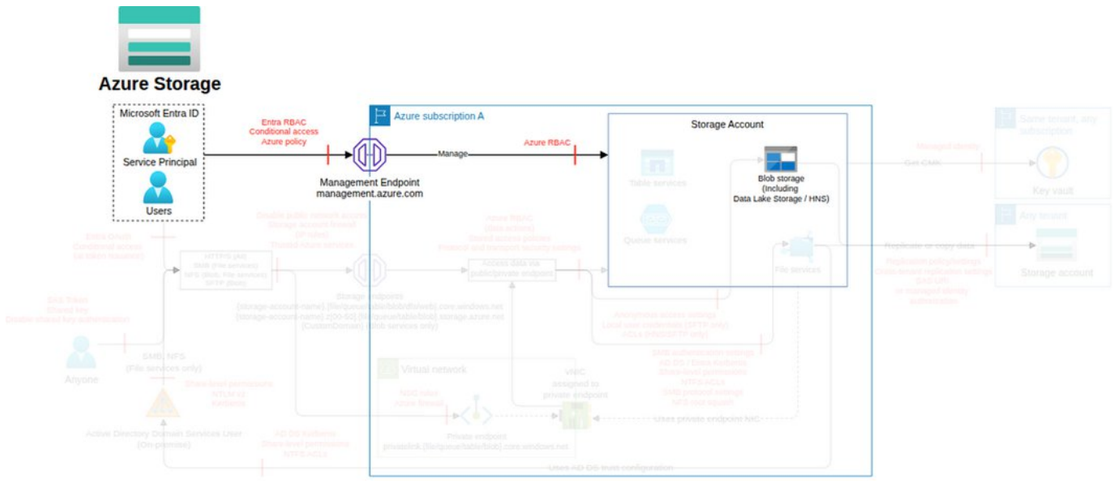
Threat Id	Storage.T55
Name	Unauthorized access to blob data by renaming a path used by Azure RBAC assignments
Description	Azure Storage can authorize blob data access through Azure RBAC assignments that evaluate blob path attributes. An attacker can rename or move a blob path to satisfy an overly permissive path-based assignment (ref) and read blob data outside the intended authorization scope.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/move/action" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Impact of disabling blob versioning

Threat Id	Storage.T40
Name	Impact of disabling blob versioning
Description	Blob versioning can be enabled for supported Azure Blob Storage accounts to preserve previous versions of blob data when blobs are modified or overwritten. An attacker can disable blob versioning on a supported storage account to prevent new versions from being created, reducing recovery options after later unauthorized modification or overwrite of blob data.
Goal	Launch another attack
MITRE ATT&CK®	TA0040
CVSS	Medium (5.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/write" }

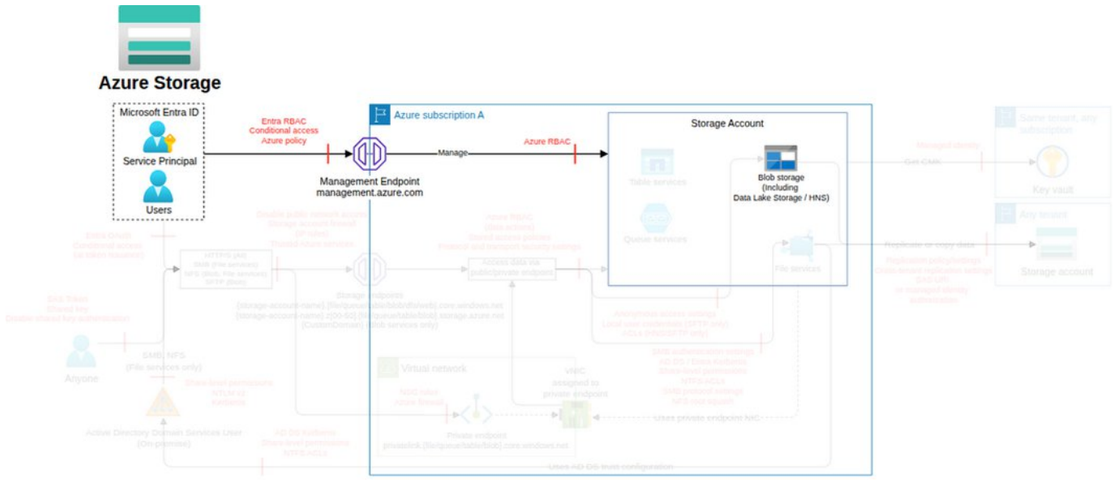


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C10 - Ensure versioning is enabled on required containers.	Medium	1	-	-
CO24 - Protect the integrity of blob storage data C144 - Maintain the list of storage accounts or blob containers that require blob versioning.	Very Low	1	-	-

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Impact by modifying blob container immutability protections

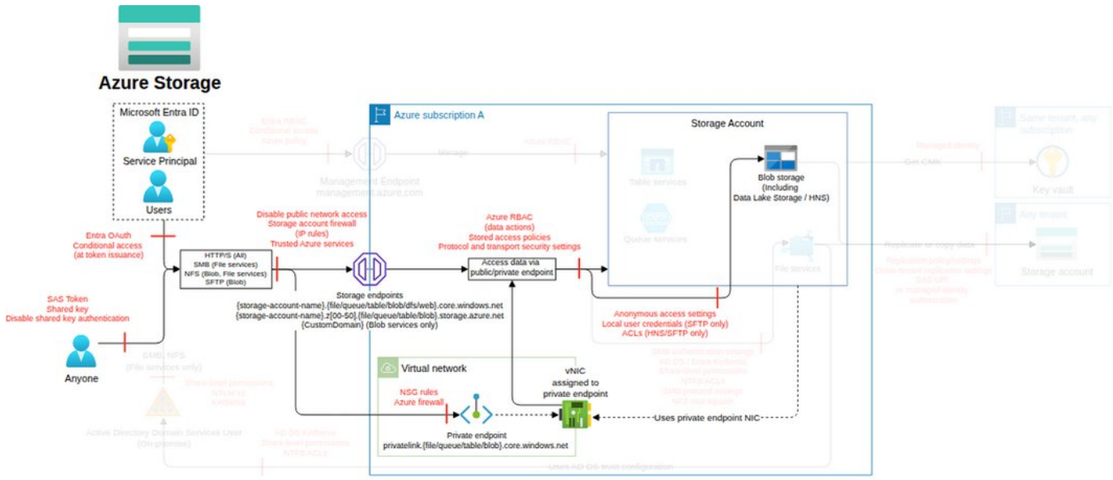
Threat Id	Storage.T56
Name	Impact by modifying blob container immutability protections
Description	Azure blob containers can use immutability policies and legal holds to prevent protected blob data from being modified or deleted. An attacker can modify immutability policy settings or legal hold state to weaken protection for blob data and enable later unauthorized modification or deletion.
Goal	Launch another attack
MITRE ATT&CK®	TA0040
CVSS	Medium (5.2)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/write", "Microsoft.Storage/storageAccounts/blobServices/containers/setLegalHold/action", "Microsoft.Storage/storageAccounts/blobServices/containers/clearLegalHold/action"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C32 - Maintain a list of storage container blobs that require immutability or soft delete. C33 - Ensure required storage container blobs have immutability or soft delete enabled.	Medium	2	-	-

Service impact through unauthorized modification of blob content

Threat Id	Storage.T12
Name	Service impact through unauthorized modification of blob content
Description	Azure Blob Storage can host files consumed by users, applications, or downstream services. An attacker can replace blob contents with malicious or unauthorized content (e.g., malware), causing consumers of the blob to retrieve harmful content from a previously trusted storage location.
Goal	Launch another attack
MITRE ATT&CK®	TA0003
CVSS	Medium (5.1)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write" }

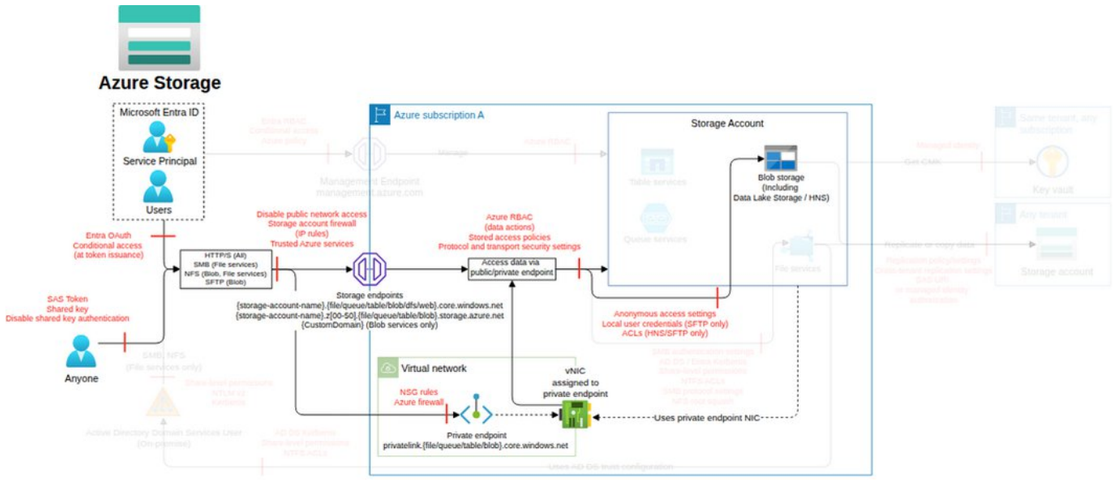


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C32 - Maintain a list of storage container blobs that require immutability or soft delete. C33 - Ensure required storage container blobs have immutability or soft delete enabled. C110 - Prevent modification or deletion of required storage accounts by using a resource lock set to read-only and/or delete, using the Azure Resource Manager ThreatModel.	Very High	2	1	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	High	8	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Medium	4	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-

CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	1	-	2
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C15 - Maintain a list of the blob storage containers that are required to have a minimum retention period enabled. C16 - Ensure required storage accounts have the soft-delete feature for blobs enabled for the defined minimum retention period. C19 - Ensure required storage accounts have the soft-delete option enabled for the containers. C46 - Ensure storage blobs are using Azure Backup, following the requirements in the Azure Backup ThreatModel. C147 - Maintain a list of storage blobs that require Azure Backup. C150 - Maintain a list of authorized replication policies and their destination storage accounts.	Low	6	-	-

Financial impact through query acceleration usage

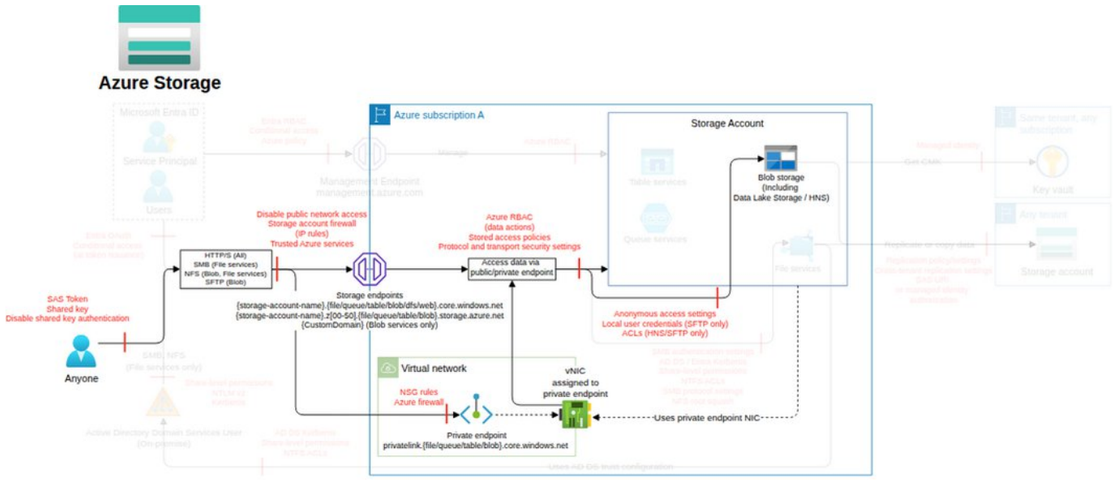
Threat Id	Storage.T34
Name	Financial impact through query acceleration usage
Description	Azure Storage query acceleration can process readable blob data using service-side query execution with a separate pricing model. An attacker can repeatedly execute query acceleration requests against readable blob data to generate unexpected processing costs.
Goal	Financial Drain
MITRE ATT&CK®	TA0040
CVSS	Medium (4.6)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token).	Very Low	1	-	-

Publish malicious content by modifying static website blobs

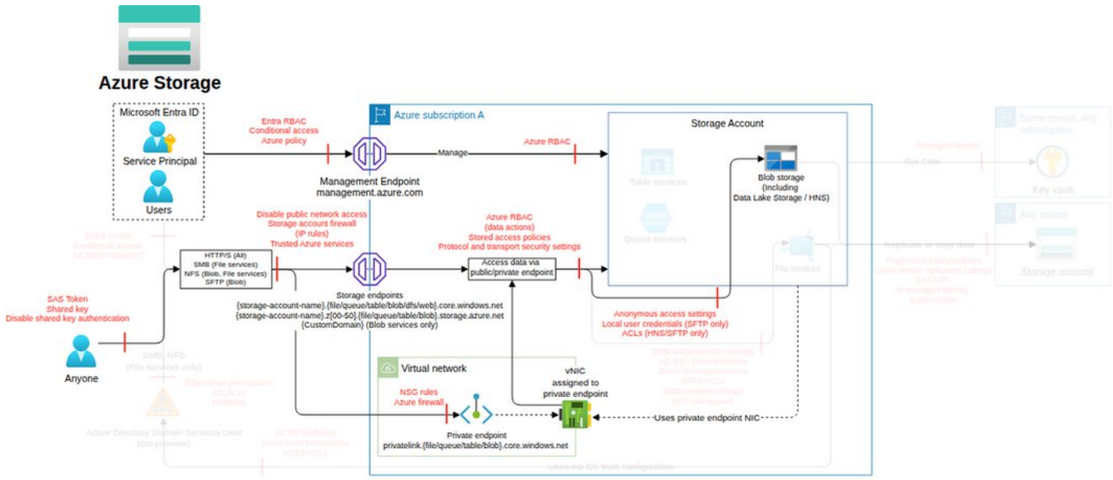
Threat Id	Storage.T22
Name	Publish malicious content by modifying static website blobs
Description	Azure Storage static websites serve content from a blob storage container, named $web, over a web endpoint. An attacker can modify files in the $web container to publish malicious or unauthorized content through a trusted storage-hosted website.
Goal	Launch another attack
MITRE ATT&CK®	TA0040
CVSS	Medium (4.6)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C90 - Maintain a list of storage accounts that require static website hosting. C91 - Ensure only storage accounts that require static website hosting have it enabled. C92 - Prevent storage accounts that do not require static website hosting from having it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/staticWebsite.enabled": true} in Deny mode).	Medium	2	1	-

Unauthorized access to blob data by enabling anonymous public access

Threat Id	Storage.T5
Name	Unauthorized access to blob data by enabling anonymous public access
Description	Azure blob containers can allow anonymous public access to blobs when account-level public access is permitted. An attacker can modify container public access settings to expose blob data outside the intended authorization path.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (4.5)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/blobServices/containers/write" }

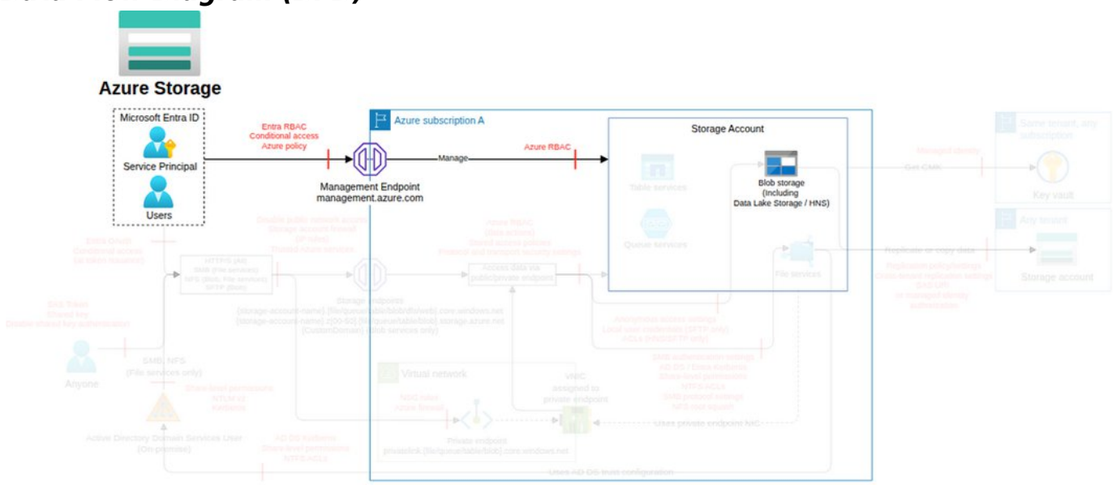


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	High	8	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C6 - Maintain the list of storage accounts authorized to have public network access enabled. C7 - Ensure only authorized storage accounts have public network access enabled. C8 - Prevent unauthorized storage accounts from having public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Deny mode). C55 - Maintain a list of blobs and containers that require anonymous access. C56 - Ensure anonymous access is set only for required blobs and containers. C57 - Ensure only required blobs and containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Deny mode).	Medium	5	2	-

Blob inventory *(subclass of Blob storage, FC10)*

The Azure Storage blob inventory feature provides an overview of your containers, blobs, snapshots, and blob versions within a storage account. Use the inventory report to understand various attributes of blobs and containers, such as the total data size, age, encryption status, immutability policy, and legal hold.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

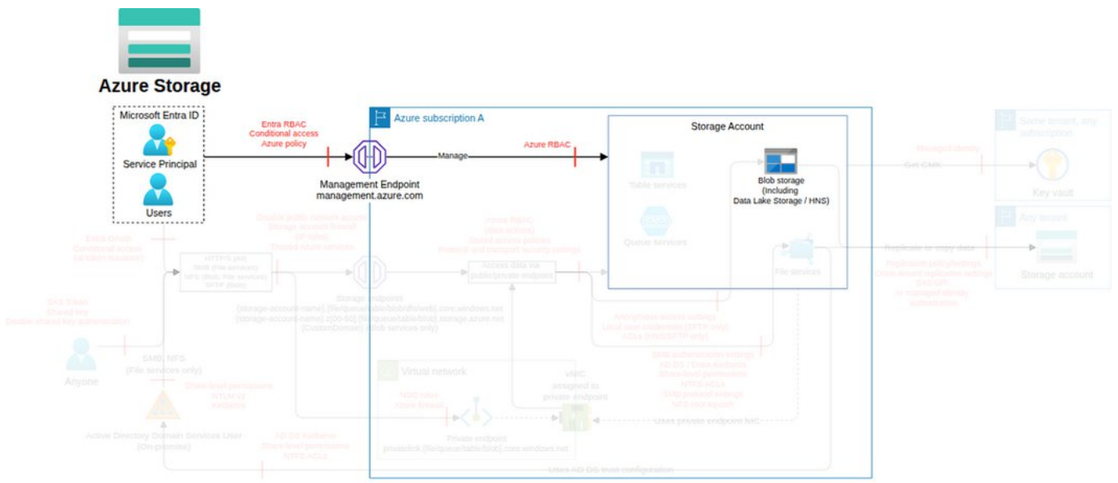
Action	IAM Permission
Sets the blob inventory policy to the specified storage account.	Microsoft.Storage/storageAccounts/inventoryPolicies/write

Threat List

Name	CVSS
Blob inventory discovery through inventory access	Low (3.5)

Blob inventory discovery through inventory access

Threat Id	Storage.T24
Name	Blob inventory discovery through inventory access
Description	Blob inventory reports list blob objects, metadata, versions, and related storage details for a storage account. An attacker can access blob inventory reports to identify stored objects, naming patterns, and data locations for later targeted access.
Goal	Launch another attack
MITRE ATT&CK®	TA0007
CVSS	Low (3.5)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/inventoryPolicies/read", "Microsoft.Storage/storageAccounts/inventoryPolicies/write"] }

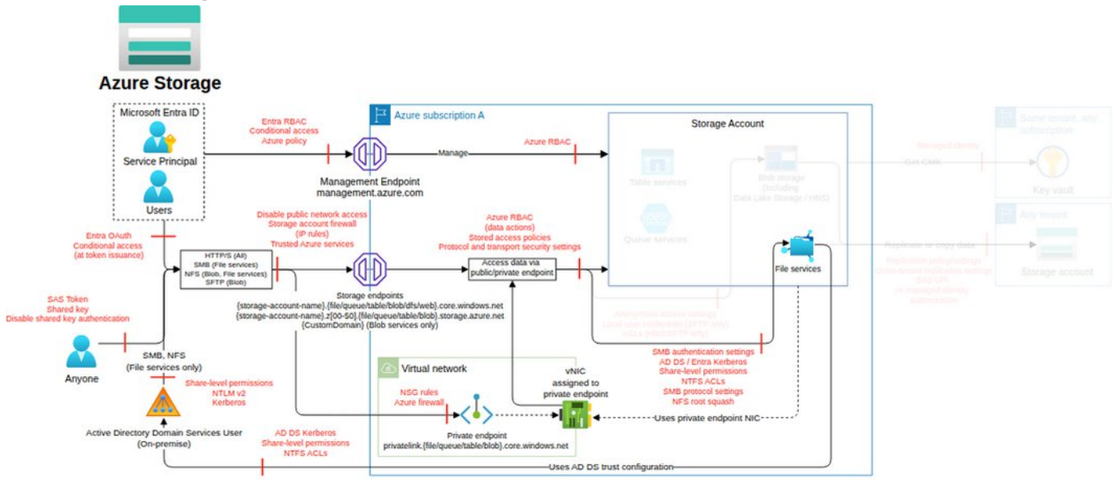


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

File shares *(subclass of Storage account, FC3)*

Azure Files offers fully governed file shares in the cloud accessible via Server Message Block (SMB) protocol, Network File System (NFS) v4.1 protocol, or the in-preview Azure Files REST API.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

Action	IAM Permission
Creates a new share under the specified account as described by the request body. The share resource includes metadata and properties for that share. It does not include a list of the files contained by the share. Updates share properties as specified in the request body. Properties not mentioned in the request will not be changed. Update fails if the specified share does not already exist.	Microsoft.Storage/storageAccounts/fileServices/shares/write

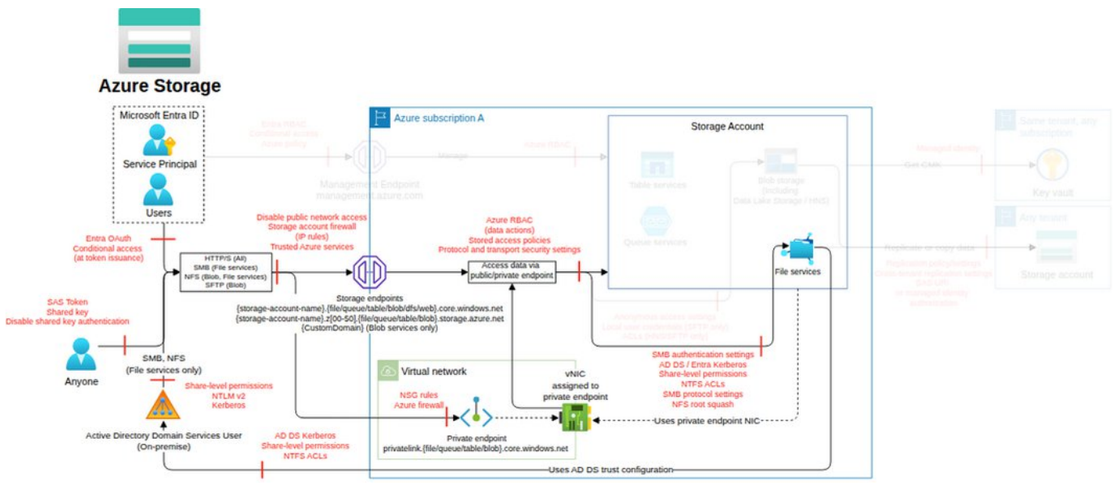
Threat List

Name	CVSS
Privilege escalation by modifying Azure file share permissions	Medium (6.9)
Service impact through recursive deletion of Azure file share contents	Medium (6.1)
Impact through unauthorized encryption of Azure file share contents	Medium (6.1)
Unauthorized access to Azure file share contents	Medium (5.7)
Distribute malicious content via unauthorized upload	Medium (5.6)
Data exposure through legacy file share protocols	Medium (4.2)
Financial impact through unauthorized upload of Azure file share contents	Low (3.5)
Data exposure through file share encryption changes	Low (2.4)

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Privilege escalation by modifying Azure file share permissions

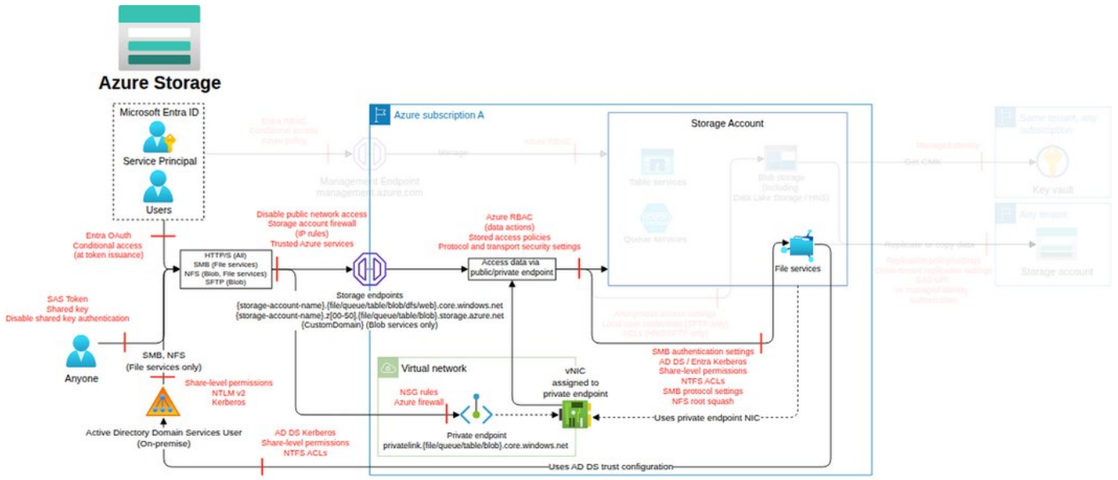
Threat Id	Storage.T17
Name	Privilege escalation by modifying Azure file share permissions
Description	Azure file shares support file and directory permissions that control access to share content. An attacker can modify file or directory permissions to grant broader access to protected share content or enable later unauthorized operations.
Goal	Launch another attack
MITRE ATT&CK®	TA0004
CVSS	Medium (6.9)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write", "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/modifyPermissions/action"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Service impact through recursive deletion of Azure file share contents

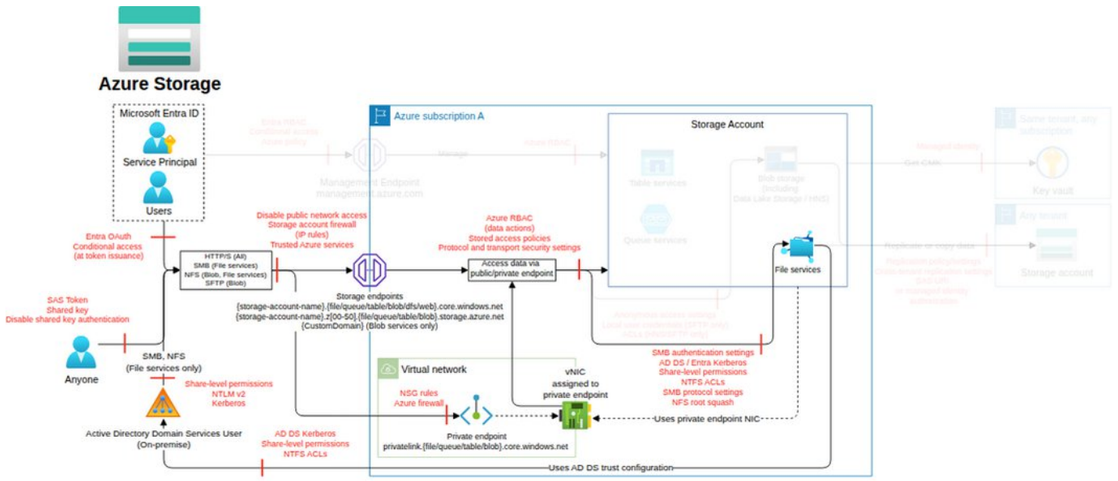
Threat Id	Storage.T18
Name	Service impact through recursive deletion of Azure file share contents
Description	Azure file shares can store files in hierarchical directory structures (e.g., project folders, user home directories, application data folders). An attacker can recursively delete directories or files in a file share, making stored content unavailable to legitimate users.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Medium (6.1)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/delete" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO24 - Protect the integrity of blob storage data C107 - Maintain the list of storage accounts required to use Azure resource locks, following the Azure Resource Manager ThreatModel. C108 - Ensure required storage accounts use Azure resource locks. C110 - Prevent modification or deletion of required storage accounts by using a resource lock set to read-only and/or delete, using the Azure Resource Manager ThreatModel.	Very High	2	1	-
CO5 - Ensure backup, replication, and recovery capabilities for storage account services C83 - Define the minimum retention period for required file shares (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/deleteRetentionPolicy.days": 7} in Deny mode). C84 - Ensure file shares have the soft-delete option enabled for at least the defined minimum retention period. C85 - Prevent the creation of file shares without the soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/shares/deleteRetentionPolicy.enabled": false} in Deny mode).	Medium	2	1	-

Impact through unauthorized encryption of Azure file share contents

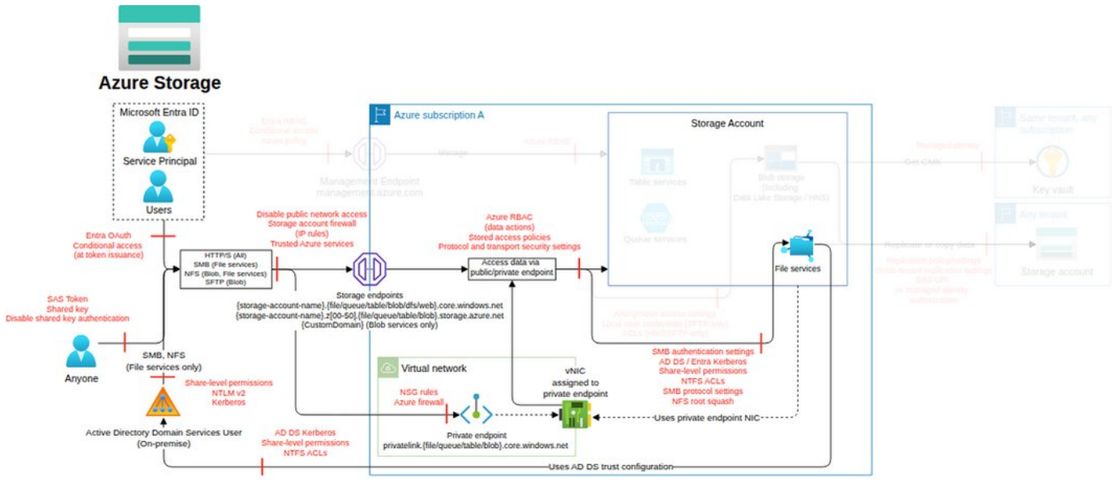
Threat Id	Storage.T19
Name	Impact through unauthorized encryption of Azure file share contents
Description	Azure file shares store file content that can be accessed and modified through authorized file access. An attacker can overwrite or encrypt file contents in a file share to make the data unusable and demand payment for recovery.
Goal	Direct Financial Gain
MITRE ATT&CK®	TA0040
CVSS	Medium (6.1)
IAM Access	{ "AND": ["Microsoft.Storage/storageAccounts/fileServices/fileshares/files/read", "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Unauthorized access to Azure file share contents

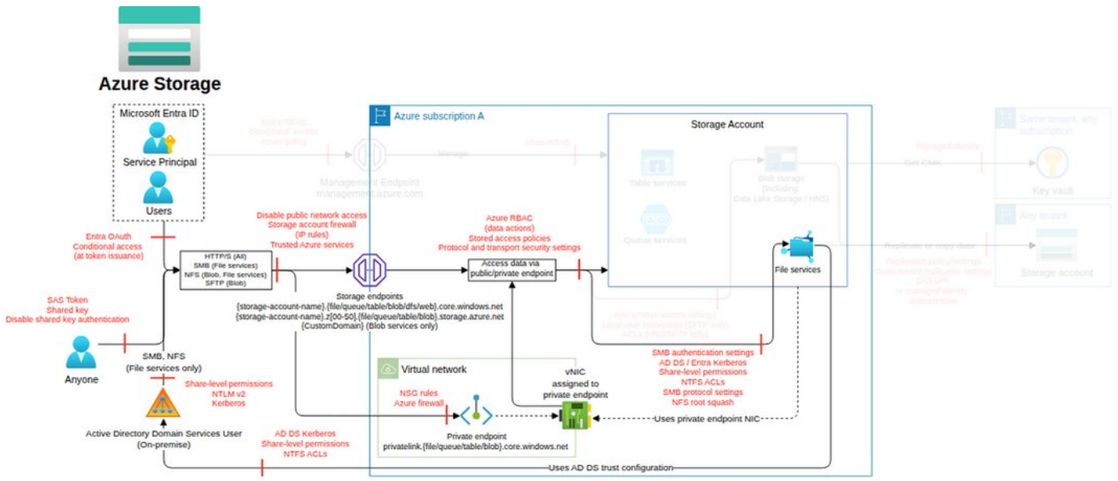
Threat Id	Storage.T15
Name	Unauthorized access to Azure file share contents
Description	Azure file shares can expose stored files through supported access methods (e.g., SMB, NFS, REST APIs). An attacker can browse exposed file shares to access and retrieve file contents (e.g., text files, Word documents, PDF files).
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.7)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/read" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C44 - Prevent storage account firewall rules from allowing unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode). C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	High	8	1	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO8 - Enforce encryption in transit C120 - Maintain a list of required file shares' security protocol settings (ideally maximum security SMB 3.1.1, Kerberos, AES-256 only). C121 - Ensure required file shares' security protocol settings are set. C176 - Maintain the authorized minimum TLS version for each storage account. C177 - Ensure storage accounts use the authorized minimum TLS version.	Medium	4	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token).	Very Low	1	-	-

Distribute malicious content via unauthorized upload

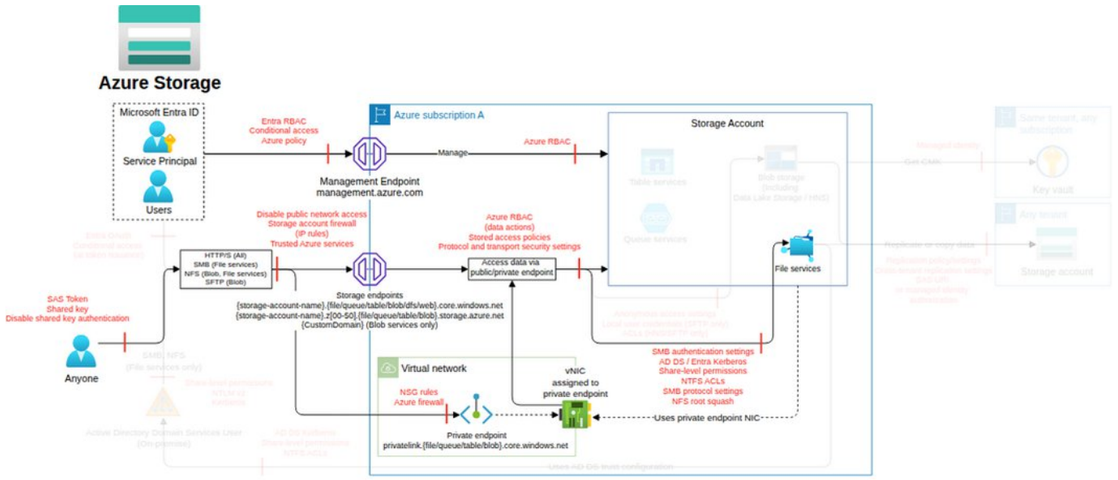
Threat Id	Storage.T20
Name	Distribute malicious content via unauthorized upload
Description	Azure file shares can host files consumed by users, applications, or mounted systems. An attacker can upload malicious or unauthorized content (e.g., scripts, executables, documents with embedded payloads), causing consumers of the file share to retrieve harmful content from a previously trusted storage location.
Goal	Launch another attack
MITRE ATT&CK®	TA0003
CVSS	Medium (5.6)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C149 - Ensure required blobs, file shares, queues, tables, and DFS are using authorized authentication methods.	High	2	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO8 - Enforce encryption in transit C120 - Maintain a list of required file shares' security protocol settings (ideally maximum security SMB 3.1.1, Kerberos, AES-256 only). C121 - Ensure required file shares' security protocol settings are set. C122 - Prevent security protocol settings from changing on required file shares (e.g., by using custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb": {"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Deny mode). C176 - Maintain the authorized minimum TLS version for each storage account. C177 - Ensure storage accounts use the authorized minimum TLS version.	Medium	4	1	-

Data exposure through legacy file share protocols

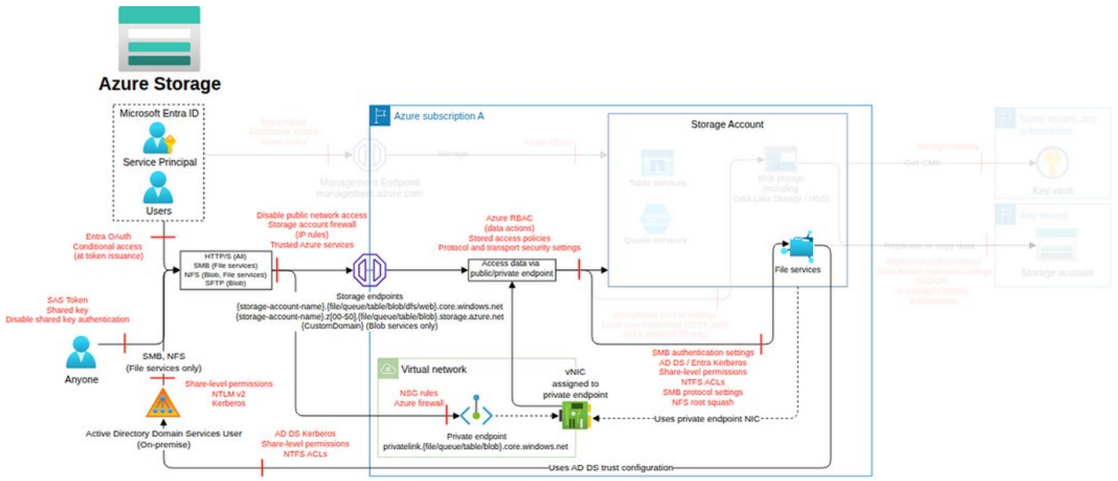
Threat Id	Storage.T61
Name	Data exposure through legacy file share protocols
Description	Azure file shares can support legacy SMB protocol versions and encryption settings for compatibility with older clients (e.g., SMB 2.1, SMB 3.x without encryption, or weaker channel encryption settings). An attacker can modify file share protocol settings to permit weaker access methods, potentially exposing file share contents to protocol-level weaknesses or interception.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (4.2)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO8 - Enforce encryption in transit C98 - Maintain a list of file shares that require NFS/SMB 2.1 to be enabled. C99 - Ensure only file shares that require NFS/SMB 2.1 have it enabled. C100 - Prevent file shares that do not require NFS/SMB 2.1 from being enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb.protocolVersions[*]": ["SMB2.1"]} in Deny mode). C101 - Monitor the creation or update of Azure Files NFS/SMB 2.1 and corresponding settings (e.g., using activity logs on properties.supportsHttpsTrafficOnly with scope "supportsHttpsTrafficOnly").	High	2	1	1
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-

Financial impact through unauthorized upload of Azure file share contents

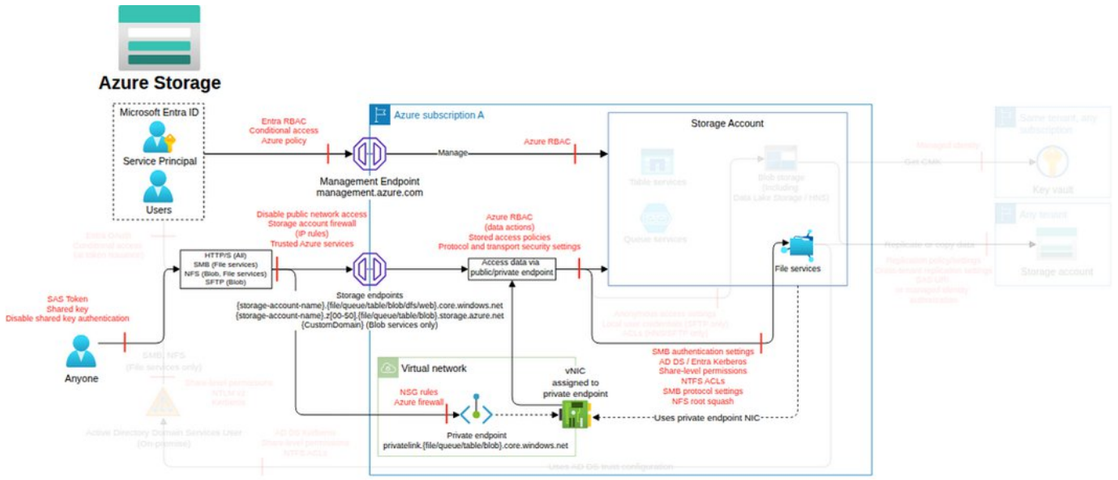
Threat Id	Storage.T16
Name	Financial impact through unauthorized upload of Azure file share contents
Description	Azure file shares can incur storage and transaction costs based on stored file data and usage. An attacker can upload excessive file content to increase storage consumption and generate unexpected costs.
Goal	Financial Drain
MITRE ATT&CK®	TA0040
CVSS	Low (3.5)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-

Data exposure through file share encryption changes

Threat Id	Storage.T60
Name	Data exposure through file share encryption changes
Description	Azure file shares can support encryption settings for data in transit. An attacker can modify file share encryption settings to permit weaker or unencrypted connections, potentially exposing file share contents to interception while they are being accessed.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Low (2.4)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/fileServices/write" }

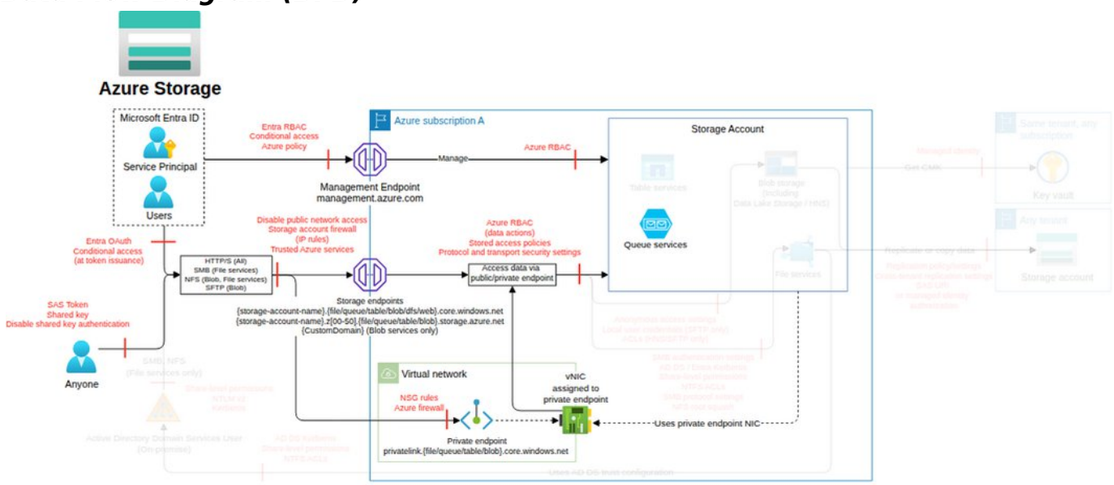


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO8 - Enforce encryption in transit C120 - Maintain a list of required file shares' security protocol settings (ideally maximum security SMB 3.1.1, Kerberos, AES-256 only). C121 - Ensure required file shares' security protocol settings are set. C122 - Prevent security protocol settings from changing on required file shares (e.g., by using custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb": {"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Deny mode). C176 - Maintain the authorized minimum TLS version for each storage account. C177 - Ensure storage accounts use the authorized minimum TLS version.	High	4	1	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-

Queues *(subclass of Storage account, FC4)*

Azure Queue Storage is a service for storing messages. Access messages via HTTP/S calls.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

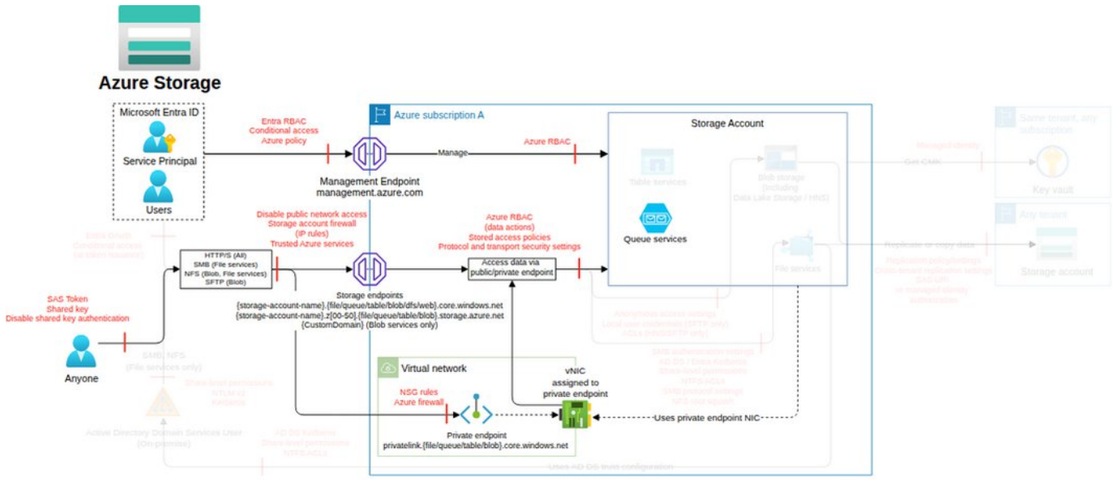
Action	IAM Permission
Creates a new queue with the specified queue name, under the specified account. Creates a new queue with the specified queue name, under the specified account.	Microsoft.Storage/storageAccounts/queueServices/queues/write

Threat List

Name	CVSS
Privilege escalation through queue access policy modification	Medium (6.2)
Service impact through queue message modification or deletion	Medium (6.1)
Unauthorized access to sensitive queue message contents	Medium (5.2)

Privilege escalation through queue access policy modification

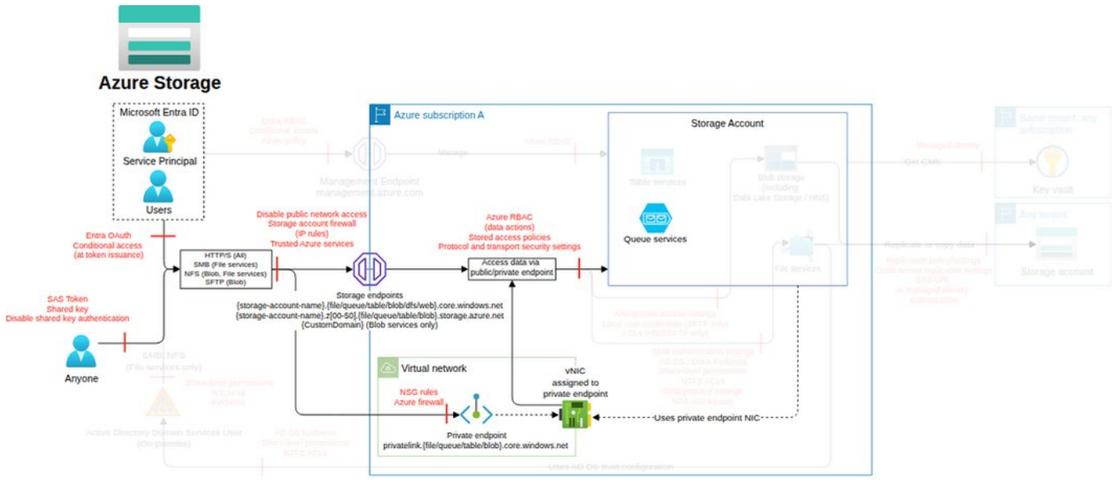
Threat Id	Storage.T27
Name	Privilege escalation through queue access policy modification
Description	Queue stored access policies can define permissions and constraints for Shared Access Signatures used with Azure Storage queues. An attacker can modify queue access policies to grant broader queue access than intended.
Goal	Launch another attack
MITRE ATT&CK®	TA0004
CVSS	Medium (6.2)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/queueServices/write", "Microsoft.Storage/storageAccounts/queueServices/queues/write"] }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements. C156 - Ensure stored access policies and their permissions are authorized for each container, file share, queue, and table.	Medium	5	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	1	-	2

Service impact through queue message modification or deletion

Threat Id	Storage.T31
Name	Service impact through queue message modification or deletion
Description	Azure Storage queues contain messages consumed by applications and downstream services. An attacker can modify, delete, or purge queue messages, causing downstream applications to process incorrect data or lose required work items.
Goal	Data manipulation
MITRE ATT&CK®	TA0040
CVSS	Medium (6.1)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/queueServices/write", "Microsoft.Storage/storageAccounts/queueServices/queues/write", "Microsoft.Storage/storageAccounts/queueServices/queues/delete"] }

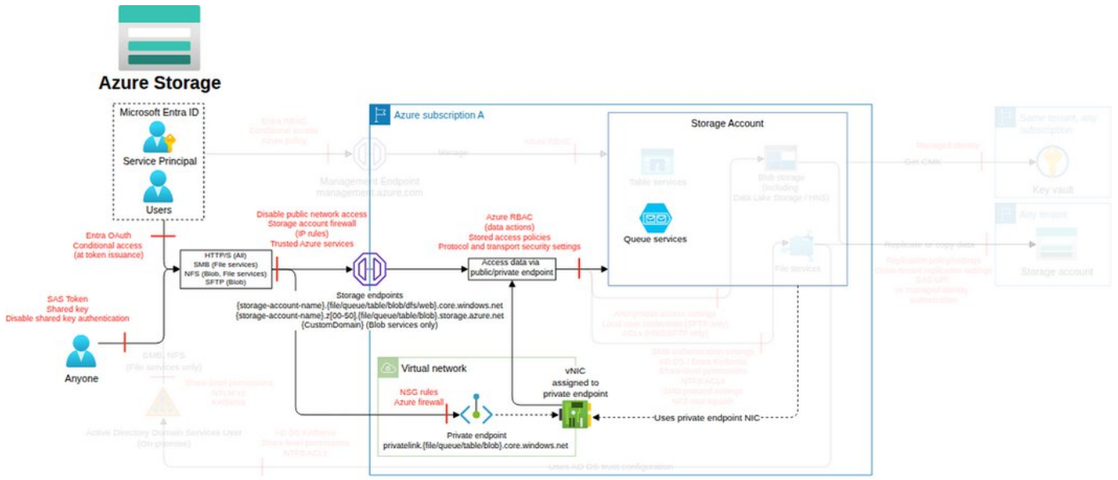


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts. C42 - Maintain a list of authorized IPs that are permitted through each storage account firewall. C43 - Ensure each storage account firewall only allows authorized IPs. C44 - Prevent storage account firewall rules from allowing unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode). C167 - Maintain the list of authorized virtual network rules for each storage account. C168 - Ensure each storage account uses only authorized virtual network rules. C170 - Maintain the list of authorized network ACL bypass configurations for each storage account. C171 - Ensure each storage account uses only authorized network ACL bypass configurations.	High	8	1	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Medium	4	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode).	Medium	1	-	2

C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).				

Unauthorized access to sensitive queue message contents

Threat Id	Storage.T67
Name	Unauthorized access to sensitive queue message contents
Description	Azure Storage queues can contain messages used by applications and downstream services. An attacker with queue message access can read, peek, or dequeue queue messages to access sensitive data, operational instructions, or workflow state outside the intended processing path.
Goal	Data theft
MITRE ATT&CK®	TA0010
CVSS	Medium (5.2)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/queueServices/queues/messages/read", "Microsoft.Storage/storageAccounts/queueServices/queues/messages/process/action"] }

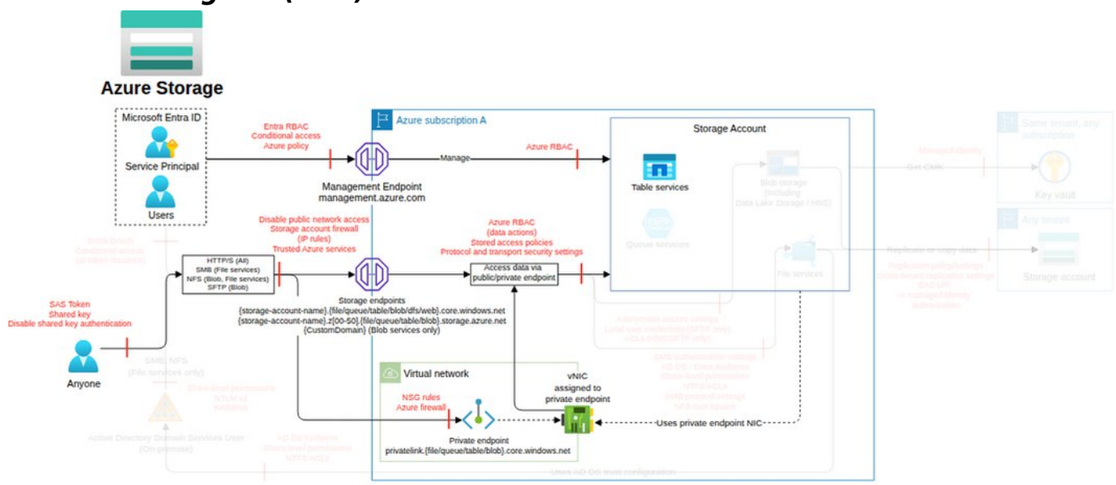


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements. C156 - Ensure stored access policies and their permissions are authorized for each container, file share, queue, and table.	Medium	5	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C164 - Maintain the list of authorized storage queues and access methods allowed to read, peek, or dequeue queue messages. C165 - Ensure only authorized identities and access methods can read, peek, or dequeue storage queue messages. C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	3	-	2

Tables *(subclass of Storage account, FC5)*

Azure table storage is a service that stores non-relational structured data (or structured NoSQL data) in the cloud, providing a key/attribute store with a schemaless design.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

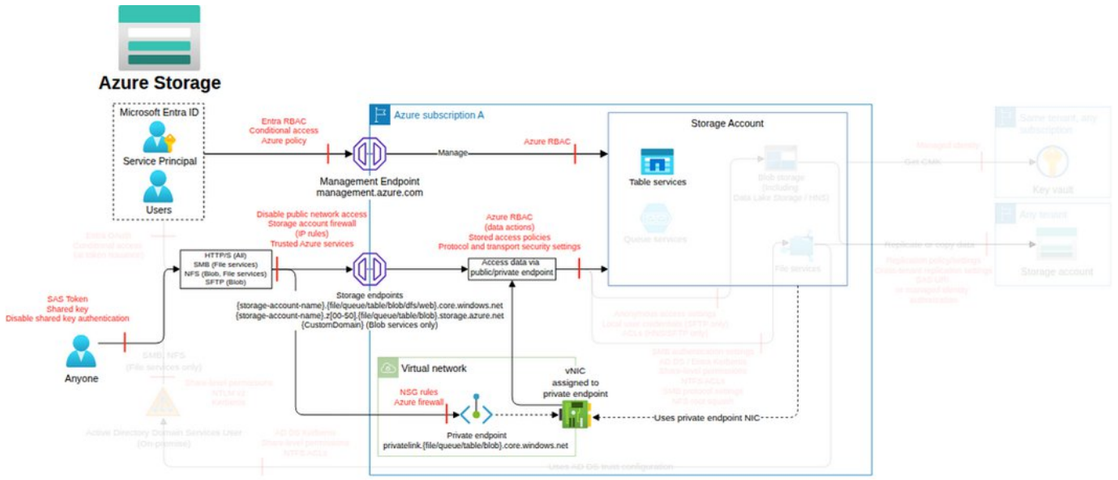
Action	IAM Permission
Creates a new table with the specified table name, under the specified account. Creates a new table with the specified table name, under the specified account.	Microsoft.Storage/storageAccounts/tableServices/tables/write

Threat List

Name	CVSS
Privilege escalation through table access policy modification	Medium (6.2)

Privilege escalation through table access policy modification

Threat Id	Storage.T28
Name	Privilege escalation through table access policy modification
Description	Table stored access policies can define permissions and constraints for Shared Access Signatures used with Azure Storage tables. An attacker can modify table access policies to grant broader table access than intended.
Goal	Launch another attack
MITRE ATT&CK®	TA0004
CVSS	Medium (6.2)
IAM Access	{ "OR": ["Microsoft.Storage/storageAccounts/tableServices/write", "Microsoft.Storage/storageAccounts/tableServices/tables/write"] }

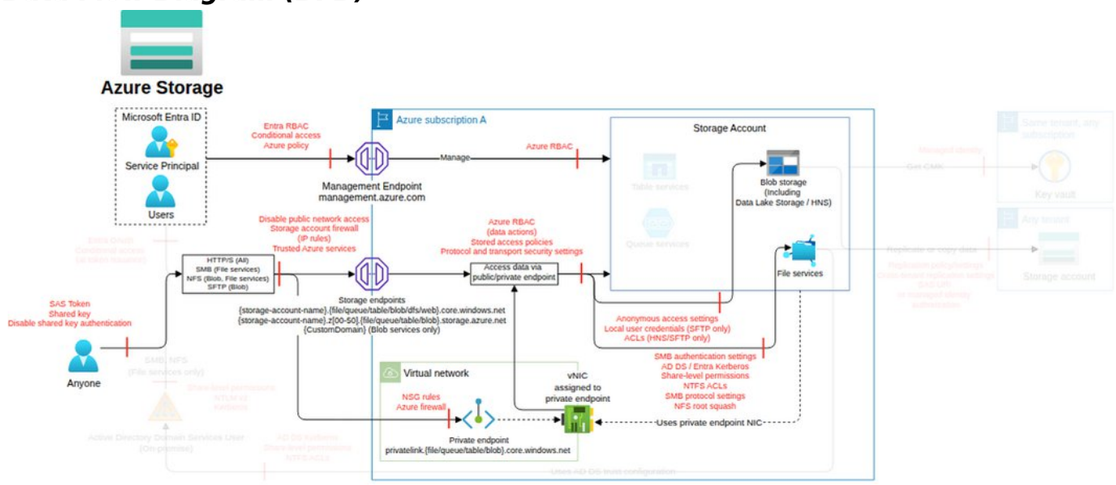


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO22 - Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use C26 - Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods. C27 - Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods. C87 - Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted. C88 - Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements. C156 - Ensure stored access policies and their permissions are authorized for each container, file share, queue, and table.	Medium	5	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C179 - Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode). C180 - Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Medium	1	-	2

Local users *(subclass of Storage account, FC11)*

Blob storage supports the SSH File Transfer Protocol (SFTP). This support lets you securely connect to blob storage via an SFTP endpoint, allowing you to use SFTP for file access, file transfer, and file management.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

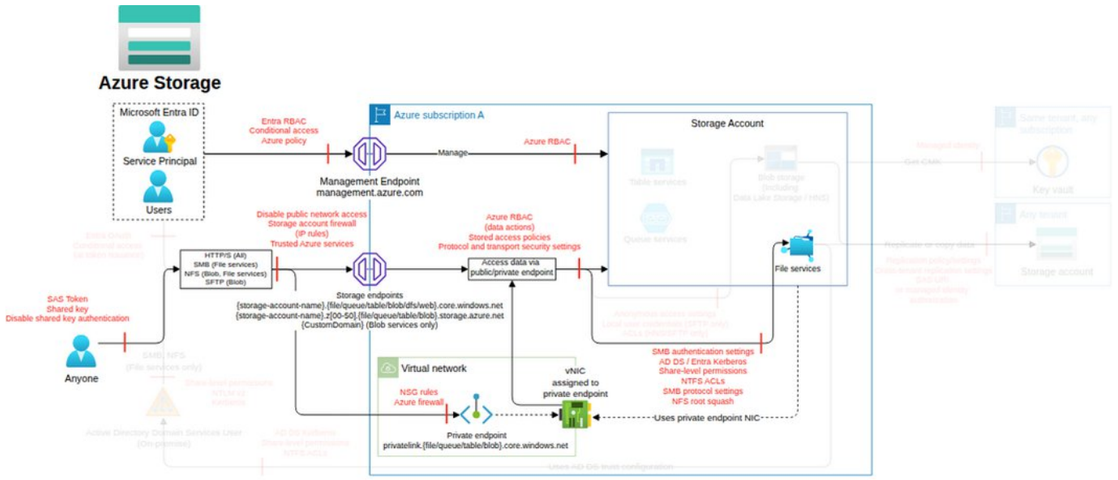
Action	IAM Permission
Create or update the properties of a local user associated with the storage account. Properties for NFSv3 enablement and extended Groups cannot be set with other properties.	Microsoft.Storage/storageAccounts/localUsers/write

Threat List

Name	CVSS
Credential access through SMB Shared Key retrieval	Medium (6.9)
File share access through SMB Shared Key	Medium (5.9)
Persistent access through SFTP local user credentials	Low (3.5)

Credential access through SMB Shared Key retrieval

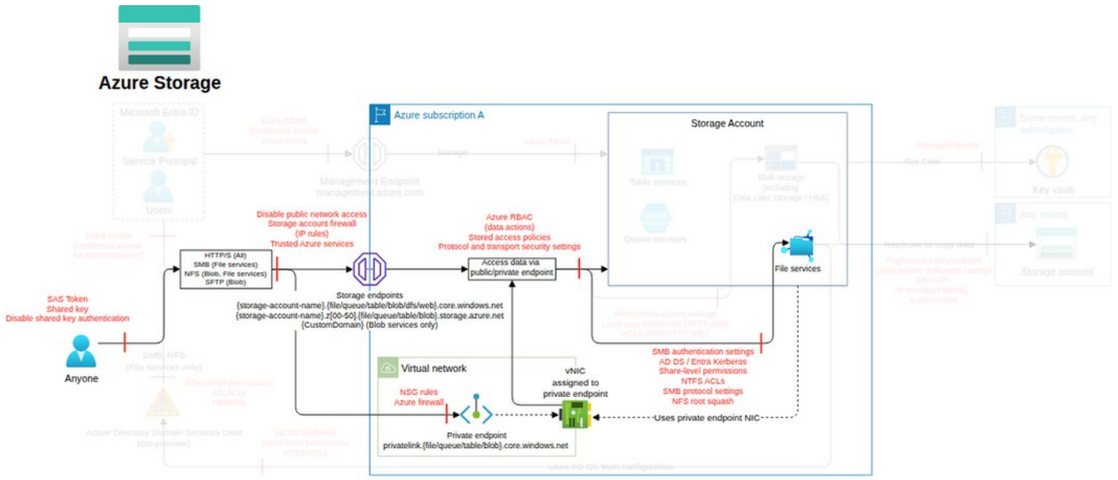
Threat Id	Storage.T64
Name	Credential access through SMB Shared Key retrieval
Description	Azure file shares can support SMB access through local user credentials that include Shared Key material. An attacker can list local user keys to obtain SMB credentials and enable later file share access outside the intended authorization path.
Goal	Launch another attack
MITRE ATT&CK®	TA0006
CVSS	Medium (6.9)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/localUsers/listKeys/action" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C161 - Maintain a list of authorized local users, their permissions, and their authentication methods (e.g., password, SSH) for each hierarchical namespace with SFTP enabled. C162 - Ensure all local users and associated permissions and authentication methods are authorized.	Medium	3	-	-
CO21 - Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model C22 - Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled.	Very Low	1	-	-

File share access through SMB Shared Key

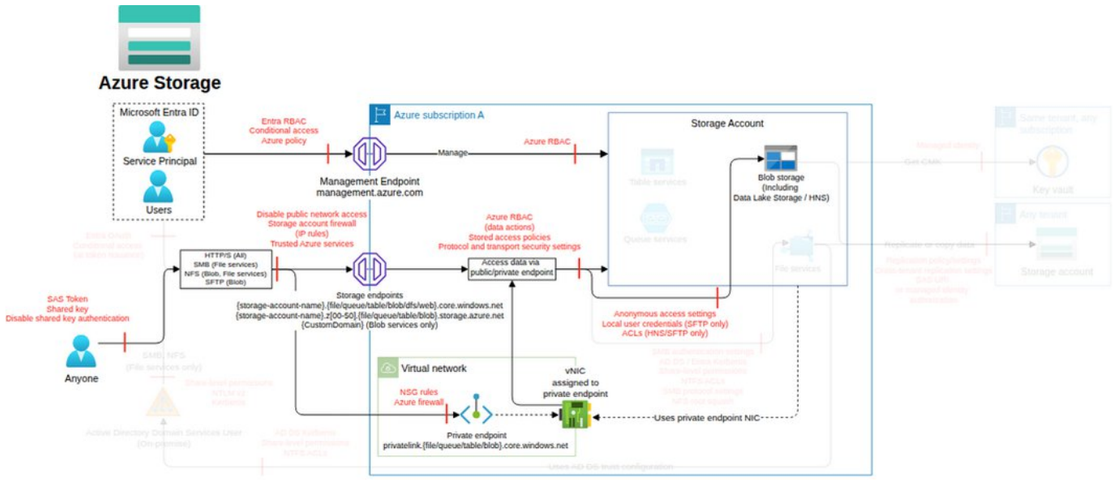
Threat Id	Storage.T65
Name	File share access through SMB Shared Key
Description	Azure file shares can support SMB authentication using Shared Key credentials. An attacker can use an obtained or compromised SMB Shared Key to mount or access the file share outside the intended authorization path.
Goal	Launch another attack
MITRE ATT&CK®	TA0001
CVSS	Medium (5.9)
IAM Access	{}



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C161 - Maintain a list of authorized local users, their permissions, and their authentication methods (e.g., password, SSH) for each hierarchical namespace with SFTP enabled. C162 - Ensure all local users and associated permissions and authentication methods are authorized.	Medium	3	-	-
CO21 - Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model C22 - Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled.	Very Low	1	-	-

Persistent access through SFTP local user credentials

Threat Id	Storage.T44
Name	Persistent access through SFTP local user credentials
Description	Azure Storage accounts can support SFTP access through local user credentials. An attacker can create or modify SFTP local user credentials to maintain an alternate access path to storage data.
Goal	Launch another attack
MITRE ATT&CK®	TA0003
CVSS	Low (3.5)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/localUsers/write" }

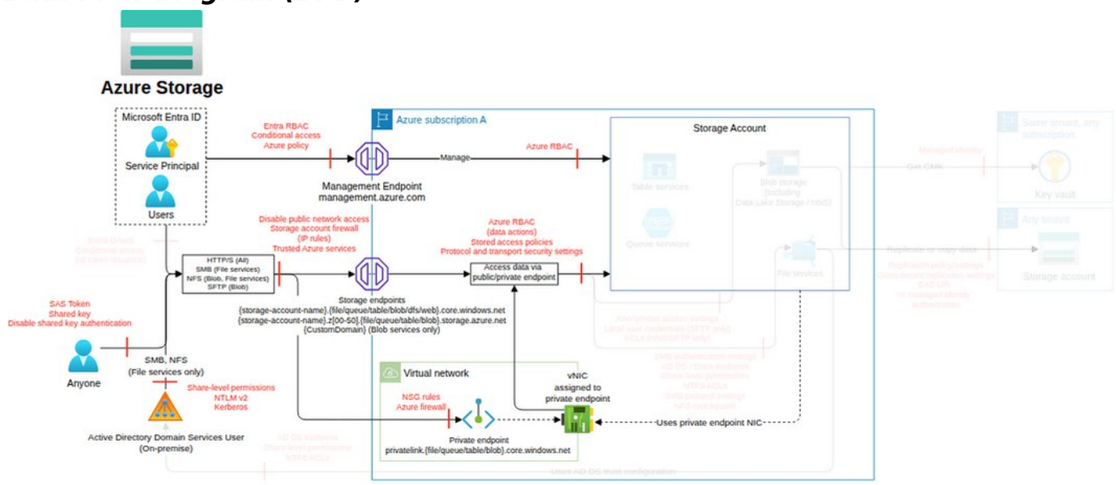


Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO4 - Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) C2 - Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token). C149 - Ensure required blobs, file shares, queues, tables, and DFS are using authorized authentication methods. C161 - Maintain a list of authorized local users, their permissions, and their authentication methods (e.g., password, SSH) for each hierarchical namespace with SFTP enabled.	High	3	-	-
CO17 - Verify the safety of objects ingested and produced C103 - Maintain the list of storage accounts that require malware scanning for uploaded blob content. C104 - Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Medium	2	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO21 - Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model C22 - Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled. C111 - Maintain a list of Data Lake Storage accounts that require SFTP enabled. C114 - Ensure only storage accounts that require SFTP have it enabled. C115 - Prevent storage accounts that do not require SFTP from having it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/isSftpEnabled": true} in Deny mode).	Medium	3	1	-

Private endpoints *(subclass of Storage account, FC12)*

Azure Storage private endpoint is a network interface that enables secure, private connectivity between Azure Storage and virtual networks using Azure Private Link, preventing exposure to the public internet and enhancing security.

Data Flow Diagram (DFD)



Actions and IAM Permissions to deny the feature

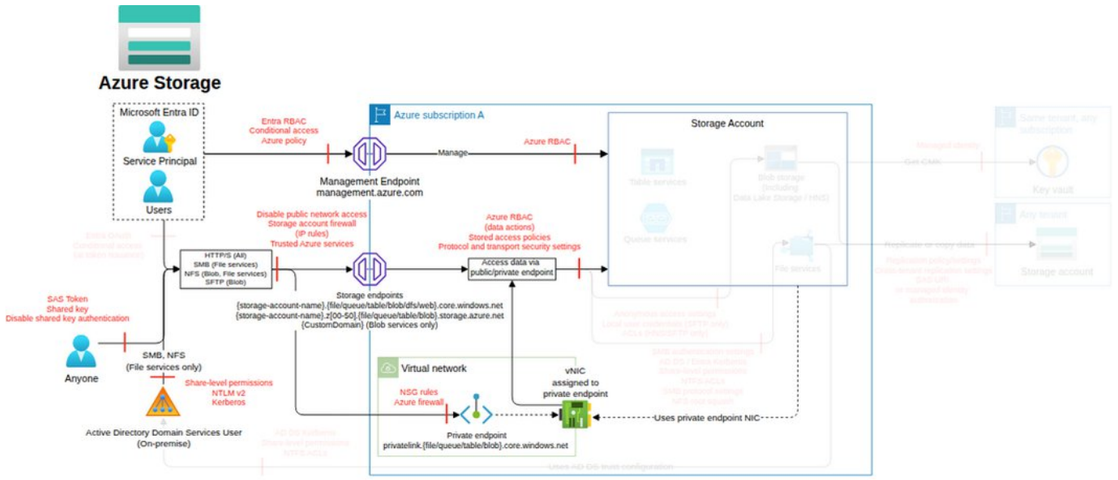
Action	IAM Permission
Update the state of a specified private endpoint connection associated with the storage account.	Microsoft.Storage/storageAccounts/privateEndpointConnections/write

Threat List

Name	CVSS
Service impact through private endpoint deletion	Low (2.4)

Service impact through private endpoint deletion

Threat Id	Storage.T62
Name	Service impact through private endpoint deletion
Description	Azure Storage accounts can use private endpoints to provide private network access to storage services. An attacker can delete a private endpoint connection, disrupting private access paths used by legitimate clients.
Goal	Disruption of Service
MITRE ATT&CK®	TA0040
CVSS	Low (2.4)
IAM Access	{ "UNIQUE": "Microsoft.Storage/storageAccounts/privateEndpointConnections/delete" }



Control Objectives	Priority	# of associated Controls		
		Directive	Preventative	Detective
CO1 - Limit the IAM entities allowed to use the IAM actions required to execute attacks C1 - Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Very High	1	-	-
CO18 - Ensure storage accounts use authorized account types C132 - Maintain the list of authorized account types (e.g., StorageV2) for each storage account. C133 - Ensure storage accounts use authorized account types.	Medium	2	-	-
CO9 - Limit network access to storage account services with private endpoints C37 - Maintain a list of authorized private endpoints on storage accounts. C38 - Ensure only authorized private endpoints are configured on storage accounts.	Medium	2	-	-

Control Implementation

Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO1]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Protect (NIST CSF)	[Storage.C1] Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Request the list of authorized IAM principals with the permissions required to launch attacks, a list of Trusted Networks and PIM settings where applicable, its review process, and its review records.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC10 Storage.FC11 Storage.FC12	Storage.T1 (Very High) Storage.T2 (Very High) Storage.T4 (Very High) Storage.T5 (Very High) Storage.T7 (Very High) Storage.T8 (Very High) Storage.T9 (Very High) Storage.T12 (Very High) Storage.T13 (Very High) Storage.T15 (Very High) Storage.T16 (Very High) Storage.T17 (Very High) Storage.T18 (Very High) Storage.T19 (Very High) Storage.T20 (Very High) Storage.T22 (Very High) Storage.T24 (Very High) Storage.T25 (Very High) Storage.T26 (Very High) Storage.T27 (Very High) Storage.T28 (Very High) Storage.T31 (Very High) Storage.T34 (Very High) Storage.T38 (Very High) Storage.T39 (Very High) Storage.T40 (Very High) Storage.T42 (Very High) Storage.T44 (Very High) Storage.T48 (Very High) Storage.T50 (Very High) Storage.T55 (Very High) Storage.T56 (Very High) Storage.T58 (Very High) Storage.T59 (Very High) Storage.T60 (Very High) Storage.T62 (Very High) Storage.T63 (Very High) Storage.T64 (Very High) Storage.T65 (Very High) Storage.T66 (Very High) Storage.T67 (Very High) Storage.T68 (Very High) Storage.T69 (Very High)	Very High

Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO4]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C2] Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token).	Request the list of storage accounts and their authorized authentication methods, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T3 (Very Low) Storage.T5 (Very Low) Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	High
Directive (COSO) Identify (NIST CSF)	[Storage.C6] Maintain the list of storage accounts authorized to have public network access enabled.	Request the list of storage accounts authorized to have public network access enabled, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T5 (Very Low) Storage.T50 (Very Low) Storage.T68 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C7, depends on Storage.C6, assured by Storage.C9] Ensure only authorized storage accounts have public network access enabled.	Request 1) the mechanism ensuring only authorized storage accounts have public network access enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C8, depends on Storage.C6] Prevent unauthorized storage accounts from having public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Deny mode).	Enable public network access on an unauthorized storage account; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C9] Verify unauthorized storage accounts do not have public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Audit mode).	Enable public network access on an unauthorized storage account; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C55] Maintain a list of blobs and containers that require anonymous access.	Request the list of authorized blobs and containers that require anonymous access, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T5 (Very Low) Storage.T50 (Very Low) Storage.T68 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C56, depends on Storage.C55, assured by Storage.C59] Ensure anonymous access is set only for required blobs and containers.	Request 1) the mechanism ensuring only required blobs/containers are anonymously accessible, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C57, depends on Storage.C55, assured by Storage.C59] Ensure only required blobs and containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Deny mode).	Create or update a non-required blob or a container to be anonymously accessible; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Assurance (COSO)	[Storage.C59]	Create or update a non-required blob or a container to	Medium	Storage.FC1	-	Medium

Detect (NIST CSF)	Verify only required blobs or containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Audit mode).	be anonymously accessible; it should be detected.		Storage.FC2		
Assurance (COSO) Detect (NIST CSF)	[Storage.C81] Verify Shared Key authorization is disabled (and optionally that OAuth is the default) for authorized storage accounts that host authorized blobs, file shares, queues, tables, and DFS (e.g., by using a custom Azure Policy to require "Microsoft.Storage/storageAccounts/allowSharedKeyAccess" = false and, optionally, "Microsoft.Storage/storageAccounts/defaultToOAuthAuthentication" = true in Audit mode).	Configure a storage account with allowSharedKeyAccess enabled and/or defaultToOAuthAuthentication disabled; it should be detected.	Medium	Storage.FC3 Storage.FC11	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C90] Maintain a list of storage accounts that require static website hosting.	Request the list of storage accounts that require static website hosting, its review process, and its review records.	Low	Storage.FC2	Storage.T22 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C91, depends on Storage.C90, assured by Storage.C93] Ensure only storage accounts that require static website hosting have it enabled.	Request 1) the mechanism ensuring only required storage accounts have the static website hosting enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T22 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C92, depends on Storage.C90] Prevent storage accounts that do not require static website hosting from having it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/staticWebsite.enabled": true} in Deny mode).	Modify an existing storage account that does not require static website hosting to be enabled; it should be denied.	Medium	Storage.FC2	Storage.T22 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C93] Verify storage accounts that do not require static website hosting do not have it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/staticWebsite.enabled": true} in Audit mode).	Modify an existing storage account that does not require static website hosting to be enabled; it should be detected.	Medium	Storage.FC2	-	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C149, depends on Storage.C2, assured by Storage.C81] Ensure required blobs, file shares, queues, tables, and DFS are using authorized authentication methods.	Request 1) the mechanism ensuring that blobs, file shares, queues, tables, and DFS use authorized authentication methods, 2) its records of execution for new storage account services, and 3) the plan to move any older storage account services.	Very Low	Storage.FC3 Storage.FC11	Storage.T20 (Medium) Storage.T44 (Medium)	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C161, depends on Storage.C22] Maintain a list of authorized local users, their permissions, and their authentication methods (e.g., password, SSH) for each hierarchical namespace with SFTP enabled.	Request the list of authorized local users, their permissions, their authentication methods, its review process, and its review records.	Low	Storage.FC11	Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C162, depends on Storage.C161, assured by Storage.C163] Ensure all local users and associated permissions and authentication methods are authorized.	Request 1) the mechanism ensuring all local users and associated permissions and authentication methods are authorized, 2) its records of execution for all new local users, and 3) the plan to move any older local users.	Low	Storage.FC11	Storage.T64 (Medium) Storage.T65 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C163] Verify local user functionality for SFTP is enabled only where local users, their permissions, and authentication	Enable local users in a hierarchical namespace with SFTP, where local users are not authorized; it should be detected.	Medium	Storage.FC11	-	Medium

	methods are authorized (e.g., by using a custom Azure Policy to detect storage accounts where "Microsoft.Storage/storageAccounts/localUsersEnabled" = true and "Microsoft.Storage/storageAccounts/isSftpEnabled" = true, in Audit mode).					
Directive (COSO) Identify (NIST CSF)	[Storage.C164] Maintain the list of authorized storage queues and access methods allowed to read, peek, or dequeue queue messages.	Request the list of authorized storage queues and access methods allowed to read, peek, or dequeue queue messages, its review process, and its review records.	Low	Storage.FC4	Storage.T67 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C165, depends on Storage.C164, assured by Storage.C166] Ensure only authorized identities and access methods can read, peek, or dequeue storage queue messages.	Request 1) the mechanism ensuring that only authorized identities and access methods can read, peek, or dequeue storage queue messages, 2) its records of execution for all new storage queues and access grants, and 3) the plan to move any older storage queues and access grants.	Medium	Storage.FC4	Storage.T67 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C166] Verify only authorized identities and access methods can read, peek, or dequeue storage queue messages.	Read, peek, or dequeue a storage queue message outside authorized access methods; it should be detected.	Medium	Storage.FC4	-	Medium
Detective (COSO) Detect (NIST CSF)	[Storage.C179] Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode).	Configure a storage account to allow Shared Key authorization; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium) Storage.T69 (Medium)	Medium
Detective (COSO) Detect (NIST CSF)	[Storage.C180] Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Configure a storage account without a SAS expiration policy; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium)	Medium

Ensure backup, replication, and recovery capabilities for storage account services [Storage.CO5]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Protect (NIST CSF)	[Storage.C10, depends on Storage.C144, assured by Storage.C11] Ensure versioning is enabled on required containers.	Request 1) the mechanism ensuring versioning is enabled on required containers, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T7 (Low) Storage.T40 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C11] Verify versioning is enabled on required containers (e.g., by using Azure built-in policy "Configure your Storage account to enable blob versioning" in Audit mode).	Remove versioning from a container; it should be detected.	Medium	Storage.FC2	-	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C12, depends on Storage.C145, assured by Storage.C13] Ensure snapshots are enabled for required file shares.	Request 1) the mechanism that ensures snapshots are enabled for required file shares, 2) its records of	Low	Storage.FC2	Storage.T7 (Low)	Low

		execution for all new storage accounts, and 3) the plan to move any older storage accounts.				
Assurance (COSO) Detect (NIST CSF)	[Storage.C13] Verify snapshots are enabled for required file shares (e.g., by using a custom Azure Policy on {"properties.shareSnapshotEnabled": false} in Audit mode).	Disable snapshotting from a required file share; it should be detected.	Medium	Storage.FC2	-	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C15] Maintain a list of the blob storage containers that are required to have a minimum retention period enabled.	Request the list of storage account containers that have a minimum retention period according to requirements, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T25 (Very Low) Storage.T39 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C16, depends on Storage.C15, assured by Storage.C18] Ensure required storage accounts have the soft-delete feature for blobs enabled for the defined minimum retention period.	Request 1) the mechanism ensuring required storage accounts have soft-delete for blobs enabled for at least the defined minimum retention, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low)	Very Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C17, depends on Storage.C15] Prevent the creation of required storage accounts without the blob soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/deleteRetentionPolicy.enabled": false} in Deny mode).	Create a required storage account without soft-delete for the blob; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (High) Storage.T25 (Low) Storage.T39 (Very Low)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C18] Verify required storage accounts have the blob soft-delete option enabled (e.g., by using custom Azure Policy in Audit mode).	Create a required storage account without soft-delete for the blob option; it should be detected.	Medium	Storage.FC2	-	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C19, depends on Storage.C15, assured by Storage.C21] Ensure required storage accounts have the soft-delete option enabled for the containers.	Request 1) the mechanism ensuring required storage accounts have soft-delete for the container enabled, 2) its records of execution for all new required storage accounts, and 3) the plan to move any older required storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T25 (Low) Storage.T39 (Very Low)	Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C20, depends on Storage.C15] Prevent the creation of containers without the soft-delete option enabled (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/blobServices/containers/softDelete" in Deny mode).	Create a container without soft-delete; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (High) Storage.T25 (Low) Storage.T39 (Very Low)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C21] Verify required storage accounts have container soft delete enabled (e.g., by using a custom Azure Policy in Audit mode).	Create a storage account without soft-delete for the container option; it should be detected.	Low	Storage.FC1 Storage.FC2	-	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C46, depends on Storage.C147] Ensure storage blobs are using Azure Backup, following the requirements in the Azure Backup ThreatModel.	Request 1) the mechanism ensuring storage blobs are using Azure Backup, 2) its records of execution for all new storage blobs, and 3) the plan to move any older storage blobs.	High	Storage.FC2	Storage.T7 (Medium) Storage.T9 (Medium) Storage.T12 (Medium)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C47] Maintain a list of storage accounts that require cross-tenant replication and their allowed destination tenants.	Request the list of storage accounts that require cross-tenant replication and their allowed destination tenants, its review process, and its review records.	Low	Storage.FC1	Storage.T13 (Very Low) Storage.T42 (Very Low) Storage.T63 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C48, depends on Storage.C47, assured by Storage.C49] Ensure only required storage accounts have cross-tenant replication enabled and the destination storage accounts	Request 1) the mechanism that ensures only required storage accounts have cross-tenant replication enabled	Low	Storage.FC1	Storage.T13 (High) Storage.T42 (High)	High

	are authorized.	and the destination storage accounts are authorized, 2) its records of execution for all new storage accounts, 3) the plan to move any older storage accounts.			Storage.T63 (Medium)	
Assurance (COSO) Detect (NIST CSF)	[Storage.C49] Verify only the storage accounts that require cross-tenant replication have it enabled (e.g., by using the Azure built-in policy StorageAccountAllowCrossTenantReplication_Audit.json).	Create a storage account with cross-tenant replication enabled; it should be detected.	Medium	Storage.FC1	-	High
Directive (COSO) Identify (NIST CSF)	[Storage.C71] Maintain a list of storage accounts that require redundancy.	Request the list of storage accounts that require redundancy, its review process, and its review records.	Low	Storage.FC1	Storage.T42 (Very Low)	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C72, depends on Storage.C71, assured by Storage.C74] Ensure required storage accounts use redundancy.	Request 1) the mechanism ensuring required storage accounts use redundancy, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T42 (Medium)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C74] Verify required storage accounts use redundancy (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/sku.name": ["Standard_GRS","Standard_ZRS"]} in Audit mode).	Remove a redundancy configuration from a storage account; it should be detected.	Medium	Storage.FC1	-	Very Low
Directive (COSO) Identify (NIST CSF)	[Storage.C75] Maintain a list of authorized storage account regions that can be used for redundancy.	Request the list of authorized storage account regions that are used for redundancy, its review process, and its review records.	Low	Storage.FC1	Storage.T59 (Very Low)	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C76, depends on Storage.C75, assured by Storage.C78] Ensure the authorized storage account region used for redundancy is configured.	Request 1) the mechanism ensuring only authorized regions for storage accounts are in use, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T59 (Medium)	Very Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C77, depends on Storage.C75] Ensure only the authorized storage account region is set for redundancy (e.g., by using a custom Azure Policy on {"location": ["eastus","westeurope"]} in Deny mode).	Create a redundancy configuration on a storage account with an unauthorized region; it should be denied.	Medium	Storage.FC1	Storage.T59 (High)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C78] Verify only the authorized storage account region is set for redundancy (e.g., by using a custom Azure Policy with {"location": ["eastus","westeurope"]} in Audit mode).	Create a storage account with an unauthorized region; it should be detected.	Medium	Storage.FC1	-	Very Low
Directive (COSO) Identify (NIST CSF)	[Storage.C83] Define the minimum retention period for required file shares (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/deleteRetentionPolicy.days": 7} in Deny mode).	For each file share, request the minimum retention from deletion, its review process, and its review records.	Medium	Storage.FC3	Storage.T18 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C84, depends on Storage.C83, assured by Storage.C86] Ensure file shares have the soft-delete option enabled for at least the defined minimum retention period.	Request 1) the mechanism ensuring file shares have soft-delete enabled for at least the defined minimum retention, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Low	Storage.FC3	Storage.T18 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C85, depends on Storage.C83] Prevent the creation of file shares without the soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/shares/deleteRetentionPolicy.enabled": false} in Deny mode).	Create a file share without soft-delete; it should be denied.	Medium	Storage.FC3	Storage.T18 (Medium)	Medium
Assurance (COSO)	[Storage.C86] Verify all file shares have the soft-delete option enabled	Create a file share without soft-delete; it should be	Medium	Storage.FC3	-	Medium

Detect (NIST CSF)	(e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/shares/deleteRetentionPolicy.enabled": false} in Audit mode).	detected.				
Directive (COSO) Identify (NIST CSF)	[Storage.C145] Maintain a list of Azure Files that require snapshots.	Request the list of Azure Files that require snapshots, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C147] Maintain a list of storage blobs that require Azure Backup.	Request the list of storage blobs that require Azure Backup, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C150] Maintain a list of authorized replication policies and their destination storage accounts.	Request the list of authorized replication policies and their destination storage accounts, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C151, depends on Storage.C150, assured by Storage.C152] Ensure replication policies and their destination storage accounts are authorized.	Request 1) the mechanism ensuring replication policies and their destination storage accounts are authorized, 2) its records of execution for all new replication policies, and 3) the plan to move any older replication policies.	Very Low	Storage.FC1	Storage.T13 (Low)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C152] Verify replication policies and their destination storage accounts are authorized (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/objectReplicationPolicies[*].destinationAccount" and "Microsoft.Storage/storageAccounts/objectReplicationPolicies[*].rules[*].sourceContainer" in Audit mode).	Add a replication policy with an unauthorized destination storage account or source container; it should be detected.	Medium	Storage.FC1	-	Medium

Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO7]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C158] Maintain a list of authorized Access Control Lists (ACLs) on Data Lake Storage containers.	Request the list of authorized Access Control Lists (ACLs) on Data Lake Storage containers, its review process, and its review records.	Low	Storage.FC2	Storage.T58 (Very Low) Storage.T66 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C159, depends on Storage.C158, assured by Storage.C160] Ensure all Access Control Lists (ACLs) on Data Lake Storage containers are authorized.	Request 1) the mechanism ensuring all Access Control Lists (ACLs) on Data Lake Storage containers are authorized, 2) its records of execution for all new ACLs, and 3) the plan to move any older ACLs.	Low	Storage.FC2	Storage.T58 (Medium) Storage.T66 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C160] Verify Azure Data Lake Storage Gen2 functionality is enabled only where Access Control Lists (ACLs) on containers are authorized (e.g., by using a custom Azure Policy to detect storage accounts where "Microsoft.Storage/storageAccounts/isHnsEnabled" = true and "Microsoft.Storage/storageAccounts/allowSharedKeyAccess" = true, in Audit mode).	Enable Shared Key authorization (allowSharedKeyAccess = true) on an HNS-enabled storage account, or disable hierarchical namespace (isHnsEnabled = false) on a storage account; it should be detected.	Medium	Storage.FC2	-	Medium

Enforce encryption in transit [Storage.CO8]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C98] Maintain a list of file shares that require NFS/SMB 2.1 to be enabled.	Request the list of file shares that require NFS/SMB 2.1 to be enabled, its review process, and its review records.	Low	Storage.FC3	Storage.T61 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C99, depends on Storage.C98, assured by Storage.C102] Ensure only file shares that require NFS/SMB 2.1 have it enabled.	Request 1) the mechanism ensuring only file shares that require NFS/SMB 2.1 have it enabled, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Very Low	Storage.FC3	Storage.T61 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C100, depends on Storage.C98] Prevent file shares that do not require NFS/SMB 2.1 from being enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb.protocolVersions[*]": ["SMB2.1"]} in Deny mode).	Modify an existing file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be denied.	Medium	Storage.FC3	Storage.T61 (High)	Medium
Detective (COSO) Detect (NIST CSF)	[Storage.C101] Monitor the creation or update of Azure Files NFS/SMB 2.1 and corresponding settings (e.g., using activity logs on properties.supportsHttpsTrafficOnly with scope "supportsHttpsTrafficOnly").	Modify or create a file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be detected.	Low	Storage.FC3	Storage.T61 (Very Low)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C102] Verify only file shares that require NFS/SMB 2.1 have it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb.protocolVersions[*]": ["SMB2.1"]} in Audit mode).	Modify an existing file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be detected.	Medium	Storage.FC3	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C120] Maintain a list of required file shares' security protocol settings (ideally maximum security SMB 3.1.1, Kerberos, AES-256 only).	Request the list of required file share security protocol settings, its review process, and its review records.	Low	Storage.FC3	Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T60 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C121, depends on Storage.C120, assured by Storage.C123] Ensure required file shares' security protocol settings are set.	Request 1) the mechanism ensuring that the required security protocol settings for file shares in storage accounts are in use, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Low	Storage.FC3	Storage.T15 (Medium) Storage.T20 (Low) Storage.T60 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C122, depends on Storage.C120] Prevent security protocol settings from changing on required file shares (e.g., by using custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb":{"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Deny mode).	Modify a required file share's security protocol settings; it should be denied.	Medium	Storage.FC3	Storage.T20 (Low) Storage.T60 (High)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C123] Verify required file shares' security protocol settings are configured (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb":{"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Audit mode).	Modify a required file share's security protocol settings; it should be detected.	Medium	Storage.FC3	-	Medium

Directive (COSO) Identify (NIST CSF)	[Storage.C176] Maintain the authorized minimum TLS version for each storage account.	Request the authorized minimum TLS version for each storage account, its review process, and its review records.	Low	Storage.FC3	Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T60 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C177, depends on Storage.C176, assured by Storage.C178] Ensure storage accounts use the authorized minimum TLS version.	Request 1) the mechanism ensuring storage accounts use the authorized minimum TLS version, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC3	Storage.T15 (Medium) Storage.T20 (Medium) Storage.T60 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C178] Verify storage accounts use the authorized minimum TLS version (e.g., by using Azure built-in policy "Storage accounts should have the specified minimum TLS version" in Audit mode).	Create or modify a storage account with a minimum TLS version below the authorized value; it should be detected.	Medium	Storage.FC3	-	Medium

Limit network access to storage account services with private endpoints [Storage.CO9]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C37] Maintain a list of authorized private endpoints on storage accounts.	Request the list of authorized private endpoints, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC12	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low) Storage.T62 (Very Low)	Very High
Directive (COSO) Protect (NIST CSF)	[Storage.C38, depends on Storage.C37, assured by Storage.C40] Ensure only authorized private endpoints are configured on storage accounts.	Request 1) the mechanism ensuring only authorized private endpoints are configured on storage accounts, 2) its records of execution for all private endpoints.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC12	Storage.T1 (Very High) Storage.T5 (Low) Storage.T12 (Medium) Storage.T15 (Low) Storage.T31 (Low) Storage.T50 (Medium) Storage.T62 (Medium)	Very High
Assurance (COSO) Detect (NIST CSF)	[Storage.C40] Verify only authorized private endpoints are configured on storage accounts (e.g., by using the built-in Azure Policy "Storage accounts should use Private Link" in Audit mode).	Configure an unauthorized private endpoint on a storage account; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC12	-	Very High
Directive (COSO) Identify (NIST CSF)	[Storage.C42] Maintain a list of authorized IPs that are permitted through each storage account firewall.	Request the list of authorized IPs, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C43, depends on Storage.C42, assured by Storage.C45] Ensure each storage account firewall only allows authorized IPs.	Request 1) the mechanism ensuring firewall rules only allow authorized IP addresses, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (High) Storage.T5 (High) Storage.T12 (High) Storage.T15 (High) Storage.T31 (High) Storage.T50 (High)	High
Preventative (COSO)	[Storage.C44, depends on Storage.C42] Prevent storage account firewall rules from allowing	Configure a storage account firewall to allow an	Very Low	Storage.FC1	Storage.T1 (Very Low)	Medium

Protect (NIST CSF)	unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode).	unauthorized IP; it should be denied.		Storage.FC3 Storage.FC4	Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	
Assurance (COSO) Detect (NIST CSF)	[Storage.C45] Verify storage account firewall rules allow only authorized IPs (e.g., by using built-in Azure Policy "Storage accounts should restrict network access" in Audit mode).	Configure a storage account firewall to allow an unauthorized IP; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	-	High
Directive (COSO) Identify (NIST CSF)	[Storage.C167] Maintain the list of authorized virtual network rules for each storage account.	Request the list of authorized virtual network rules for each storage account, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C168, depends on Storage.C167, assured by Storage.C169] Ensure each storage account uses only authorized virtual network rules.	Request 1) the mechanism ensuring each storage account uses only authorized virtual network rules, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Medium) Storage.T5 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C169] Verify each storage account uses only authorized virtual network rules (e.g., by using Azure built-in policy "Storage accounts should restrict network access using virtual network rules" in Audit mode).	Configure a storage account with an unauthorized virtual network rule; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C170] Maintain the list of authorized network ACL bypass configurations for each storage account.	Request the list of authorized network ACL bypass configurations for each storage account, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C171, depends on Storage.C170, assured by Storage.C172] Ensure each storage account uses only authorized network ACL bypass configurations.	Request 1) the mechanism ensuring each storage account uses only authorized network ACL bypass configurations, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Medium) Storage.T5 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C172] Verify each storage account uses only authorized network ACL bypass configurations (e.g., by using Azure built-in policy "Storage Accounts should restrict network access through network ACL bypass configuration only" in Audit mode).	Configure a storage account with an unauthorized network ACL bypass configuration; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	-	Medium

Enable enhanced monitoring and notifications for storage accounts [Storage.CO11]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C50] Maintain a list of storage accounts that require diagnostic	Request the list of storage accounts that require diagnostic settings, its review process, and its review	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low)	Very Low

	settings to be enabled and their respective log destinations.	records.			Storage.T9 (Very Low) Storage.T13 (Very Low) Storage.T42 (Very Low)	
Directive (COSO) Protect (NIST CSF)	[Storage.C51, depends on Storage.C50, assured by Storage.C54] Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Request 1) the mechanism ensuring only authorized diagnostic settings destinations are enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T13 (Very Low) Storage.T42 (Very Low)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C54] Verify storage accounts have diagnostic settings configured (e.g., by using built-in Azure Policy "Configure diagnostic settings for storage accounts to Log Analytics workspace" in Audit mode).	Create a storage account with unauthorized diagnostic settings options; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Very Low

Enforce encryption on data at rest and protect encryption keys [Storage.CO12]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C60] Maintain a list of authorized customer-managed keys used by storage accounts.	Request the list of authorized customer-managed keys used by storage accounts, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Very Low) Storage.T38 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C61, depends on Storage.C60, assured by Storage.C65] Ensure only authorized customer-managed keys are configured for storage accounts.	Request 1) the mechanism ensuring only authorized customer-managed keys are used to encrypt each storage account, 2) its records of execution for all new storage accounts, and 3) the plan to move any older authorized storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Low) Storage.T38 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C63, depends on Storage.C61, assured by Storage.C65] Prevent storage accounts that require customer-managed keys from using service-managed keys (e.g., by using built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Deny mode).	Create a storage account requiring CMK with a service-managed key; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (Medium) Storage.T38 (High)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C65] Verify only authorized customer-managed keys are configured for storage accounts (e.g., by using the built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Audit mode).	Modify a required storage account to use an unauthorized CMK; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C173] Maintain a list of storage accounts that require Azure infrastructure encryption (i.e., double encryption).	Request the list of storage accounts that require Azure infrastructure encryption (double encryption), its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Very Low) Storage.T38 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C174, depends on Storage.C173, assured by Storage.C175] Ensure required storage accounts use Azure infrastructure encryption.	Request 1) the mechanism ensuring required storage accounts use Azure infrastructure encryption, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (Medium) Storage.T38 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C175] Verify required storage accounts use Azure infrastructure encryption (e.g., by using Azure built-in policy "Storage accounts should have infrastructure encryption" in Audit mode).	Create or modify a required storage account without Azure infrastructure encryption; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium

Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO14]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C94] Maintain a list of authorized CORS per endpoint, trusted origins, and corresponding settings.	Request the list of authorized storage accounts with CORS trusted origins and corresponding settings, its review process, and its review records.	Low	Storage.FC1	Storage.T26 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C95, depends on Storage.C94, assured by Storage.C97] Ensure only authorized storage accounts have CORS-trusted origins and corresponding settings configured.	Request 1) the mechanism ensuring only authorized storage accounts have CORS trusted origins and corresponding settings configured, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T26 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C96, depends on Storage.C94] Prevent unauthorized storage accounts from using CORS trusted origins and corresponding settings (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/cors.allowedOrigins[*]": ["https://myapp.contoso.com"]} in Deny mode).	Create a storage account with untrusted CORS settings; it should be denied.	Medium	Storage.FC1	Storage.T26 (High)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C97] Verify only authorized CORS trusted origins and corresponding settings are configured (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/cors.allowedOrigins[*]": ["https://myapp.contoso.com"]} in Audit mode).	Create a storage account with untrusted CORS settings; it should be detected.	Medium	Storage.FC1	-	Medium

Verify the safety of objects ingested and produced [Storage.CO17]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C103] Maintain the list of storage accounts that require malware scanning for uploaded blob content.	Request the list of storage accounts that require malware scanning for uploaded blob content, its review process, and its review records.	Low	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C104, depends on Storage.C103, assured by Storage.C105] Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Request 1) the mechanism ensuring required storage accounts use malware scanning for uploaded blob content, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T8 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T20 (Medium) Storage.T27 (Medium) Storage.T28 (Medium)	Medium

					Storage.T31 (Medium) Storage.T34 (Medium) Storage.T44 (Medium) Storage.T64 (Medium) Storage.T65 (Medium)	
Assurance (COSO) Detect (NIST CSF)	[Storage.C105] Verify required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage malware scanning).	Upload malware test content to a required storage account; it should be detected.	Medium	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	-	Medium

Ensure storage accounts use authorized account types [Storage.CO18]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C132] Maintain the list of authorized account types (e.g., StorageV2) for each storage account.	Request the list of authorized account types for each storage account, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11 Storage.FC12	Storage.T5 (Very Low) Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T25 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T39 (Very Low) Storage.T42 (Very Low) Storage.T44 (Very Low) Storage.T50 (Very Low) Storage.T59 (Very Low) Storage.T60 (Very Low) Storage.T61 (Very Low) Storage.T62 (Very Low) Storage.T63 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low) Storage.T66 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C133, depends on Storage.C132, assured by Storage.C134] Ensure storage accounts use authorized account types.	Request 1) the mechanism ensuring storage accounts use authorized account types, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11 Storage.FC12	Storage.T5 (Medium) Storage.T7 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T13 (Medium) Storage.T15 (Medium) Storage.T20 (Medium) Storage.T25 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T34 (Medium)	Medium

					Storage.T39 (Medium) Storage.T42 (Medium) Storage.T44 (Medium) Storage.T50 (Medium) Storage.T59 (Medium) Storage.T60 (Medium) Storage.T61 (Medium) Storage.T62 (Medium) Storage.T63 (Medium) Storage.T64 (Medium) Storage.T65 (Medium) Storage.T66 (Medium)	
Assurance (COSO) Detect (NIST CSF)	[Storage.C134] Verify storage accounts use authorized account types (e.g., by using Azure built-in policy "Storage accounts should be migrated to new Azure Resource Manager resources" in Audit mode).	Create or modify a storage account with an unauthorized account type; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11 Storage.FC12	-	Medium

Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO21]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C22, depends on Storage.C2] Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled.	Request the list of storage accounts that require hierarchical namespace, its review process, and its review records.	Low	Storage.FC2 Storage.FC11	Storage.T7 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C23, depends on Storage.C22, assured by Storage.C24] Ensure all required storage accounts have hierarchical namespace (i.e., Data Lake Storage) enabled.	Request 1) the mechanism ensuring that required storage accounts have hierarchical namespace enabled, 2) its records of execution for all new required storage accounts, 3) the plan to move any older storage accounts with hierarchical namespace not meeting the requirement.	Low	Storage.FC2	Storage.T7 (Low)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C24] Verify only required storage accounts have the hierarchical namespace (HNS) option enabled (e.g., by using a custom Azure Policy on {"isHnsEnabled": "true"} in Audit mode).	Create a non-required storage account with the hierarchical namespace (HNS) option enabled; it should be detected.	Medium	Storage.FC2	-	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C111, depends on Storage.C22] Maintain a list of Data Lake Storage accounts that require SFTP enabled.	Request the list of storage accounts that require SFTP, its review process, and its review records.	Low	Storage.FC11	Storage.T44 (Very Low)	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C114, depends on Storage.C111, assured by Storage.C116] Ensure only storage accounts that require SFTP have it enabled.	Request 1) the mechanism ensuring storage accounts that require SFTP have it enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Very Low	Storage.FC11	Storage.T44 (Low)	Very Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C115, depends on Storage.C111] Prevent storage accounts that do not require SFTP from having it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/isSftpEnabled":	Modify a storage account that does not require SFTP to use SFTP; it should be denied.	Medium	Storage.FC11	Storage.T44 (Medium)	Very Low

	true} in Deny mode).					
Assurance (COSO) Detect (NIST CSF)	[Storage.C116] Verify only storage accounts that require SFTP have it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/isSftpEnabled": true} in Audit mode).	Modify a storage account that does not require SFTP to use SFTP; it should be detected.	Medium	Storage.FC11	-	Very Low

Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use [Storage.CO22]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C26, depends on Storage.C2] Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods.	Request the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C27, depends on Storage.C26, assured by Storage.C28] Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods.	Request 1) the mechanism ensuring SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods, 2) its records of execution for all new SAS token usage, and 3) the plan to move any older SAS token usage.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium) Storage.T69 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C28] Verify SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods.	Configure or use a SAS token outside authorized storage accounts, services, permissions, expiration, or access methods; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C87] Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted.	Request the authorized use, key rotation, and Shared Key authorization requirements for storage account keys, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C88, depends on Storage.C87, assured by Storage.C89] Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Request 1) the mechanism ensuring storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium) Storage.T69 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C89] Verify storage account keys and Shared Key authorization are managed according to authorized use, key rotation,	Configure or use a storage account key or Shared Key authorization path outside authorized use, key rotation, or Shared Key authorization requirements; it should be	Medium	Storage.FC1 Storage.FC2 Storage.FC4	-	Medium

	and Shared Key authorization requirements.	detected.		Storage.FC5		
Directive (COSO) Protect (NIST CSF)	[Storage.C156, depends on Storage.C26, assured by Storage.C157] Ensure stored access policies and their permissions are authorized for each container, file share, queue, and table.	Request 1) the mechanism ensuring stored access policies and their permissions are authorized for each container, file share, queue, and table, 2) its records of execution for all new containers, file shares, queues, and tables, and 3) the plan to move any older containers, file shares, queues, and tables.	Low	Storage.FC4 Storage.FC5	Storage.T27 (Low) Storage.T28 (Low) Storage.T67 (Low)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C157] Verify stored access policies and their permissions are authorized for each container, file share, queue, and table.	Modify an existing stored access policy or its permissions outside authorized values; it should be detected.	Low	Storage.FC4 Storage.FC5	-	Low

Protect the integrity of blob storage data [Storage.CO24]

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Identify (NIST CSF)	[Storage.C32] Maintain a list of storage container blobs that require immutability or soft delete.	Request the list of storage container blobs that require immutability, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T56 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C33, depends on Storage.C32, assured by Storage.C146] Ensure required storage container blobs have immutability or soft delete enabled.	Request 1) the mechanism ensuring that required storage container blobs have immutability enabled, 2) its records of execution for all new storage container blobs, and 3) the plan to move any older storage container blobs.	Low	Storage.FC2	Storage.T8 (Very High) Storage.T9 (Very High) Storage.T12 (Medium) Storage.T56 (Medium)	High
Directive (COSO) Identify (NIST CSF)	[Storage.C107] Maintain the list of storage accounts required to use Azure resource locks, following the Azure Resource Manager ThreatModel.	Request the list of storage accounts that require resource locks, its review process, and its review records.	Low	Storage.FC1 Storage.FC3	Storage.T4 (Very Low) Storage.T18 (Very Low) Storage.T63 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C108, depends on Storage.C107, assured by Storage.C109] Ensure required storage accounts use Azure resource locks.	Request 1) the mechanism ensuring that required storage accounts have resource locks applied, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Very Low	Storage.FC1 Storage.FC3	Storage.T4 (High) Storage.T18 (Very High) Storage.T63 (High)	High
Assurance (COSO) Detect (NIST CSF)	[Storage.C109] Verify required storage accounts use Azure resource locks (e.g., by using built-in Azure Policy "Resource locks should be applied" in Audit mode).	Remove a resource lock from a storage account; it should be detected.	Medium	Storage.FC1 Storage.FC3	-	High
Preventative (COSO) Protect (NIST CSF)	[Storage.C110] Prevent modification or deletion of required storage accounts by using a resource lock set to read-only and/or delete, using the Azure Resource Manager ThreatModel.	Modify or delete a required storage account; it should be denied.	Very Low	Storage.FC1 Storage.FC2 Storage.FC3	Storage.T12 (Very High) Storage.T18 (High) Storage.T63 (Very High)	High
Directive (COSO) Identify (NIST CSF)	[Storage.C144] Maintain the list of storage accounts or blob containers that require blob versioning.	Request the list of storage accounts or blob containers that require blob versioning, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T40 (Very Low)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C146] Verify required storage container blobs have immutability policies enabled (e.g., by using the built-in Azure Policy "Immutable blob storage should be	Disable or remove the immutability policy from a storage container that is required to have immutability enabled; it should be detected.	Low	Storage.FC2	-	High

	enabled" in Audit mode).					

Compliance Mapping

PCI DSS v4

PCI DSS v4	Control Objectives	Controls				
		Very High	High	Medium	Low	Very Low
1.1 1.2.1	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO8] Enforce encryption in transit [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	Storage.C122	Storage.C101

1.2.3	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.2.4 1.2.5	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95	-	-

				Storage.C96 Storage.C97 Storage.C144		
1.2.6	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.2.7	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173	-	Storage.C50 Storage.C51 Storage.C54

				Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
1.2.8	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.2.10	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1 Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C144		
1.3	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.3.1 1.3.2	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO8] Enforce encryption in transit [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C44 Storage.C167 Storage.C168	Storage.C122	Storage.C101

				Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97		
1.3.3	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.4	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96	-	-

				Storage.C97		
1.4.1	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
1.4.2	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
1.4.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues)	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180	-	-
1.4.4	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95	-	-

				Storage.C96 Storage.C97		
1.4.5	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	-	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144	-	-
1.5 1.5.1 2.2.1	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89		
2.2.2	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO8] Enforce encryption in transit [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101

				Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97		
2.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO8] Enforce encryption in transit [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88	Storage.C122 Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C101 Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C89		
2.3.1	[Storage.CO8] Enforce encryption in transit [Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175	Storage.C122	Storage.C101
2.3.2	[Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175	-	-
3.3.2	[Storage.CO24] Protect the integrity of blob storage data	-	Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C144	-	-
3.5.1	[Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175	-	-
3.5.1.1 3.5.1.2 3.5.1.3 3.6.1 3.6.1.1 3.6.1.2 3.6.1.3 3.6.1.4 3.7.1 3.7.2	[Storage.CO8] Enforce encryption in transit [Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101

3.7.3 3.7.4				Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175		
3.7.5	[Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175	-	-
3.7.6 3.7.7 3.7.9	[Storage.CO8] Enforce encryption in transit	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101
4.1 4.2.1	[Storage.CO12] Enforce encryption on data at rest and protect encryption keys	-	-	Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175	-	-
4.2.1.1	[Storage.CO8] Enforce encryption in transit	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101
4.2.1.2 4.2.2	[Storage.CO17] Verify the safety of objects ingested and produced	-	-	Storage.C103 Storage.C104 Storage.C105	-	-
5.1.2 5.2.1 5.2.2 5.2.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

5.2.3.1 5.3.1 5.3.2 5.3.2.1 5.3.3 5.3.4 5.3.5	[Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use			Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89		
6.5.2	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	-	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159	-	-

				Storage.C160 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
6.5.6	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C144	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116
7.1 7.2	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C144		
7.2.1 7.2.2	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C144	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116
7.2.3	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks	Storage.C1	-	-	-	-
7.2.4	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues)	Storage.C1	Storage.C2 Storage.C32 Storage.C33	Storage.C6 Storage.C7 Storage.C8	-	-

	[Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data		Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
7.2.5	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160	Storage.C23 Storage.C24	Storage.C50 Storage.C51 Storage.C54 Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C144		
7.2.6	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C144	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116
7.3 7.3.1 7.3.2 7.3.3	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22		
8.1 8.2	[Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model	-	-	Storage.C158 Storage.C159 Storage.C160	-	-
8.2.1	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116
8.2.2	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59	-	-

				Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160		
8.2.3	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116
8.2.4	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs,	Storage.C1	Storage.C2	Storage.C6 Storage.C7	-	-

	queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model			Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160		
8.2.6	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C87 Storage.C88 Storage.C89		
8.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116
8.3.1	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO8] Enforce encryption in transit [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	-	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162	Storage.C122	Storage.C101

				Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
8.3.2	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89		
8.3.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116
8.3.5 8.3.6 8.3.7	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22		
8.3.8	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116
8.3.9	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks	Storage.C1	Storage.C2	Storage.C6	-	-

	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model			Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160		
8.3.10	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C28 Storage.C87 Storage.C88 Storage.C89		
8.3.10.1 8.3.11	[Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model	-	-	Storage.C158 Storage.C159 Storage.C160	-	-
8.4 8.4.1 8.4.2 8.4.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	-	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144	-	-
8.5	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22		
8.5.1	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144	-	-
8.6	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs,	Storage.C1	Storage.C2 Storage.C32	Storage.C6 Storage.C7	Storage.C23 Storage.C24	Storage.C111 Storage.C114

	queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use [Storage.CO24] Protect the integrity of blob storage data		Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89 Storage.C144	Storage.C156 Storage.C157	Storage.C115 Storage.C116
8.6.1	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO22] Ensure storage account keys, Shared Key authorization, and SAS tokens are restricted to authorized use	-	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163	Storage.C23 Storage.C24 Storage.C156 Storage.C157	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22 Storage.C26 Storage.C27 Storage.C28 Storage.C87 Storage.C88 Storage.C89		
8.6.2 8.6.3	[Storage.CO24] Protect the integrity of blob storage data	-	Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C144	-	-
9.4 9.4.1	[Storage.CO5] Ensure backup, replication, and recovery capabilities for storage account services	-	Storage.C47 Storage.C48 Storage.C49	Storage.C10 Storage.C11 Storage.C15 Storage.C17 Storage.C20 Storage.C83 Storage.C84 Storage.C85 Storage.C86 Storage.C150 Storage.C151 Storage.C152	Storage.C12 Storage.C13 Storage.C19 Storage.C21 Storage.C46 Storage.C75 Storage.C77 Storage.C145 Storage.C147	Storage.C16 Storage.C18 Storage.C71 Storage.C72 Storage.C74 Storage.C76 Storage.C78
9.4.1.1 9.4.1.2	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
10.1	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C1 Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81	Storage.C23 Storage.C24	Storage.C50 Storage.C51 Storage.C54 Storage.C111 Storage.C114 Storage.C115 Storage.C116

	[Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data		Storage.C110 Storage.C146	Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C144		
10.2	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model [Storage.CO24] Protect the integrity of blob storage data	Storage.C1	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162	Storage.C23 Storage.C24	Storage.C50 Storage.C51 Storage.C54 Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C22 Storage.C144		
10.2.1 10.2.1.1 10.2.1.2 10.2.1.3 10.2.1.4 10.2.1.5 10.2.1.6 10.2.1.7 10.2.2	[Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	-	-	-	-	Storage.C50 Storage.C51 Storage.C54
10.3 10.3.1 10.3.2 10.3.3	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
10.3.4 10.4 10.4.1 10.4.1.1 10.4.2 10.4.2.1 10.4.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	-	Storage.C2 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93	-	Storage.C50 Storage.C51 Storage.C54

				Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
10.6 10.6.1 10.6.2 10.6.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168	-	Storage.C50 Storage.C51 Storage.C54

				Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
10.7 10.7.1 10.7.2 10.7.3	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO8] Enforce encryption in transit [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO12] Enforce encryption on data at rest and protect encryption keys [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts [Storage.CO24] Protect the integrity of blob storage data	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45 Storage.C32 Storage.C33 Storage.C107 Storage.C108 Storage.C109 Storage.C110 Storage.C146	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C44	Storage.C122	Storage.C101 Storage.C50 Storage.C51 Storage.C54

				Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C60 Storage.C61 Storage.C63 Storage.C65 Storage.C173 Storage.C174 Storage.C175 Storage.C94 Storage.C95 Storage.C96 Storage.C97 Storage.C144		
11.2	[Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO8] Enforce encryption in transit [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C2 Storage.C42 Storage.C43 Storage.C45	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91 Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178 Storage.C44 Storage.C167	Storage.C122	Storage.C101

				Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97		
11.2.1 11.2.2	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
11.4.5 11.4.6	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
11.5	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	Storage.C50 Storage.C51 Storage.C54
11.5.1	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
11.5.1.1 11.5.2	[Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	-	-	Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-

12.4.2	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
12.5.2	[Storage.CO8] Enforce encryption in transit	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101
12.10.1	[Storage.CO8] Enforce encryption in transit [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101 Storage.C50 Storage.C51 Storage.C54
12.10.5	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1	-	Storage.C158 Storage.C159 Storage.C160 Storage.C22	Storage.C23 Storage.C24	Storage.C50 Storage.C51 Storage.C54 Storage.C111 Storage.C114 Storage.C115 Storage.C116
15.1	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-

A1.1.4	[Storage.CO8] Enforce encryption in transit	-	-	Storage.C98 Storage.C99 Storage.C100 Storage.C102 Storage.C120 Storage.C121 Storage.C123 Storage.C176 Storage.C177 Storage.C178	Storage.C122	Storage.C101
A2.1 A2.1.1 A2.1.2	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO14] Restrict and manage external websites (origins) that can make browser-based requests to storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C94 Storage.C95 Storage.C96 Storage.C97	-	-
A3.2.1 A3.2.4	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1 Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C158 Storage.C159 Storage.C160 Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172 Storage.C22	Storage.C23 Storage.C24	Storage.C50 Storage.C51 Storage.C54 Storage.C111 Storage.C114 Storage.C115 Storage.C116
A3.2.6.1	[Storage.CO9] Limit network access to storage account services with private endpoints [Storage.CO11] Enable enhanced monitoring and notifications for storage accounts	Storage.C37 Storage.C38 Storage.C40	Storage.C42 Storage.C43 Storage.C45	Storage.C44 Storage.C167 Storage.C168 Storage.C169 Storage.C170 Storage.C171 Storage.C172	-	Storage.C50 Storage.C51 Storage.C54
A3.3.1	[Storage.CO1] Limit the IAM entities allowed to use the IAM actions required to execute attacks [Storage.CO4] Restrict access to storage accounts and their services (e.g., Azure Files, containers, blobs, queues) [Storage.CO7] Restrict access to Data Lake Storage containers by integrating ACLs into the IAM Operating Model [Storage.CO21] Restrict access to Data Lake Storage, including SFTP, by integrating the IAM Operating Model	Storage.C1	Storage.C2	Storage.C6 Storage.C7 Storage.C8 Storage.C9 Storage.C55 Storage.C56 Storage.C57 Storage.C59 Storage.C81 Storage.C90 Storage.C91	Storage.C23 Storage.C24	Storage.C111 Storage.C114 Storage.C115 Storage.C116

				Storage.C92 Storage.C93 Storage.C149 Storage.C161 Storage.C162 Storage.C163 Storage.C164 Storage.C165 Storage.C166 Storage.C179 Storage.C180 Storage.C158 Storage.C159 Storage.C160 Storage.C22		

The Control Objectives are mapped to the Secure Controls Framework (SCF), provided under Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0). Compliance mappings are not designed to fully ensure compliance with a specific governance or compliance standard. You are responsible for making your own assessment of whether your use of the Services meets applicable legal and regulatory requirements.

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Appendixes

Appendix 1 - Prioritized list for control implementation

Type	Control	Testing	Effort	Feature Class(es)	Threat(s) and Impact	CVSS-weighted Priority
Directive (COSO) Protect (NIST CSF)	[Storage.C1] Limit and monitor the access to the IAM actions required to perform attacks using Azure RBAC and local privileges, limiting them to Trusted Networks with Conditional Access and Just-in-Time access using Privileged Identity Management when possible, following the Entra ID, Azure Resource Manager, and Azure Management ThreatModels.	Request the list of authorized IAM principals with the permissions required to launch attacks, a list of Trusted Networks and PIM settings where applicable, its review process, and its review records.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC10 Storage.FC11 Storage.FC12	Storage.T1 (Very High) Storage.T2 (Very High) Storage.T4 (Very High) Storage.T5 (Very High) Storage.T7 (Very High) Storage.T8 (Very High) Storage.T9 (Very High) Storage.T12 (Very High) Storage.T13 (Very High) Storage.T15 (Very High) Storage.T16 (Very High) Storage.T17 (Very High) Storage.T18 (Very High) Storage.T19 (Very High) Storage.T20 (Very High) Storage.T22 (Very High) Storage.T24 (Very High) Storage.T25 (Very High) Storage.T26 (Very High) Storage.T27 (Very High) Storage.T28 (Very High) Storage.T31 (Very High) Storage.T34 (Very High) Storage.T38 (Very High) Storage.T39 (Very High) Storage.T40 (Very High) Storage.T42 (Very High) Storage.T44 (Very High) Storage.T48 (Very High) Storage.T50 (Very High) Storage.T55 (Very High) Storage.T56 (Very High) Storage.T58 (Very High) Storage.T59 (Very High) Storage.T60 (Very High) Storage.T62 (Very High) Storage.T63 (Very High) Storage.T64 (Very High) Storage.T65 (Very High) Storage.T66 (Very High) Storage.T67 (Very High) Storage.T68 (Very High) Storage.T69 (Very High)	Very High
Directive (COSO) Identify (NIST CSF)	[Storage.C37] Maintain a list of authorized private endpoints on storage accounts.	Request the list of authorized private endpoints, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T1 (Very Low) Storage.T5 (Very Low)	Very High

				Storage.FC3 Storage.FC4 Storage.FC12	Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low) Storage.T62 (Very Low)	
Directive (COSO) Protect (NIST CSF)	[Storage.C38, depends on Storage.C37, assured by Storage.C40] Ensure only authorized private endpoints are configured on storage accounts.	Request 1) the mechanism ensuring only authorized private endpoints are configured on storage accounts, 2) its records of execution for all private endpoints.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC12	Storage.T1 (Very High) Storage.T5 (Low) Storage.T12 (Medium) Storage.T15 (Low) Storage.T31 (Low) Storage.T50 (Medium) Storage.T62 (Medium)	Very High
Assurance (COSO) Detect (NIST CSF)	[Storage.C40] Verify only authorized private endpoints are configured on storage accounts (e.g., by using the built-in Azure Policy "Storage accounts should use Private Link" in Audit mode).	Configure an unauthorized private endpoint on a storage account; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC12	-	Very High
Directive (COSO) Identify (NIST CSF)	[Storage.C2] Maintain a list of storage accounts and their authorized authentication methods enabled according to requirements (e.g., Entra ID, SAS token).	Request the list of storage accounts and their authorized authentication methods, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T3 (Very Low) Storage.T5 (Very Low) Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	High
Directive (COSO) Identify (NIST CSF)	[Storage.C47] Maintain a list of storage accounts that require cross-tenant replication and their allowed destination tenants.	Request the list of storage accounts that require cross-tenant replication and their allowed destination tenants, its review process, and its review records.	Low	Storage.FC1	Storage.T13 (Very Low) Storage.T42 (Very Low) Storage.T63 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C48, depends on Storage.C47, assured by Storage.C49] Ensure only required storage accounts have cross-tenant replication enabled and the destination storage accounts are authorized.	Request 1) the mechanism that ensures only required storage accounts have cross-tenant replication enabled and the destination storage accounts are authorized, 2) its records of execution for all new storage accounts, 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T13 (High) Storage.T42 (High) Storage.T63 (Medium)	High
Assurance (COSO) Detect (NIST CSF)	[Storage.C49] Verify only the storage accounts that require cross-tenant replication have it enabled (e.g., by using the Azure built-in policy StorageAccountAllowCrossTenantReplication_Audit.json).	Create a storage account with cross-tenant replication enabled; it should be detected.	Medium	Storage.FC1	-	High
Directive (COSO) Identify (NIST CSF)	[Storage.C42] Maintain a list of authorized IPs that are permitted through each storage account firewall.	Request the list of authorized IPs, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low)	High

					Storage.T31 (Very Low) Storage.T50 (Very Low)	
Directive (COSO) Protect (NIST CSF)	[Storage.C43, depends on Storage.C42, assured by Storage.C45] Ensure each storage account firewall only allows authorized IPs.	Request 1) the mechanism ensuring firewall rules only allow authorized IP addresses, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (High) Storage.T5 (High) Storage.T12 (High) Storage.T15 (High) Storage.T31 (High) Storage.T50 (High)	High
Assurance (COSO) Detect (NIST CSF)	[Storage.C45] Verify storage account firewall rules allow only authorized IPs (e.g., by using built-in Azure Policy "Storage accounts should restrict network access" in Audit mode).	Configure a storage account firewall to allow an unauthorized IP; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	-	High
Directive (COSO) Identify (NIST CSF)	[Storage.C32] Maintain a list of storage container blobs that require immutability or soft delete.	Request the list of storage container blobs that require immutability, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T56 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C33, depends on Storage.C32, assured by Storage.C146] Ensure required storage container blobs have immutability or soft delete enabled.	Request 1) the mechanism ensuring that required storage container blobs have immutability enabled, 2) its records of execution for all new storage container blobs, and 3) the plan to move any older storage container blobs.	Low	Storage.FC2	Storage.T8 (Very High) Storage.T9 (Very High) Storage.T12 (Medium) Storage.T56 (Medium)	High
Directive (COSO) Identify (NIST CSF)	[Storage.C107] Maintain the list of storage accounts required to use Azure resource locks, following the Azure Resource Manager ThreatModel.	Request the list of storage accounts that require resource locks, its review process, and its review records.	Low	Storage.FC1 Storage.FC3	Storage.T4 (Very Low) Storage.T18 (Very Low) Storage.T63 (Very Low)	High
Directive (COSO) Protect (NIST CSF)	[Storage.C108, depends on Storage.C107, assured by Storage.C109] Ensure required storage accounts use Azure resource locks.	Request 1) the mechanism ensuring that required storage accounts have resource locks applied, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Very Low	Storage.FC1 Storage.FC3	Storage.T4 (High) Storage.T18 (Very High) Storage.T63 (High)	High
Assurance (COSO) Detect (NIST CSF)	[Storage.C109] Verify required storage accounts use Azure resource locks (e.g., by using built-in Azure Policy "Resource locks should be applied" in Audit mode).	Remove a resource lock from a storage account; it should be detected.	Medium	Storage.FC1 Storage.FC3	-	High
Preventative (COSO) Protect (NIST CSF)	[Storage.C110] Prevent modification or deletion of required storage accounts by using a resource lock set to read-only and/or delete, using the Azure Resource Manager ThreatModel.	Modify or delete a required storage account; it should be denied.	Very Low	Storage.FC1 Storage.FC2 Storage.FC3	Storage.T12 (Very High) Storage.T18 (High) Storage.T63 (Very High)	High
Assurance (COSO) Detect (NIST CSF)	[Storage.C146] Verify required storage container blobs have immutability policies enabled (e.g., by using the built-in Azure Policy "Immutable blob storage should be enabled" in Audit mode).	Disable or remove the immutability policy from a storage container that is required to have immutability enabled; it should be detected.	Low	Storage.FC2	-	High
Directive (COSO) Identify (NIST CSF)	[Storage.C6] Maintain the list of storage accounts authorized to have public network access enabled.	Request the list of storage accounts authorized to have public network access enabled, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T5 (Very Low) Storage.T50 (Very Low) Storage.T68 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C7, depends on Storage.C6, assured by Storage.C9] Ensure only authorized storage accounts have public network access enabled.	Request 1) the mechanism ensuring only authorized storage accounts have public network access enabled, 2) its records of execution for all new storage accounts,	Low	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium

		and 3) the plan to move any older storage accounts.				
Preventative (COSO) Protect (NIST CSF)	[Storage.C8, depends on Storage.C6] Prevent unauthorized storage accounts from having public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Deny mode).	Enable public network access on an unauthorized storage account; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C9] Verify unauthorized storage accounts do not have public network access enabled (e.g., by using Azure built-in policy "Storage accounts should disable public network access" in Audit mode).	Enable public network access on an unauthorized storage account; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C55] Maintain a list of blobs and containers that require anonymous access.	Request the list of authorized blobs and containers that require anonymous access, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T5 (Very Low) Storage.T50 (Very Low) Storage.T68 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C56, depends on Storage.C55, assured by Storage.C59] Ensure anonymous access is set only for required blobs and containers.	Request 1) the mechanism ensuring only required blobs/containers are anonymously accessible, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C57, depends on Storage.C55, assured by Storage.C59] Ensure only required blobs and containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Deny mode).	Create or update a non-required blob or a container to be anonymously accessible; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T5 (Medium) Storage.T50 (Medium) Storage.T68 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C59] Verify only required blobs or containers are anonymously accessible (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/containers/publicAccess": "Blob"} in Audit mode).	Create or update a non-required blob or a container to be anonymously accessible; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C81] Verify Shared Key authorization is disabled (and optionally that OAuth is the default) for authorized storage accounts that host authorized blobs, file shares, queues, tables, and DFS (e.g., by using a custom Azure Policy to require "Microsoft.Storage/storageAccounts/allowSharedKeyAccess" = false and, optionally, "Microsoft.Storage/storageAccounts/defaultToOAuthAuthentication" = true in Audit mode).	Configure a storage account with allowSharedKeyAccess enabled and/or defaultToOAuthAuthentication disabled; it should be detected.	Medium	Storage.FC3 Storage.FC11	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C90] Maintain a list of storage accounts that require static website hosting.	Request the list of storage accounts that require static website hosting, its review process, and its review records.	Low	Storage.FC2	Storage.T22 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C91, depends on Storage.C90, assured by Storage.C93] Ensure only storage accounts that require static website hosting have it enabled.	Request 1) the mechanism ensuring only required storage accounts have the static website hosting enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T22 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C92, depends on Storage.C90] Prevent storage accounts that do not require static website hosting from having it enabled (e.g., by using a custom Azure Policy on	Modify an existing storage account that does not require static website hosting to be enabled; it should be denied.	Medium	Storage.FC2	Storage.T22 (Medium)	Medium

	{"Microsoft.Storage/storageAccounts/staticWebsite.enabled": true} in Deny mode).					
Assurance (COSO) Detect (NIST CSF)	[Storage.C93] Verify storage accounts that do not require static website hosting do not have it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/staticWebsite.enabled": true} in Audit mode).	Modify an existing storage account that does not require static website hosting to be enabled; it should be detected.	Medium	Storage.FC2	-	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C149, depends on Storage.C2, assured by Storage.C81] Ensure required blobs, file shares, queues, tables, and DFS are using authorized authentication methods.	Request 1) the mechanism ensuring that blobs, file shares, queues, tables, and DFS use authorized authentication methods, 2) its records of execution for new storage account services, and 3) the plan to move any older storage account services.	Very Low	Storage.FC3 Storage.FC11	Storage.T20 (Medium) Storage.T44 (Medium)	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C161, depends on Storage.C22] Maintain a list of authorized local users, their permissions, and their authentication methods (e.g., password, SSH) for each hierarchical namespace with SFTP enabled.	Request the list of authorized local users, their permissions, their authentication methods, its review process, and its review records.	Low	Storage.FC11	Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C162, depends on Storage.C161, assured by Storage.C163] Ensure all local users and associated permissions and authentication methods are authorized.	Request 1) the mechanism ensuring all local users and associated permissions and authentication methods are authorized, 2) its records of execution for all new local users, and 3) the plan to move any older local users.	Low	Storage.FC11	Storage.T64 (Medium) Storage.T65 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C163] Verify local user functionality for SFTP is enabled only where local users, their permissions, and authentication methods are authorized (e.g., by using a custom Azure Policy to detect storage accounts where "Microsoft.Storage/storageAccounts/localUsersEnabled" = true and "Microsoft.Storage/storageAccounts/isSftpEnabled" = true, in Audit mode).	Enable local users in a hierarchical namespace with SFTP, where local users are not authorized; it should be detected.	Medium	Storage.FC11	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C164] Maintain the list of authorized storage queues and access methods allowed to read, peek, or dequeue queue messages.	Request the list of authorized storage queues and access methods allowed to read, peek, or dequeue queue messages, its review process, and its review records.	Low	Storage.FC4	Storage.T67 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C165, depends on Storage.C164, assured by Storage.C166] Ensure only authorized identities and access methods can read, peek, or dequeue storage queue messages.	Request 1) the mechanism ensuring that only authorized identities and access methods can read, peek, or dequeue storage queue messages, 2) its records of execution for all new storage queues and access grants, and 3) the plan to move any older storage queues and access grants.	Medium	Storage.FC4	Storage.T67 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C166] Verify only authorized identities and access methods can read, peek, or dequeue storage queue messages.	Read, peek, or dequeue a storage queue message outside authorized access methods; it should be detected.	Medium	Storage.FC4	-	Medium
Detective (COSO) Detect (NIST CSF)	[Storage.C179] Monitor for storage accounts that allow Shared Key authorization (e.g., by monitoring storage account configuration where "properties.allowSharedKeyAccess == true" in Audit mode).	Configure a storage account to allow Shared Key authorization; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium)	Medium

					Storage.T69 (Medium)	
Detective (COSO) Detect (NIST CSF)	[Storage.C180] Monitor for storage accounts without a SAS expiration policy (e.g., by monitoring storage account configuration where "properties.sasPolicy.sasExpirationPeriod" is not configured in Audit mode).	Configure a storage account without a SAS expiration policy; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C10, depends on Storage.C144, assured by Storage.C11] Ensure versioning is enabled on required containers.	Request 1) the mechanism ensuring versioning is enabled on required containers, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T7 (Low) Storage.T40 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C11] Verify versioning is enabled on required containers (e.g., by using Azure built-in policy "Configure your Storage account to enable blob versioning" in Audit mode).	Remove versioning from a container; it should be detected.	Medium	Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C15] Maintain a list of the blob storage containers that are required to have a minimum retention period enabled.	Request the list of storage account containers that have a minimum retention period according to requirements, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T25 (Very Low) Storage.T39 (Very Low)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C17, depends on Storage.C15] Prevent the creation of required storage accounts without the blob soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/deleteRetentionPolicy.enabled": false} in Deny mode).	Create a required storage account without soft-delete for the blob; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (High) Storage.T25 (Low) Storage.T39 (Very Low)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C20, depends on Storage.C15] Prevent the creation of containers without the soft-delete option enabled (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/blobServices/containers/softDelete" in Deny mode).	Create a container without soft-delete; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (High) Storage.T25 (Low) Storage.T39 (Very Low)	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C83] Define the minimum retention period for required file shares (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/deleteRetentionPolicy.days": 7} in Deny mode).	For each file share, request the minimum retention from deletion, its review process, and its review records.	Medium	Storage.FC3	Storage.T18 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C84, depends on Storage.C83, assured by Storage.C86] Ensure file shares have the soft-delete option enabled for at least the defined minimum retention period.	Request 1) the mechanism ensuring file shares have soft-delete enabled for at least the defined minimum retention, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Low	Storage.FC3	Storage.T18 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C85, depends on Storage.C83] Prevent the creation of file shares without the soft-delete option enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/shares/deleteRetentionPolicy.enabled": false} in Deny mode).	Create a file share without soft-delete; it should be denied.	Medium	Storage.FC3	Storage.T18 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C86] Verify all file shares have the soft-delete option enabled (e.g., by using a custom Azure Policy on	Create a file share without soft-delete; it should be detected.	Medium	Storage.FC3	-	Medium

	{"Microsoft.Storage/storageAccounts/fileServices/shares/deleteRetentionPolicy.enabled": false} in Audit mode).					
Directive (COSO) Identify (NIST CSF)	[Storage.C150] Maintain a list of authorized replication policies and their destination storage accounts.	Request the list of authorized replication policies and their destination storage accounts, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C151, depends on Storage.C150, assured by Storage.C152] Ensure replication policies and their destination storage accounts are authorized.	Request 1) the mechanism ensuring replication policies and their destination storage accounts are authorized, 2) its records of execution for all new replication policies, and 3) the plan to move any older replication policies.	Very Low	Storage.FC1	Storage.T13 (Low)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C152] Verify replication policies and their destination storage accounts are authorized (e.g., by using a custom Azure Policy on "Microsoft.Storage/storageAccounts/objectReplicationPolicies[*].destinationAccount" and "Microsoft.Storage/storageAccounts/objectReplicationPolicies[*].rules[*].sourceContainer" in Audit mode).	Add a replication policy with an unauthorized destination storage account or source container; it should be detected.	Medium	Storage.FC1	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C158] Maintain a list of authorized Access Control Lists (ACLs) on Data Lake Storage containers.	Request the list of authorized Access Control Lists (ACLs) on Data Lake Storage containers, its review process, and its review records.	Low	Storage.FC2	Storage.T58 (Very Low) Storage.T66 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C159, depends on Storage.C158, assured by Storage.C160] Ensure all Access Control Lists (ACLs) on Data Lake Storage containers are authorized.	Request 1) the mechanism ensuring all Access Control Lists (ACLs) on Data Lake Storage containers are authorized, 2) its records of execution for all new ACLs, and 3) the plan to move any older ACLs.	Low	Storage.FC2	Storage.T58 (Medium) Storage.T66 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C160] Verify Azure Data Lake Storage Gen2 functionality is enabled only where Access Control Lists (ACLs) on containers are authorized (e.g., by using a custom Azure Policy to detect storage accounts where "Microsoft.Storage/storageAccounts/isHnsEnabled" = true and "Microsoft.Storage/storageAccounts/allowSharedKeyAccess" = true, in Audit mode).	Enable Shared Key authorization (allowSharedKeyAccess = true) on an HNS-enabled storage account, or disable hierarchical namespace (isHnsEnabled = false) on a storage account; it should be detected.	Medium	Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C98] Maintain a list of file shares that require NFS/SMB 2.1 to be enabled.	Request the list of file shares that require NFS/SMB 2.1 to be enabled, its review process, and its review records.	Low	Storage.FC3	Storage.T61 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C99, depends on Storage.C98, assured by Storage.C102] Ensure only file shares that require NFS/SMB 2.1 have it enabled.	Request 1) the mechanism ensuring only file shares that require NFS/SMB 2.1 have it enabled, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Very Low	Storage.FC3	Storage.T61 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C100, depends on Storage.C98] Prevent file shares that do not require NFS/SMB 2.1 from being enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb.protocolVersions[*]": ["SMB2.1"]} in Deny mode).	Modify an existing file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be denied.	Medium	Storage.FC3	Storage.T61 (High)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C102] Verify only file shares that require NFS/SMB 2.1 have it	Modify an existing file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be detected.	Medium	Storage.FC3	-	Medium

	enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb.protocolVersions[*]": ["SMB2.1"]} in Audit mode).					
Directive (COSO) Identify (NIST CSF)	[Storage.C120] Maintain a list of required file shares' security protocol settings (ideally maximum security SMB 3.1.1, Kerberos, AES-256 only).	Request the list of required file share security protocol settings, its review process, and its review records.	Low	Storage.FC3	Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T60 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C121, depends on Storage.C120, assured by Storage.C123] Ensure required file shares' security protocol settings are set.	Request 1) the mechanism ensuring that the required security protocol settings for file shares in storage accounts are in use, 2) its records of execution for all new file shares, and 3) the plan to move any older file shares.	Low	Storage.FC3	Storage.T15 (Medium) Storage.T20 (Low) Storage.T60 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C123] Verify required file shares' security protocol settings are configured (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb":{"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Audit mode).	Modify a required file share's security protocol settings; it should be detected.	Medium	Storage.FC3	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C176] Maintain the authorized minimum TLS version for each storage account.	Request the authorized minimum TLS version for each storage account, its review process, and its review records.	Low	Storage.FC3	Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T60 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C177, depends on Storage.C176, assured by Storage.C178] Ensure storage accounts use the authorized minimum TLS version.	Request 1) the mechanism ensuring storage accounts use the authorized minimum TLS version, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC3	Storage.T15 (Medium) Storage.T20 (Medium) Storage.T60 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C178] Verify storage accounts use the authorized minimum TLS version (e.g., by using Azure built-in policy "Storage accounts should have the specified minimum TLS version" in Audit mode).	Create or modify a storage account with a minimum TLS version below the authorized value; it should be detected.	Medium	Storage.FC3	-	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C44, depends on Storage.C42] Prevent storage account firewall rules from allowing unauthorized IPs (e.g., by using the Azure built-in policy "Storage accounts should restrict network access" in Deny mode).	Configure a storage account firewall to allow an unauthorized IP; it should be denied.	Very Low	Storage.FC1 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C167] Maintain the list of authorized virtual network rules for each storage account.	Request the list of authorized virtual network rules for each storage account, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C168, depends on Storage.C167, assured by Storage.C169] Ensure each storage account uses only authorized virtual network rules.	Request 1) the mechanism ensuring each storage account uses only authorized virtual network rules, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Medium) Storage.T5 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C169] Verify each storage account uses only authorized virtual network rules (e.g., by using Azure built-in policy	Configure a storage account with an unauthorized virtual network rule; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium

	"Storage accounts should restrict network access using virtual network rules" in Audit mode).			Storage.FC3 Storage.FC4		
Directive (COSO) Identify (NIST CSF)	[Storage.C170] Maintain the list of authorized network ACL bypass configurations for each storage account.	Request the list of authorized network ACL bypass configurations for each storage account, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Very Low) Storage.T5 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T31 (Very Low) Storage.T50 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C171, depends on Storage.C170, assured by Storage.C172] Ensure each storage account uses only authorized network ACL bypass configurations.	Request 1) the mechanism ensuring each storage account uses only authorized network ACL bypass configurations, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	Storage.T1 (Medium) Storage.T5 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T31 (Medium) Storage.T50 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C172] Verify each storage account uses only authorized network ACL bypass configurations (e.g., by using Azure built-in policy "Storage Accounts should restrict network access through network ACL bypass configuration only" in Audit mode).	Configure a storage account with an unauthorized network ACL bypass configuration; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C60] Maintain a list of authorized customer-managed keys used by storage accounts.	Request the list of authorized customer-managed keys used by storage accounts, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Very Low) Storage.T38 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C61, depends on Storage.C60, assured by Storage.C65] Ensure only authorized customer-managed keys are configured for storage accounts.	Request 1) the mechanism ensuring only authorized customer-managed keys are used to encrypt each storage account, 2) its records of execution for all new storage accounts, and 3) the plan to move any older authorized storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Low) Storage.T38 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C63, depends on Storage.C61, assured by Storage.C65] Prevent storage accounts that require customer-managed keys from using service-managed keys (e.g., by using built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Deny mode).	Create a storage account requiring CMK with a service-managed key; it should be denied.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (Medium) Storage.T38 (High)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C65] Verify only authorized customer-managed keys are configured for storage accounts (e.g., by using the built-in Azure Policy "Storage accounts should use customer-managed key for encryption" in Audit mode).	Modify a required storage account to use an unauthorized CMK; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C173] Maintain a list of storage accounts that require Azure infrastructure encryption (i.e., double encryption).	Request the list of storage accounts that require Azure infrastructure encryption (double encryption), its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T9 (Very Low) Storage.T38 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C174, depends on Storage.C173, assured by Storage.C175] Ensure required storage accounts use Azure infrastructure encryption.	Request 1) the mechanism ensuring required storage accounts use Azure infrastructure encryption, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2	Storage.T9 (Medium) Storage.T38 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C175] Verify required storage accounts use Azure infrastructure encryption (e.g., by using Azure built-in policy "Storage accounts should have infrastructure encryption" in Audit mode).	Create or modify a required storage account without Azure infrastructure encryption; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Medium

Directive (COSO) Identify (NIST CSF)	[Storage.C94] Maintain a list of authorized CORS per endpoint, trusted origins, and corresponding settings.	Request the list of authorized storage accounts with CORS trusted origins and corresponding settings, its review process, and its review records.	Low	Storage.FC1	Storage.T26 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C95, depends on Storage.C94, assured by Storage.C97] Ensure only authorized storage accounts have CORS-trusted origins and corresponding settings configured.	Request 1) the mechanism ensuring only authorized storage accounts have CORS trusted origins and corresponding settings configured, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T26 (Medium)	Medium
Preventative (COSO) Protect (NIST CSF)	[Storage.C96, depends on Storage.C94] Prevent unauthorized storage accounts from using CORS trusted origins and corresponding settings (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/cors.allowedOrigins[*]": ["https://myapp.contoso.com"]} in Deny mode).	Create a storage account with untrusted CORS settings; it should be denied.	Medium	Storage.FC1	Storage.T26 (High)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C97] Verify only authorized CORS trusted origins and corresponding settings are configured (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/blobServices/cors.allowedOrigins[*]": ["https://myapp.contoso.com"]} in Audit mode).	Create a storage account with untrusted CORS settings; it should be detected.	Medium	Storage.FC1	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C103] Maintain the list of storage accounts that require malware scanning for uploaded blob content.	Request the list of storage accounts that require malware scanning for uploaded blob content, its review process, and its review records.	Low	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C104, depends on Storage.C103, assured by Storage.C105] Ensure required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage).	Request 1) the mechanism ensuring required storage accounts use malware scanning for uploaded blob content, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	Storage.T8 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T15 (Medium) Storage.T20 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T34 (Medium) Storage.T44 (Medium) Storage.T64 (Medium) Storage.T65 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C105] Verify required storage accounts use malware scanning for uploaded blob content (e.g., by using Microsoft Defender for Storage malware scanning).	Upload malware test content to a required storage account; it should be detected.	Medium	Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	-	Medium
Directive (COSO)	[Storage.C132]	Request the list of authorized account types for each	Low	Storage.FC1	Storage.T5 (Very Low)	Medium

Identify (NIST CSF)	Maintain the list of authorized account types (e.g., StorageV2) for each storage account.	storage account, its review process, and its review records.		Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11 Storage.FC12	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low) Storage.T15 (Very Low) Storage.T20 (Very Low) Storage.T25 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T34 (Very Low) Storage.T39 (Very Low) Storage.T42 (Very Low) Storage.T44 (Very Low) Storage.T50 (Very Low) Storage.T59 (Very Low) Storage.T60 (Very Low) Storage.T61 (Very Low) Storage.T62 (Very Low) Storage.T63 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low) Storage.T66 (Very Low)	
Directive (COSO) Protect (NIST CSF)	[Storage.C133, depends on Storage.C132, assured by Storage.C134] Ensure storage accounts use authorized account types.	Request 1) the mechanism ensuring storage accounts use authorized account types, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11 Storage.FC12	Storage.T5 (Medium) Storage.T7 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T13 (Medium) Storage.T15 (Medium) Storage.T20 (Medium) Storage.T25 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T34 (Medium) Storage.T39 (Medium) Storage.T42 (Medium) Storage.T44 (Medium) Storage.T50 (Medium) Storage.T59 (Medium) Storage.T60 (Medium) Storage.T61 (Medium) Storage.T62 (Medium) Storage.T63 (Medium) Storage.T64 (Medium) Storage.T65 (Medium) Storage.T66 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C134] Verify storage accounts use authorized account types (e.g., by using Azure built-in policy "Storage accounts should be migrated to new Azure Resource Manager resources" in Audit mode).	Create or modify a storage account with an unauthorized account type; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC3 Storage.FC4 Storage.FC5 Storage.FC11	-	Medium

				Storage.FC12		
Directive (COSO) Identify (NIST CSF)	[Storage.C22, depends on Storage.C2] Maintain a list of storage accounts that require hierarchical namespace (i.e., Data Lake Storage) and SSH enabled.	Request the list of storage accounts that require hierarchical namespace, its review process, and its review records.	Low	Storage.FC2 Storage.FC11	Storage.T7 (Very Low) Storage.T44 (Very Low) Storage.T64 (Very Low) Storage.T65 (Very Low)	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C26, depends on Storage.C2] Maintain the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods.	Request the list of storage accounts authorized to use SAS tokens, including authorized services, permissions, expiration, and access methods, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C27, depends on Storage.C26, assured by Storage.C28] Ensure SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods.	Request 1) the mechanism ensuring SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods, 2) its records of execution for all new SAS token usage, and 3) the plan to move any older SAS token usage.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium) Storage.T69 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C28] Verify SAS tokens are used only for authorized storage accounts, services, permissions, expiration, and access methods.	Configure or use a SAS token outside authorized storage accounts, services, permissions, expiration, or access methods; it should be detected.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C87] Define the requirements for authorized use of storage account keys and Shared Key authorization, including key rotation requirements and when Shared Key authorization is permitted.	Request the authorized use, key rotation, and Shared Key authorization requirements for storage account keys, its review process, and its review records.	Low	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T27 (Very Low) Storage.T28 (Very Low) Storage.T31 (Very Low) Storage.T67 (Very Low) Storage.T69 (Very Low)	Medium
Directive (COSO) Protect (NIST CSF)	[Storage.C88, depends on Storage.C87, assured by Storage.C89] Ensure storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Request 1) the mechanism ensuring storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Medium	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	Storage.T3 (Medium) Storage.T9 (Medium) Storage.T12 (Medium) Storage.T27 (Medium) Storage.T28 (Medium) Storage.T31 (Medium) Storage.T67 (Medium) Storage.T69 (Medium)	Medium
Assurance (COSO) Detect (NIST CSF)	[Storage.C89] Verify storage account keys and Shared Key authorization are managed according to authorized use, key rotation, and Shared Key authorization requirements.	Configure or use a storage account key or Shared Key authorization path outside authorized use, key rotation, or Shared Key authorization requirements; it should be detected.	Medium	Storage.FC1 Storage.FC2 Storage.FC4 Storage.FC5	-	Medium
Directive (COSO) Identify (NIST CSF)	[Storage.C144] Maintain the list of storage accounts or blob containers that require blob versioning.	Request the list of storage accounts or blob containers that require blob versioning, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T40 (Very Low)	Medium
Directive (COSO)	[Storage.C12, depends on Storage.C145, assured by Storage.C13] Ensure snapshots are enabled for required file shares.	Request 1) the mechanism that ensures snapshots are	Low	Storage.FC2	Storage.T7 (Low)	Low

Protect (NIST CSF)		enabled for required file shares, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.				
Assurance (COSO) Detect (NIST CSF)	[Storage.C13] Verify snapshots are enabled for required file shares (e.g., by using a custom Azure Policy on {"properties.shareSnapshotEnabled": false} in Audit mode).	Disable snapshotting from a required file share; it should be detected.	Medium	Storage.FC2	-	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C19, depends on Storage.C15, assured by Storage.C21] Ensure required storage accounts have the soft-delete option enabled for the containers.	Request 1) the mechanism ensuring required storage accounts have soft-delete for the container enabled, 2) its records of execution for all new required storage accounts, and 3) the plan to move any older required storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T25 (Low) Storage.T39 (Very Low)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C21] Verify required storage accounts have container soft delete enabled (e.g., by using a custom Azure Policy in Audit mode).	Create a storage account without soft-delete for the container option; it should be detected.	Low	Storage.FC1 Storage.FC2	-	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C46, depends on Storage.C147] Ensure storage blobs are using Azure Backup, following the requirements in the Azure Backup ThreatModel.	Request 1) the mechanism ensuring storage blobs are using Azure Backup, 2) its records of execution for all new storage blobs, and 3) the plan to move any older storage blobs.	High	Storage.FC2	Storage.T7 (Medium) Storage.T9 (Medium) Storage.T12 (Medium)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C75] Maintain a list of authorized storage account regions that can be used for redundancy.	Request the list of authorized storage account regions that are used for redundancy, its review process, and its review records.	Low	Storage.FC1	Storage.T59 (Very Low)	Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C77, depends on Storage.C75] Ensure only the authorized storage account region is set for redundancy (e.g., by using a custom Azure Policy on {"location": ["eastus","westeurope"]} in Deny mode).	Create a redundancy configuration on a storage account with an unauthorized region; it should be denied.	Medium	Storage.FC1	Storage.T59 (High)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C145] Maintain a list of Azure Files that require snapshots.	Request the list of Azure Files that require snapshots, its review process, and its review records.	Low	Storage.FC2	Storage.T7 (Very Low)	Low
Directive (COSO) Identify (NIST CSF)	[Storage.C147] Maintain a list of storage blobs that require Azure Backup.	Request the list of storage blobs that require Azure Backup, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low) Storage.T13 (Very Low)	Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C122, depends on Storage.C120] Prevent security protocol settings from changing on required file shares (e.g., by using custom Azure Policy on {"Microsoft.Storage/storageAccounts/fileServices/protocolSettings.smb": {"kerberosTicketEncryption":"AES256","channelEncryption":"Required"}} in Deny mode).	Modify a required file share's security protocol settings; it should be denied.	Medium	Storage.FC3	Storage.T20 (Low) Storage.T60 (High)	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C23, depends on Storage.C22, assured by Storage.C24] Ensure all required storage accounts have hierarchical namespace (i.e., Data Lake Storage) enabled.	Request 1) the mechanism ensuring that required storage accounts have hierarchical namespace enabled, 2) its records of execution for all new required storage accounts, 3) the plan to move any older storage accounts with hierarchical namespace not meeting the requirement.	Low	Storage.FC2	Storage.T7 (Low)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C24] Verify only required storage accounts have the	Create a non-required storage account with the hierarchical namespace (HNS) option enabled; it should	Medium	Storage.FC2	-	Low

	hierarchical namespace (HNS) option enabled (e.g., by using a custom Azure Policy on {"isHnsEnabled": "true"} in Audit mode).	be detected.				
Directive (COSO) Protect (NIST CSF)	[Storage.C156, depends on Storage.C26, assured by Storage.C157] Ensure stored access policies and their permissions are authorized for each container, file share, queue, and table.	Request 1) the mechanism ensuring stored access policies and their permissions are authorized for each container, file share, queue, and table, 2) its records of execution for all new containers, file shares, queues, and tables, and 3) the plan to move any older containers, file shares, queues, and tables.	Low	Storage.FC4 Storage.FC5	Storage.T27 (Low) Storage.T28 (Low) Storage.T67 (Low)	Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C157] Verify stored access policies and their permissions are authorized for each container, file share, queue, and table.	Modify an existing stored access policy or its permissions outside authorized values; it should be detected.	Low	Storage.FC4 Storage.FC5	-	Low
Directive (COSO) Protect (NIST CSF)	[Storage.C16, depends on Storage.C15, assured by Storage.C18] Ensure required storage accounts have the soft-delete feature for blobs enabled for the defined minimum retention period.	Request 1) the mechanism ensuring required storage accounts have soft-delete for blobs enabled for at least the defined minimum retention, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC2	Storage.T7 (Very Low) Storage.T9 (Very Low) Storage.T12 (Very Low)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C18] Verify required storage accounts have the blob soft-delete option enabled (e.g., by using custom Azure Policy in Audit mode).	Create a required storage account without soft-delete for the blob option; it should be detected.	Medium	Storage.FC2	-	Very Low
Directive (COSO) Identify (NIST CSF)	[Storage.C71] Maintain a list of storage accounts that require redundancy.	Request the list of storage accounts that require redundancy, its review process, and its review records.	Low	Storage.FC1	Storage.T42 (Very Low)	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C72, depends on Storage.C71, assured by Storage.C74] Ensure required storage accounts use redundancy.	Request 1) the mechanism ensuring required storage accounts use redundancy, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T42 (Medium)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C74] Verify required storage accounts use redundancy (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/sku.name": ["Standard_GRS","Standard_ZRS"]} in Audit mode).	Remove a redundancy configuration from a storage account; it should be detected.	Medium	Storage.FC1	-	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C76, depends on Storage.C75, assured by Storage.C78] Ensure the authorized storage account region used for redundancy is configured.	Request 1) the mechanism ensuring only authorized regions for storage accounts are in use, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1	Storage.T59 (Medium)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C78] Verify only the authorized storage account region is set for redundancy (e.g., by using a custom Azure Policy with {"location": ["eastus","westeurope"]} in Audit mode).	Create a storage account with an unauthorized region; it should be detected.	Medium	Storage.FC1	-	Very Low
Detective (COSO) Detect (NIST CSF)	[Storage.C101] Monitor the creation or update of Azure Files NFS/SMB 2.1 and corresponding settings (e.g., using activity logs on properties.supportsHttpsTrafficOnly with scope "supportsHttpsTrafficOnly").	Modify or create a file share that does not require NFS/SMB 2.1 to use NFS/SMB 2.1; it should be detected.	Low	Storage.FC3	Storage.T61 (Very Low)	Very Low
Directive (COSO) Identify (NIST CSF)	[Storage.C50] Maintain a list of storage accounts that require diagnostic settings to be enabled and their respective log destinations.	Request the list of storage accounts that require diagnostic settings, its review process, and its review records.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T13 (Very Low)	Very Low

					Storage.T42 (Very Low)	
Directive (COSO) Protect (NIST CSF)	[Storage.C51, depends on Storage.C50, assured by Storage.C54] Ensure diagnostic settings are enabled on storage accounts that require it, and their respective log destinations are authorized.	Request 1) the mechanism ensuring only authorized diagnostic settings destinations are enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Low	Storage.FC1 Storage.FC2	Storage.T7 (Very Low) Storage.T8 (Very Low) Storage.T9 (Very Low) Storage.T13 (Very Low) Storage.T42 (Very Low)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C54] Verify storage accounts have diagnostic settings configured (e.g., by using built-in Azure Policy "Configure diagnostic settings for storage accounts to Log Analytics workspace" in Audit mode).	Create a storage account with unauthorized diagnostic settings options; it should be detected.	Medium	Storage.FC1 Storage.FC2	-	Very Low
Directive (COSO) Identify (NIST CSF)	[Storage.C111, depends on Storage.C22] Maintain a list of Data Lake Storage accounts that require SFTP enabled.	Request the list of storage accounts that require SFTP, its review process, and its review records.	Low	Storage.FC11	Storage.T44 (Very Low)	Very Low
Directive (COSO) Protect (NIST CSF)	[Storage.C114, depends on Storage.C111, assured by Storage.C116] Ensure only storage accounts that require SFTP have it enabled.	Request 1) the mechanism ensuring storage accounts that require SFTP have it enabled, 2) its records of execution for all new storage accounts, and 3) the plan to move any older storage accounts.	Very Low	Storage.FC11	Storage.T44 (Low)	Very Low
Preventative (COSO) Protect (NIST CSF)	[Storage.C115, depends on Storage.C111] Prevent storage accounts that do not require SFTP from having it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/isSftpEnabled": true} in Deny mode).	Modify a storage account that does not require SFTP to use SFTP; it should be denied.	Medium	Storage.FC11	Storage.T44 (Medium)	Very Low
Assurance (COSO) Detect (NIST CSF)	[Storage.C116] Verify only storage accounts that require SFTP have it enabled (e.g., by using a custom Azure Policy on {"Microsoft.Storage/storageAccounts/isSftpEnabled": true} in Audit mode).	Modify a storage account that does not require SFTP to use SFTP; it should be detected.	Medium	Storage.FC11	-	Very Low

Appendix 2 - List of all Actions and their details

Id	Description	Feature Class ID	IAM Permission	Event	API	Threats
Storage.A1	Registers the subscription for the storage resource provider and enables the creation of storage accounts.	Storage.FC1	Microsoft.Storage/register/action	Microsoft.Storage/register/action	Providers_Register	-
Storage.A2	Notifies Azure Storage that a virtual network or subnet is being deleted.	Storage.FC1	Microsoft.Storage/locations/deleteVirtualNetworkOrSubnets/action	Microsoft.Storage/locations/deleteVirtualNetworkOrSubnets/action	-	-
Storage.A3	List blob services of a storage account. It returns a collection of one object named default.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/read	-	BlobServices_List	-
Storage.A4	Retrieves a user delegation key for the blob service. This is only a valid operation when using bearer token authentication.	Storage.FC1	Microsoft.Storage/storageAccounts/blobServices/generateUserDelegationKey/action	Microsoft.Storage/storageAccounts/blobServices/generateUserDelegationKey/action	Service_GetUserDelegationKey	-
Storage.A5	Sets the properties of a storage account's blob service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/write	Microsoft.Storage/storageAccounts/blobServices/write	BlobServices_SetServiceProperties	T40
Storage.A6	Gets the properties of a storage account's blob service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/read	-	BlobServices_GetServiceProperties	-
Storage.A7	Returns a list of blobs under the specified container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read	-	Blobs_List	T34, T48, T58
Storage.A8	Creates a new block, page, or append blob, or updates the content of an existing block blob. The Put blob operation will overwrite all contents of an existing blob with the same name.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Put_Blob	T8, T9, T12, T22, T66
Storage.A9	Deletes the specified blob or snapshot. Note that in order to delete a blob, you must delete all of its snapshots. You can delete both at the same time with the "Delete blob" operation.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete	Delete_Blob	T7
Storage.A10	Deletes a specific version of a blob without affecting other versions or the current blob.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/deleteblobVersion/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/deleteblobVersion/action	Delete_Blob	-
Storage.A11	This operation allows for the permanent deletion of soft-deleted blob snapshots or versions before the retention period expires.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/permanentDelete/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/permanentDelete/action	Delete_Blob	-
Storage.A12	Allows you to create a new blob or overwrite an existing blob within a container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/add/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/add/action	Put_Blob	T8
Storage.A13	Allows you to search for blobs within a container based on their assigned tags.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/filter/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/filter/action	Find_BlobsByTags	-
Storage.A14	Allows you to copy an existing blob to a new location within the same storage account or across different storage accounts.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/move/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/move/action	Copy_Blob	T55
Storage.A15	Allows you to set system properties on the blob, such as cache control, content type, and metadata.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/manageOwnership/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/manageOwnership/action	Set_BlobProperties	-

Storage.A16	Manages access control and permissions for blobs within Azure Storage.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/modifyPermissions/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/modifyPermissions/action	Set_BlobProperties	-
Storage.A17	Run as super user.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/runAsSuperUser/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/runAsSuperUser/action	-	-
Storage.A18	This operation migrates a blob container from container-level WORM to an object-level immutability enabled container. Prerequisites require a container-level immutability policy either in a locked or unlocked state, account-level versioning must be enabled and there should be no legal hold on the container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/migrate/action	Microsoft.Storage/storageAccounts/blobServices/containers/migrate/action	BlobContainers_ObjectLevelWorm	-
Storage.A19	Updates container properties as specified in the request body. Properties not mentioned in the request will be unchanged. The update fails if the specified container doesn't already exist.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/write	Microsoft.Storage/storageAccounts/blobServices/containers/write	BlobContainers_Update	T5, T66
Storage.A20	Deletes the specified container under its account.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/delete	Microsoft.Storage/storageAccounts/blobServices/containers/delete	BlobContainers_Delete	T7
Storage.A21	Gets the properties of a specified container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/read	-	BlobContainers_Get	-
Storage.A22	Lists all containers and does not support a prefix like data plane. Also SRP today does not return a continuation token.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/read	-	BlobContainers_List	-
Storage.A23	The Lease container operation establishes and manages a lock on a container for delete operations. The lock duration can be 15 to 60 seconds, or can be infinite.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/lease/action	Microsoft.Storage/storageAccounts/blobServices/containers/lease/action	BlobContainers_Lease	-
Storage.A24	Creates a new container under the specified account as described by the request body. The container resource includes metadata and properties for that container. It does not include a list of the blobs contained by the container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/write	Microsoft.Storage/storageAccounts/blobServices/containers/write	BlobContainers_Create	T5, T66
Storage.A25	Clears legal hold tags. Clearing the same or non-existing tag results in an idempotent operation. ClearLegalHold clears out only the specified tags in the request.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/clearLegalHold/action	Microsoft.Storage/storageAccounts/blobServices/containers/clearLegalHold/action	BlobContainers_ClearLegalHold	T56
Storage.A26	Sets legal hold tags. Setting the same tag results in an idempotent operation. SetLegalHold follows an append pattern and does not clear out the existing tags that are not specified in the request.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/setLegalHold/action	Microsoft.Storage/storageAccounts/blobServices/containers/setLegalHold/action	BlobContainers_SetLegalHold	T56
Storage.A27	Extends the immutabilityPeriodSinceCreationInDays of a locked ImmutabilityPolicy. The only action allowed on a locked policy will be this action. ETag in If-Match is required for this operation.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/extend/action	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/extend/action	BlobContainers_ExtendImmutabilityPolicy	-
Storage.A28	Aborts an unlocked immutability policy. The response of delete has immutabilityPeriodSinceCreationInDays set to 0. ETag in If-Match is required for this operation. Deleting a locked immutability policy is not allowed, the only way is to delete the container after deleting	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/delete	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/delete	BlobContainers_DeleteImmutabilityPolicy	-

	all expired blobs inside the policy-locked container.					
Storage.A29	Creates or updates an unlocked immutability policy. ETag in If-Match is honored if given, but not required for this operation.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/write	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/write	BlobContainers_CreateOrUpdateImmutabilityPolicy	T56
Storage.A30	Sets the ImmutabilityPolicy to locked state. The only action allowed on a locked policy is the ExtendImmutabilityPolicy action. ETag in If-Match is required for this operation.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/lock/action	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/lock/action	BlobContainers_LockImmutabilityPolicy	-
Storage.A31	Gets the existing immutability policy along with the corresponding ETag in response headers and body.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/immutabilityPolicies/read	-	BlobContainers_GetImmutabilityPolicy	-
Storage.A32	List all queue services for the storage account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/read	-	QueueServices_List	-
Storage.A33	Gets the properties of a storage account's queue service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/read	-	QueueServices_GetServiceProperties	-
Storage.A34	Sets the properties of a storage account's queue service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/write	Microsoft.Storage/storageAccounts/queueServices/write	QueueServices_SetServiceProperties	T27, T31
Storage.A35	Creates a new queue with the specified queue name, under the specified account. Creates a new queue with the specified queue name, under the specified account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/write	Microsoft.Storage/storageAccounts/queueServices/queues/write	Queue_Create	T27, T31
Storage.A36	Gets the queue with the specified queue name, under the specified account if it exists. Gets a list of all the queues under the specified storage account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/read	-	Queue_Get	-
Storage.A37	Creates a new queue with the specified queue name, under the specified account. Creates a new queue with the specified queue name, under the specified account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/write	Microsoft.Storage/storageAccounts/queueServices/queues/write	Queue_Create	T27, T31
Storage.A38	Deletes the queue with the specified queue name, under the specified account if it exists.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/delete	Microsoft.Storage/storageAccounts/queueServices/queues/delete	Queue_Delete	T31
Storage.A39	Retrieves one or more messages from the front of the queue.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/messages/read	-	Get_Messages	T67
Storage.A40	Adds a new message to the back of the message queue. A visibility time-out can also be specified to make the message invisible until the visibility time-out expires. A message must be in a format that can be included in an XML request with UTF-8 encoding. The encoded message can be up to 64 kibibytes (KiB) in size for version 2011-08-18 and later, or 8 KiB for earlier versions.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/messages/write	Microsoft.Storage/storageAccounts/queueServices/queues/messages/write	Put_Messages	-
Storage.A41	Deletes the specified message from the queue.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/messages/delete	Microsoft.Storage/storageAccounts/queueServices/queues/messages/delete	Delete_Messages	-
Storage.A42	This operation allows you to add a new message to the back of a specified queue.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/message	Microsoft.Storage/storageAccounts/queueServices/queues/message	Put_Messages	-

			s/add/action	s/add/action		
Storage.A43	Handles operations that involve processing messages in Azure Queue Storage. This encompasses actions such as retrieving messages from a queue, updating their visibility, and deleting them after processing.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/messages/process/action	Microsoft.Storage/storageAccounts/queueServices/queues/messages/process/action	-	T67
Storage.A44	The update operation can be used to update the SKU, encryption, access tier, or tags for a storage account. It can also be used to map the account to a custom domain. Only one custom domain is supported per storage account; the replacement/change of custom domain is not supported.	Storage.FC1	Microsoft.Storage/storageAccounts/updateInternalProperties/action	Microsoft.Storage/storageAccounts/updateInternalProperties/action	StorageAccounts_Update	-
Storage.A45	Live migration of storage account to enable HNS.	Storage.FC1	Microsoft.Storage/storageAccounts/hnsonmigration/action	Microsoft.Storage/storageAccounts/hnsonmigration/action	StorageAccounts_HierarchicalNamespaceMigration	-
Storage.A46	Live migration of storage account to enable HNS.	Storage.FC1	Microsoft.Storage/storageAccounts/hnsonmigration/action	Microsoft.Storage/storageAccounts/hnsonmigration/action	StorageAccounts_HierarchicalNamespaceMigration	-
Storage.A47	Restore blobs in the specified blob ranges.	Storage.FC2	Microsoft.Storage/storageAccounts/restoreBlobRanges/action	Microsoft.Storage/storageAccounts/restoreBlobRanges/action	StorageAccounts_RestoreBlobRanges	-
Storage.A48	Used for approving or rejecting private endpoint connections for Azure Storage accounts.	Storage.FC12	Microsoft.Storage/storageAccounts/PrivateEndpointConnectionsApproval/action	Microsoft.Storage/storageAccounts/PrivateEndpointConnectionsApproval/action	PrivateEndpointConnections_Put	-
Storage.A49	A failover request can be triggered for a storage account in the event a primary endpoint becomes unavailable for any reason. The failover occurs from the storage account's primary cluster to the secondary cluster for RA-GRS accounts. The secondary cluster will become primary after failover and the account is converted to LRS. In the case of a Planned failover, the primary and secondary clusters are swapped after failover and the account remains geo-replicated. Failover should continue to be used in the event of availability issues as Planned failover is only available while the primary and secondary endpoints are available. The primary use case of a Planned failover is Disaster Recovery testing drills. This type of failover is invoked by setting failoverType parameter to "Planned". Learn more about the failover options here- "https://learn.microsoft.com/en-us/azure/storage/common/storage-disaster-recovery-guidance".	Storage.FC1	Microsoft.Storage/storageAccounts/failover/action	Microsoft.Storage/storageAccounts/failover/action	StorageAccounts_Failover	T59
Storage.A50	Lists the access keys or Kerberos keys (if Active Directory enabled) for the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/listKeys/action	Microsoft.Storage/storageAccounts/listKeys/action	StorageAccounts_ListKeys	T1
Storage.A51	Regenerates one of the access keys or Kerberos keys for the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/regenerateKey/action	Microsoft.Storage/storageAccounts/regenerateKey/action	StorageAccounts_RegenerateKey	T1, T2
Storage.A52	Used as part of the regenerate keys process.	Storage.FC1	Microsoft.Storage/storageAccounts/rotateKey/action	Microsoft.Storage/storageAccounts/rotateKey/action	StorageAccounts_RegenerateKey	T1, T2
Storage.A53	Revoke user delegation keys.	Storage.FC1	Microsoft.Storage/storageAccounts/revokeUserDelegationKeys/action	Microsoft.Storage/storageAccounts/revokeUserDelegationKeys/action	StorageAccounts_RevokeUserDelegationKeys	-
Storage.A54	Deletes a storage account in Microsoft Azure.	Storage.FC1	Microsoft.Storage/storageAccount	Microsoft.Storage/storageAccount	StorageAccounts_Delete	T4

			s/delete	s/delete		
Storage.A55	Lists all the storage accounts available under the subscription.	Storage.FC1	Microsoft.Storage/storageAccounts/read	-	StorageAccounts_List	-
Storage.A56	List SAS credentials of a storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/ListAccountSas/action	Microsoft.Storage/storageAccounts/ListAccountSas/action	StorageAccounts_ListAccountSAS	-
Storage.A57	List service SAS credentials of a specific resource.	Storage.FC1	Microsoft.Storage/storageAccounts/ListServiceSas/action	Microsoft.Storage/storageAccounts/ListServiceSas/action	StorageAccounts_ListServiceSAS	-
Storage.A58	Asynchronously creates a new storage account with the specified parameters. If an account is already created and a subsequent create request is issued with different properties, the account properties will be updated. If an account is already created and a subsequent create or update request is issued with the exact same set of properties, the request will succeed. The update operation can be used to update the SKU, encryption, access tier, or tags for a storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/write	Microsoft.Storage/storageAccounts/write	StorageAccounts_Create	T13, T39, T42, T50, T63, T68, T69
Storage.A59	Configure monitoring and logging settings through Azure Resource Manager (ARM) APIs.	Storage.FC1	Microsoft.Storage/storageAccounts/services/diagnosticsettings/write	Microsoft.Storage/storageAccounts/services/diagnosticsettings/write	StorageAccounts_CreateOrUpdateDiagnosticSettings	-
Storage.A60	Get list of Azure Storage metrics definitions.	Storage.FC1	Microsoft.Storage/storageAccounts/providers/Microsoft.Insights/metricDefinitions/read	-	List_MetricDefinitions	-
Storage.A61	Gets the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/providers/Microsoft.Insights/diagnosticsettings/read	-	Get_MetricDefinitions	-
Storage.A62	Creates or updates the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/providers/Microsoft.Insights/diagnosticsettings/write	Microsoft.Storage/storageAccounts/providers/Microsoft.Insights/diagnosticsettings/write	CreateOrUpdate_DiagnosticSettings	-
Storage.A63	Get a list of Azure Storage metrics definitions.	Storage.FC1	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/metricDefinitions/read	-	List_MetricDefinitions	-
Storage.A64	Gets the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/diagnosticsettings/read	-	Get_MetricDefinitions	-
Storage.A65	Creates or updates the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/diagnosticsettings/write	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/diagnosticsettings/write	CreateOrUpdate_DiagnosticSettings	-
Storage.A66	Get list of Azure Storage metrics definitions.	Storage.FC1	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/metricDefinitions/read	-	List_MetricDefinitions	-
Storage.A67	Gets the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/diagnosticsettings/read	-	Get_MetricDefinitions	-
Storage.A68	Creates or updates the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/diagnosticsettings/write	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/diagnosticsettings/write	CreateOrUpdate_DiagnosticSettings	-
Storage.A69	Get a list of Azure Storage metrics definitions.	Storage.FC1	Microsoft.Storage/storageAccount	-	List_MetricDefinitions	-

			s/fileServices/providers/Microsoft.Insights/metricDefinitions/read			
Storage.A70	Gets the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/fileServices/providers/Microsoft.Insights/diagnosticsettings/read	-	Get_MetricDefinitions	-
Storage.A71	Creates or updates the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/fileServices/providers/Microsoft.Insights/diagnosticsettings/write	Microsoft.Storage/storageAccounts/fileServices/providers/Microsoft.Insights/diagnosticsettings/write	CreateOrUpdate_DiagnosticSettings	-
Storage.A72	Get a list of Azure Storage metrics definitions.	Storage.FC1	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/metricDefinitions/read	-	List_MetricDefinitions	-
Storage.A73	Gets the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/diagnosticsettings/read	-	Get_MetricDefinitions	-
Storage.A74	Creates or updates the diagnostic setting for the resource.	Storage.FC1	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/diagnosticsettings/write	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/diagnosticsettings/write	CreateOrUpdate_DiagnosticSettings	-
Storage.A75	Gets the log definition for the table.	Storage.FC1	Microsoft.Storage/storageAccounts/tableServices/providers/Microsoft.Insights/logDefinitions/read	-	Get_LogDefinitions	-
Storage.A76	Gets the log definition for blob.	Storage.FC1	Microsoft.Storage/storageAccounts/blobServices/providers/Microsoft.Insights/logDefinitions/read	-	Get_LogDefinitions	-
Storage.A77	Gets the log definition for a file.	Storage.FC1	Microsoft.Storage/storageAccounts/fileServices/providers/Microsoft.Insights/logDefinitions/read	-	Get_LogDefinitions	-
Storage.A78	Gets the log definition for the queue.	Storage.FC1	Microsoft.Storage/storageAccounts/queueServices/providers/Microsoft.Insights/logDefinitions/read	-	Get_LogDefinitions	-
Storage.A79	Lists the available SKUs supported by Microsoft.Storage for a given subscription.	Storage.FC1	Microsoft.Storage/skus/read	-	Skus_List	-
Storage.A80	Lists all of the available storage REST API operations.	Storage.FC1	Microsoft.Storage/operations/read	-	Operations_List	-
Storage.A81	Checks that the storage account name is valid and is not already in use.	Storage.FC1	Microsoft.Storage/checkNameAvailability/read	-	StorageAccounts_CheckNameAvailability	-
Storage.A82	Gets the current usage count and the limit for the resources in the location under the subscription.	Storage.FC1	Microsoft.Storage/locations/usages/readMicrosoft.Storage/locations/usages/read	-	Usages_ListByLocation	-
Storage.A83	Gets the current usage count and the limit for the resources under the subscription.	Storage.FC1	Microsoft.Storage/usages/read	-	Usage_List	-
Storage.A84	The get tags operation enables users to get the tags associated with a blob.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/tags/read	-	Blob_GetTags	-
Storage.A85	The set tags operation enables users to set the tags associated with a blob.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/tags/write	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/tags/write	Blob_SetTags	-

Storage.A86	Deletes the management policy associated with the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/managementPolicies/delete	Microsoft.Storage/storageAccounts/managementPolicies/delete	ManagementPolicies_Delete	-
Storage.A87	Gets the management policy associated with the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/managementPolicies/read	-	ManagementPolicies_Get	-
Storage.A88	Sets the management policy to the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/managementPolicies/write	Microsoft.Storage/storageAccounts/managementPolicies/write	ManagementPolicies_CreateOrUpdate	T25
Storage.A89	Used for operations that manage Azure file shares within a storage account.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/action	Microsoft.Storage/storageAccounts/fileServices/shares/action	-	-
Storage.A90	List all file services in storage accounts.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/read	-	FileServices_List	-
Storage.A91	Sets the properties of file services in storage accounts, including CORS (Cross-Origin Resource Sharing) rules.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/write	Microsoft.Storage/storageAccounts/fileServices/write	FileServices_SetServiceProperties	T60, T61
Storage.A92	Gets the properties of file services in storage accounts, including CORS (Cross-Origin Resource Sharing) rules.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/read	-	FileServices_GetServiceProperties	-
Storage.A93	List all table services for the storage account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/read	-	TableServices_List	-
Storage.A94	Gets the properties of a storage account's table service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/read	-	TableServices_GetServiceProperties	-
Storage.A95	Sets the properties of a storage account's table service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/write	Microsoft.Storage/storageAccounts/tableServices/write	TableServices_SetServiceProperties	T28
Storage.A96	Reads or downloads a file from the system, including its metadata and properties.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/read	-	Get_File	T15, T19
Storage.A97	Creates a new file or replaces a file. When you call "Create File", you only initialize the file. To add content to a file, you call the "Put Range" operation.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/write	Create_File	T16, T17, T19, T20
Storage.A98	The Delete file operation immediately removes the file from the storage account.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/delete	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/delete	Delete_File	T18
Storage.A99	The Set file properties operation sets system properties on the file.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/modifyPermissions/action	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/modifyPermissions/action	Set_FileProperties	T17
Storage.A100	The Get file properties operation returns all user-defined metadata, standard HTTP properties, and system properties for the file. It doesn't return the content of the file.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/actassuperuser/action	Microsoft.Storage/storageAccounts/fileServices/fileshares/files/actassuperuser/action	Get_FileProperties	-
Storage.A101	Get a private endpoint connection proxy.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnectionProxies/read	-	PrivateEndpointConnectionProxies_Get	-
Storage.A102	Delete private endpoint connection proxies.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnectionProxies/delete	Microsoft.Storage/storageAccounts/privateEndpointConnectionProxies/delete	PrivateEndpointConnectionProxies_Delete	-
Storage.A103	List all private endpoint connection proxies.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnectionProxies/write	Microsoft.Storage/storageAccounts/privateEndpointConnectionProxies/write	PrivateEndpointConnectionProxies_List	-
Storage.A104	List all the private endpoint connections associated	Storage.FC12	Microsoft.Storage/storageAccount	-	PrivateEndpointConnections_List	-

	with the storage account.		s/privateEndpointConnections/read			
Storage.A105	Deletes the specified private endpoint connection associated with the storage account.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnections/delete	Microsoft.Storage/storageAccounts/privateEndpointConnections/delete	PrivateEndpointConnections_Delete	T62
Storage.A106	Gets the specified private endpoint connection associated with the storage account.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnections/read	-	PrivateEndpointConnections_Get	-
Storage.A107	Update the state of a specified private endpoint connection associated with the storage account.	Storage.FC12	Microsoft.Storage/storageAccounts/privateEndpointConnections/write	Microsoft.Storage/storageAccounts/privateEndpointConnections/write	PrivateEndpointConnections_Put	-
Storage.A108	Gets the Private Link resources that need to be created for a storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/privateLinkResources/read	-	PrivateLinkResources_ListByStorageAccount	-
Storage.A109	Checks that the storage account name is valid and is not already in use.	Storage.FC1	Microsoft.Storage/locations/checkNameAvailability/read	-	StorageAccounts_CheckNameAvailability	-
Storage.A110	Deletes the specified share under its account.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/delete	Microsoft.Storage/storageAccounts/fileServices/shares/delete	FileShares_Delete	-
Storage.A111	Lists all shares. Gets properties of a specified share.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/read	-	FileShares_List	-
Storage.A113	Creates a new share under the specified account as described by the request body. The share resource includes metadata and properties for that share. It does not include a list of the files contained by the share. Updates share properties as specified in the request body. Properties not mentioned in the request will not be changed. Update fails if the specified share does not already exist.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/write	Microsoft.Storage/storageAccounts/fileServices/shares/write	FileShares_Create	-
Storage.A114	Returns the properties for the specified encryption scope.	Storage.FC1	Microsoft.Storage/storageAccounts/encryptionScopes/read	-	EncryptionScopes_Get	-
Storage.A115	Synchronously creates or updates an encryption scope under the specified storage account. If an encryption scope is already created and a subsequent request is issued with different properties, the encryption scope properties will be updated per the specified request. Update encryption scope properties as specified in the request body. Update fails if the specified encryption scope does not already exist.	Storage.FC1	Microsoft.Storage/storageAccounts/encryptionScopes/write	Microsoft.Storage/storageAccounts/encryptionScopes/write	EncryptionScopes_Put	T38
Storage.A116	Deletes the object replication policy associated with the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/objectReplicationPolicies/delete	Microsoft.Storage/storageAccounts/objectReplicationPolicies/delete	ObjectReplicationPolicies_Delete	-
Storage.A117	Get the object replication policy of the storage account by policy ID.	Storage.FC1	Microsoft.Storage/storageAccounts/objectReplicationPolicies/read	-	ObjectReplicationPolicies_Get	-
Storage.A118	List the object replication policies associated with the storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/objectReplicationPolicies/read	-	ObjectReplicationPolicies_List	-
Storage.A119	Create or update the object replication policy of the storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/objectReplicationPolicies/write	Microsoft.Storage/storageAccounts/objectReplicationPolicies/write	ObjectReplicationPolicies_CreateOrUpdate	-
Storage.A120	Removes a data share policy.	Storage.FC1	Microsoft.Storage/storageAccounts/dataSharePolicies/delete	Microsoft.Storage/storageAccounts/dataSharePolicies/delete	Delete_DataSharePolicy	-

Storage.A121	Gets a data share policy for a specific account.	Storage.FC1	Microsoft.Storage/storageAccounts/dataSharePolicies/read	-	Get_DataSharePolicy	-
Storage.A122	Creates or updates a data share policy.	Storage.FC1	Microsoft.Storage/storageAccounts/dataSharePolicies/write	Microsoft.Storage/storageAccounts/dataSharePolicies/write	CreateOrUpdate_DataSharePolicy	-
Storage.A123	Deletes the local user associated with the specified storage account.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/delete	Microsoft.Storage/storageAccounts/localUsers/delete	LocalUsers_Delete	-
Storage.A124	Regenerate the local user SSH password.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/regeneratePassword/action	Microsoft.Storage/storageAccounts/localUsers/regeneratePassword/action	LocalUsers_RegeneratePassword	-
Storage.A125	List SSH authorized keys and Shared Key of the local user.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/listKeys/action	Microsoft.Storage/storageAccounts/localUsers/listKeys/action	LocalUsers_ListKeys	T64
Storage.A126	List the local users associated with the storage account.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/read	-	LocalUsers_List	-
Storage.A127	Get the local user of the storage account by username.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/read	-	LocalUsers_Get	-
Storage.A128	Create or update the properties of a local user associated with the storage account. Properties for NFSv3 enablement and extended Groups cannot be set with other properties.	Storage.FC11	Microsoft.Storage/storageAccounts/localUsers/write	Microsoft.Storage/storageAccounts/localUsers/write	LocalUsers_CreateOrUpdate	T44
Storage.A129	Gets the table with the specified table name, under the specified account if it exists. Gets a list of all the tables under the specified storage account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/read	-	Table_Get	-
Storage.A130	Creates a new table with the specified table name, under the specified account. Creates a new table with the specified table name, under the specified account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/write	Microsoft.Storage/storageAccounts/tableServices/tables/write	Table_Create	T28
Storage.A131	Deletes the table with the specified table name, under the specified account if it exists.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/delete	Microsoft.Storage/storageAccounts/tableServices/tables/delete	Table_Delete	-
Storage.A132	Deletes the blob inventory policy associated with the specified storage account.	Storage.FC10	Microsoft.Storage/storageAccounts/inventoryPolicies/delete	Microsoft.Storage/storageAccounts/inventoryPolicies/delete	BlobInventoryPolicies_Delete	-
Storage.A133	Gets the blob inventory policy associated with the specified storage account.	Storage.FC10	Microsoft.Storage/storageAccounts/inventoryPolicies/read	-	BlobInventoryPolicies_Get	T24
Storage.A134	Sets the blob inventory policy to the specified storage account.	Storage.FC10	Microsoft.Storage/storageAccounts/inventoryPolicies/write	Microsoft.Storage/storageAccounts/inventoryPolicies/write	BlobInventoryPolicies_CreateOrUpdate	T24
Storage.A135	Part of the Azure Management resource lock.	Storage.FC1	Microsoft.Storage/storageAccounts/accountLocks/deleteLock/action	Microsoft.Storage/storageAccounts/accountLocks/deleteLock/action	-	-
Storage.A136	Lock read.	Storage.FC1	Microsoft.Storage/storageAccounts/accountLocks/read	-	ManagementLock_List	-
Storage.A137	Lock write.	Storage.FC1	Microsoft.Storage/storageAccounts/accountLocks/write	Microsoft.Storage/storageAccounts/accountLocks/write	ManagementLock_CreateOrUpdate	-
Storage.A138	Lock delete.	Storage.FC1	Microsoft.Storage/storageAccounts/accountLocks/delete	Microsoft.Storage/storageAccounts/accountLocks/delete	ManagementLock_Delete	-
Storage.A139	Data share policy read.	Storage.FC1	Microsoft.Storage/storageAccounts/consumerdataSharePolicies/read	-	-	-
Storage.A140	Data share policy write.	Storage.FC1	Microsoft.Storage/storageAccounts/consumerdataSharePolicies/writ	Microsoft.Storage/storageAccounts/consumerdataSharePolicies/writ	-	-

			e	e		
Storage.A141	Query table entities.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/entities/read	-	-	-
Storage.A142	Insert, merge, or replace table entities.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/entities/write	Microsoft.Storage/storageAccounts/tableServices/tables/entities/write	-	-
Storage.A143	Delete table entities.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/entities/delete	Microsoft.Storage/storageAccounts/tableServices/tables/entities/delete	-	-
Storage.A144	Insert table entities.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/entities/add/action	Microsoft.Storage/storageAccounts/tableServices/tables/entities/add/action	-	-
Storage.A145	Merge or update table entities.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/entities/update/action	Microsoft.Storage/storageAccounts/tableServices/tables/entities/update/action	-	-
Storage.A146	Run as super user.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/immutableStorage/runAsSuperUser/action	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/immutableStorage/runAsSuperUser/action	-	-
Storage.A147	Create object replication restore point marker.	Storage.FC1	Microsoft.Storage/storageAccounts/objectReplicationPolicies/restorePointMarkers/write	Microsoft.Storage/storageAccounts/objectReplicationPolicies/restorePointMarkers/write	-	-
Storage.A148	Delete object replication restore point.	Storage.FC1	Microsoft.Storage/storageAccounts/restorePoints/delete	Microsoft.Storage/storageAccounts/restorePoints/delete	-	-
Storage.A149	List object replication restore points.	Storage.FC1	Microsoft.Storage/storageAccounts/restorePoints/read	-	-	-
Storage.A150	List object replication restore points.	Storage.FC1	Microsoft.Storage/storageAccounts/restorePoints/read	-	-	-
Storage.A151	Gets the status of the ongoing migration for the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/accountMigrations/read	-	StorageAccounts_GetCustomerInitiatedMigration	-
Storage.A152	Customer is able to update their storage account redundancy for increased resiliency.	Storage.FC1	Microsoft.Storage/storageAccounts/accountMigrations/write	Microsoft.Storage/storageAccounts/accountMigrations/write	-	-
Storage.A153	Lists all containers and does not support a prefix like data plane. Also SRP today does not return a continuation token. Gets properties of a specified container.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/read	-	BlobContainers_List	-
Storage.A154	Creates a new container under the specified account. If the container with the same name already exists, the operation fails returns all user-defined metadata and system properties for the specified container. The data returned does not include the container's list of blobs operation marks the specified container for deletion. The container and any blobs contained within it are later deleted during garbage collection restype.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/write	Microsoft.Storage/storageAccounts/blobServices/containers/write	Container_Create	T5, T66
Storage.A155	Set properties for the filesystem. This operation supports conditional HTTP requests. For more information, see Specifying conditional headers for	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	FileSystem_SetProperties	T8, T9, T12, T22, T66

	blob service operations.					
Storage.A156	List filesystem paths and their properties.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read	-	FileSystem_ListPaths	T34, T48, T58
Storage.A157	List blob services of a storage account. It returns a collection of one object named default. Gets the properties of a storage account's blob service, including properties for storage Analytics and CORS (Cross-Origin Resource Sharing) rules.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/read	-	BlobServices_GetServiceProperties	-
Storage.A158	Deletes the specified container under its account.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/delete	Microsoft.Storage/storageAccounts/blobServices/containers/delete	BlobContainers_Delete	T7
Storage.A159	Create or rename a file or directory. By default, the destination is overwritten, and if the destination already exists and has a lease, the lease is broken. This operation supports conditional HTTP requests. For more information, see Specifying conditional headers for blob service operations. To fail if the destination already exists, use a conditional request with If-None-Match: "*".	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Path_Create	T8, T9, T12, T22, T66
Storage.A160	Uploads data to be appended to a file, flushes (writes) previously uploaded data to a file, sets properties for a file or directory, or sets access control for a file or directory. Data can only be appended to a file. Concurrent writes to the same file using multiple clients are not supported. This operation supports conditional HTTP requests. For more information, see Specifying conditional headers for blob service operations.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write	Path_Update	T8, T9, T12, T22, T66
Storage.A161	The Lease container operation establishes and manages a lock on a container for delete operations. The lock duration can be 15 to 60 seconds, or can be infinite.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/lease/action	Microsoft.Storage/storageAccounts/blobServices/containers/lease/action	BlobContainers_Lease	-
Storage.A162	The "Download" operation reads or downloads a blob from the system, including its metadata and properties. You can also call "Download" to read a snapshot.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read	-	Blob_Download	T34, T48, T58
Storage.A163	The Get properties operation returns all user-defined metadata, standard HTTP properties, and system properties for the blob. It does not return the content of the blob.	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read	-	Blob_GetProperties	T34, T48, T58
Storage.A164	If the storage account's soft delete feature is disabled, then when a blob is deleted, it is permanently removed from the storage account. If the storage account's soft delete feature is enabled, then when a blob is deleted, it is marked for deletion and becomes inaccessible immediately. However, the blob service retains the blob or snapshot for the number of days specified by the DeleteRetentionPolicy section of [storage service properties] (Set-blob-Service-Properties.md). After the specified number of days has passed, the blob's data is	Storage.FC2	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete	Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete	Blob_Delete	T7

	permanently removed from the storage account. Note that you continue to be charged for the soft-deleted blob's storage until it is permanently removed. Use the "List blobs" API and specify the "include=deleted" query parameter to discover which blobs and snapshots have been soft deleted. You can then use the "Undelete Blob" API to restore a soft-deleted blob. All other operations on a soft-deleted blob or snapshot cause the service to return an HTTP status code of 404 (ResourceNotFound).					
Storage.A165	Gets the current usage count and the limit for the resources under the subscription.	Storage.FC1	Microsoft.Storage/usages/read	-	Usage_List	-
Storage.A166	Lists all shares.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/read	-	FileShares_List	-
Storage.A167	Updates share properties as specified in the request body. Properties not mentioned in the request will not be changed. Update fails if the specified share does not already exist.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/write	Microsoft.Storage/storageAccounts/fileServices/shares/write	FileShares_Update	-
Storage.A168	Gets properties of a specified share.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/read	-	FileShares_Get	-
Storage.A169	Restore a file share within valid retention days if share soft delete is enabled.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/restore/action	Microsoft.Storage/storageAccounts/fileServices/shares/restore/action	FileShares_Restore	-
Storage.A170	The "Lease Share" operation establishes and manages a lock on a share for delete operations. The lock duration can be 15 to 60 seconds, or can be infinite.	Storage.FC3	Microsoft.Storage/storageAccounts/fileServices/shares/lease/action	Microsoft.Storage/storageAccounts/fileServices/shares/lease/action	FileShares_Lease	-
Storage.A171	Gets list of effective NetworkSecurityPerimeterConfiguration for storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/networkSecurityPerimeterConfigurations/read	-	NetworkSecurityPerimeterConfigurations_List	-
Storage.A172	Gets the effective NetworkSecurityPerimeterConfiguration for association.	Storage.FC1	Microsoft.Storage/storageAccounts/networkSecurityPerimeterConfigurations/read	-	NetworkSecurityPerimeterConfigurations_Get	-
Storage.A173	Refreshes any information about the association.	Storage.FC1	Microsoft.Storage/storageAccounts/networkSecurityPerimeterConfigurations/reconcile/action	Microsoft.Storage/storageAccounts/networkSecurityPerimeterConfigurations/reconcile/action	NetworkSecurityPerimeterConfigurations_Reconcile	-
Storage.A174	Gets the queue with the specified queue name, under the specified account if it exists.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/read	-	Queue_Get	-
Storage.A175	Gets a list of all the queues under the specified storage account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/read	-	Queue_List	-
Storage.A176	Creates a new queue with the specified queue name, under the specified account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/write	Microsoft.Storage/storageAccounts/queueServices/queues/write	Queue_Create	T27, T31
Storage.A177	Creates a new queue with the specified queue name, under the specified account.	Storage.FC4	Microsoft.Storage/storageAccounts/queueServices/queues/write	Microsoft.Storage/storageAccounts/queueServices/queues/write	Queue_Update	T27, T31
Storage.A178	Lists deleted accounts under the subscription.	Storage.FC1	Microsoft.Storage/deletedAccounts/read	-	DeletedAccounts_List	-
Storage.A179	Get properties of the specified deleted account resource.	Storage.FC1	Microsoft.Storage/locations/deletedAccounts/read	-	DeletedAccounts_Get	-

Storage.A180	Gets the current usage count and the limit for the resources of the location under the subscription.	Storage.FC1	Microsoft.Storage/locations/usages/read	-	Usage_ListByLocation	-
Storage.A181	Lists all the storage accounts available under the given resource group. Note that storage keys are not returned; use the ListKeys operation for this.	Storage.FC1	Microsoft.Storage/storageAccounts/read	-	StorageAccounts_ListByResourceGroup	-
Storage.A182	Returns the properties for the specified storage account including but not limited to name, SKU name, location, and account status. The ListKeys operation should be used to retrieve storage keys.	Storage.FC1	Microsoft.Storage/storageAccounts/read	-	StorageAccounts_GetProperties	-
Storage.A183	Lists all the encryption scopes available under the specified storage account.	Storage.FC1	Microsoft.Storage/storageAccounts/encryptionScopes/read	-	EncryptionScopes_List	-
Storage.A184	Update encryption scope properties as specified in the request body. Update fails if the specified encryption scope does not already exist.	Storage.FC1	Microsoft.Storage/storageAccounts/encryptionScopes/write	Microsoft.Storage/storageAccounts/encryptionScopes/write	EncryptionScopes_Patch	T38
Storage.A185	Gets the blob inventory policy associated with the specified storage account.	Storage.FC10	Microsoft.Storage/storageAccounts/inventoryPolicies/read	-	BlobInventoryPolicies_List	T24
Storage.A186	Abort live migration of storage account to enable HNS.	Storage.FC1	Microsoft.Storage/storageAccounts/aborthnsonmigration/action	Microsoft.Storage/storageAccounts/aborthnsonmigration/action	StorageAccounts_AbortHierarchicalNamespaceMigration	-
Storage.A187	Account migration request can be triggered for a storage account to change its redundancy level. The migration updates the non-zonal redundant storage account to a zonal redundant account or vice-versa in order to have better reliability and availability. Zone-redundant storage (ZRS) replicates your storage account synchronously across three Azure Availability Zones in the primary region.	Storage.FC1	Microsoft.Storage/storageAccounts/startAccountMigration/action	Microsoft.Storage/storageAccounts/startAccountMigration/action	StorageAccounts_CustomerInitiatedMigration	-
Storage.A188	List all the storage task assignments in an account.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/read	-	StorageTaskAssignments_List	-
Storage.A189	Get the storage task assignment properties.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/read	-	StorageTaskAssignments_Get	-
Storage.A190	Asynchronously creates a new storage task assignment sub-resource with the specified parameters. If a storage task assignment is already created and a subsequent create request is issued with different properties, the storage task assignment properties will be updated. If a storage task assignment is already created and a subsequent create or update request is issued with the exact same set of properties, the request will succeed.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/write	Microsoft.Storage/storageAccounts/storageTaskAssignments/write	StorageTaskAssignments_Create	-
Storage.A191	Update storage task assignment properties.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/write	Microsoft.Storage/storageAccounts/storageTaskAssignments/write	StorageTaskAssignments_Update	-
Storage.A192	Delete the storage task assignment sub-resource.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/delete	Microsoft.Storage/storageAccounts/storageTaskAssignments/delete	StorageTaskAssignments_Delete	-
Storage.A193	Fetch the report summary of all the storage task assignments and instances in an account.	Storage.FC1	Microsoft.Storage/storageAccounts/reports/read	-	StorageTaskAssignmentsInstancesReport_List	-
Storage.A194	Fetch the report summary of a single storage task assignment's instances.	Storage.FC1	Microsoft.Storage/storageAccounts/storageTaskAssignments/reports/read	-	StorageTaskAssignmentInstancesReport_List	-

Storage.A195	Gets a list of all the tables under the specified storage account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/read	-	Table_List	-
Storage.A196	Gets the table with the specified table name, under the specified account if it exists.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/read	-	Table_Get	-
Storage.A197	Creates a new table with the specified table name, under the specified account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/write	Microsoft.Storage/storageAccounts/tableServices/tables/write	Table_Create	T28
Storage.A198	Creates a new table with the specified table name, under the specified account.	Storage.FC5	Microsoft.Storage/storageAccounts/tableServices/tables/write	Microsoft.Storage/storageAccounts/tableServices/tables/write	Table_Update	T28
Storage.A199	A failover request can be triggered for a storage account in the event a primary endpoint becomes unavailable for any reason. The failover occurs from the storage account's primary cluster to the secondary cluster for RA-GRS accounts. The secondary cluster will become primary after failover and the account is converted to LRS. In the case of a Planned failover, the primary and secondary clusters are swapped after failover and the account remains geo-replicated. Failover should continue to be used in the event of availability issues as Planned failover is only available while the primary and secondary endpoints are available. The primary use case of a Planned failover is Disaster Recovery testing drills. This type of failover is invoked by setting "FailoverType" parameter to "Planned". Learn more about the failover options here ref.	Storage.FC1	Microsoft.Storage/storageAccounts/failover/action	Microsoft.Storage/storageAccounts/failover/action	StorageAccounts_Failover	T59
Storage.A200	Delete a storage connector.	Storage.FC1	Microsoft.Storage/storageAccounts/connectors/delete	-	-	-
Storage.A201	Returns the list of storage connectors or gets the properties of the specified storage connector.	Storage.FC1	Microsoft.Storage/storageAccounts/connectors/read	-	-	-
Storage.A202	Test the connection of an existing storage connector.	Storage.FC1	Microsoft.Storage/storageAccounts/connectors/testExistingConnection/action	-	-	-
Storage.A203	Creates or updates a storage connector.	Storage.FC1	Microsoft.Storage/storageAccounts/connectors/write	-	-	-
Storage.A204	Delete a storage data share.	Storage.FC1	Microsoft.Storage/storageAccounts/dataShares/delete	-	-	-
Storage.A205	Returns the list of storage data shares or gets the properties of a specified storage data share.	Storage.FC1	Microsoft.Storage/storageAccounts/dataShares/read	-	-	-
Storage.A206	Creates or updates a storage data share.	Storage.FC1	Microsoft.Storage/storageAccounts/dataShares/write	-	-	-
Storage.A207	Revert file share.	Storage.FC1	Microsoft.Storage/storageAccounts/fileServices/shares/revert/action	-	-	-